



University Of Tripoli
Faculty Of Engineering

Department Of Petroleum Engineering



Spring-2024
Part- 2

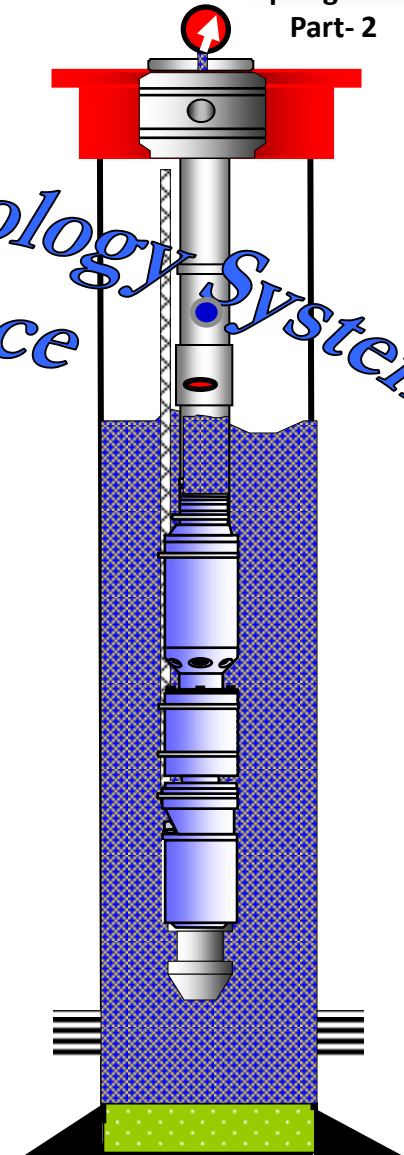
Electrical Submersible Pump Technology System *Installation & Performance* **(ESP)**

Production Engineering Design #2
PE473

Instructor :

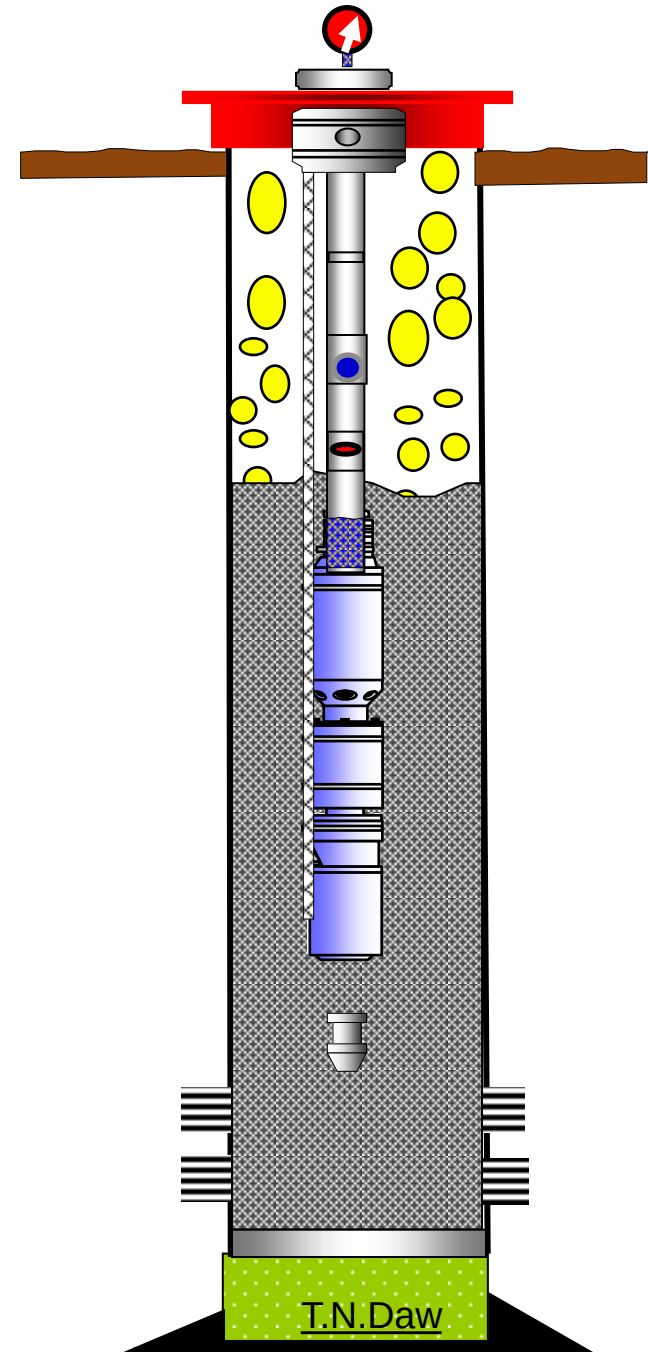
Tarek . N . A . Daw

Production Operations & Workover Engineer Specialist



Typical ESP Installation ?

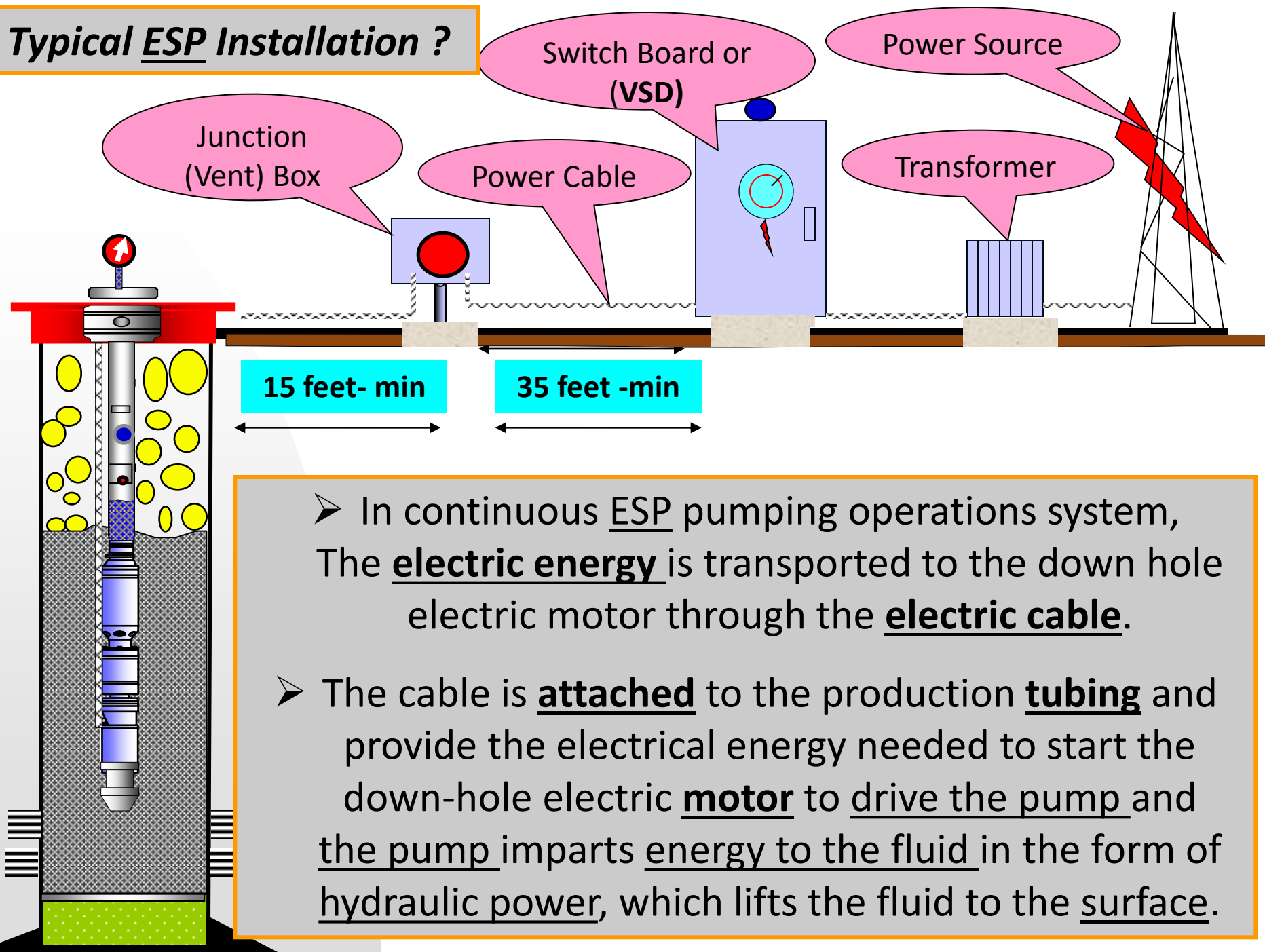
- The **ESP- unit** is run on tubing string and is submersed in well fluids.
- The electric submersible **Motor** is at the bottom of the unit, it is connected to the **Protector** (seal section).
- On top of the protector a **Pump Intake or Gas Separator** is situated which allows well fluids to enter the **Centrifugal Pump** that is located at the top of the ESP string.
- The **submersible motor** is supplied with three-phase AC current via an Electric Cable running from the surface all along the tubing string.
- Produced fluids flow through the tubing string to the surface, where a special Wellhead ensures feeding of the electric cable into the surface.
- A **Junction Box** connects the down-hole and surface electric cables and a control unit called Switchboard that provides measurement and control functions.
- The ESP unit is powered by AC electricity from a set of Transformers which supply the required voltage by stepping up or down the voltage available from the surface electric network. But it can be provided by a portable engine generator unit.



Course Objectives

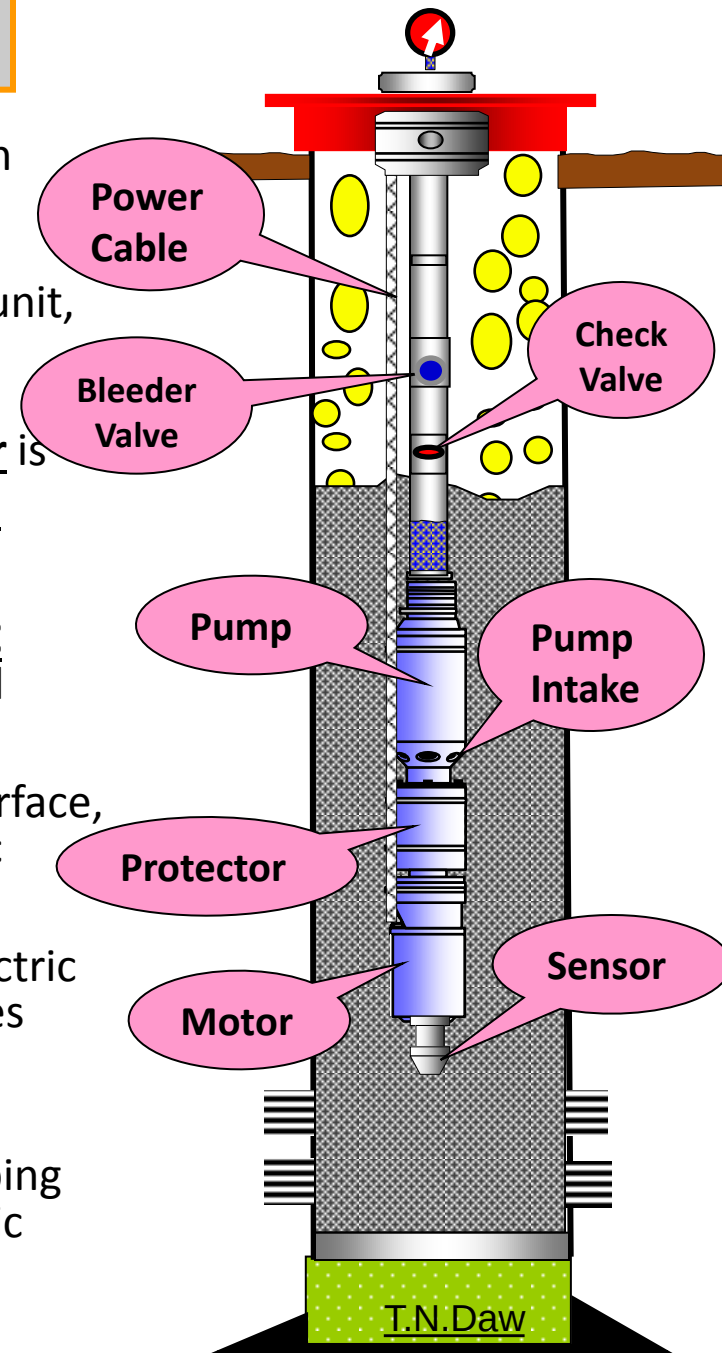
- 1-** List the major components of surface & sub-surface of an ESP system.
- 2-** Overview of accessory equipment and each components function.
- 3-** Completely understand ESP performance curves.
- 4-** Explain the causes & remedies of ESP failures.
- 5-** Understand TDH & the head-volume relationship.
- 6-** Describe how the pumping unit operates.

Typical ESP Installation ?



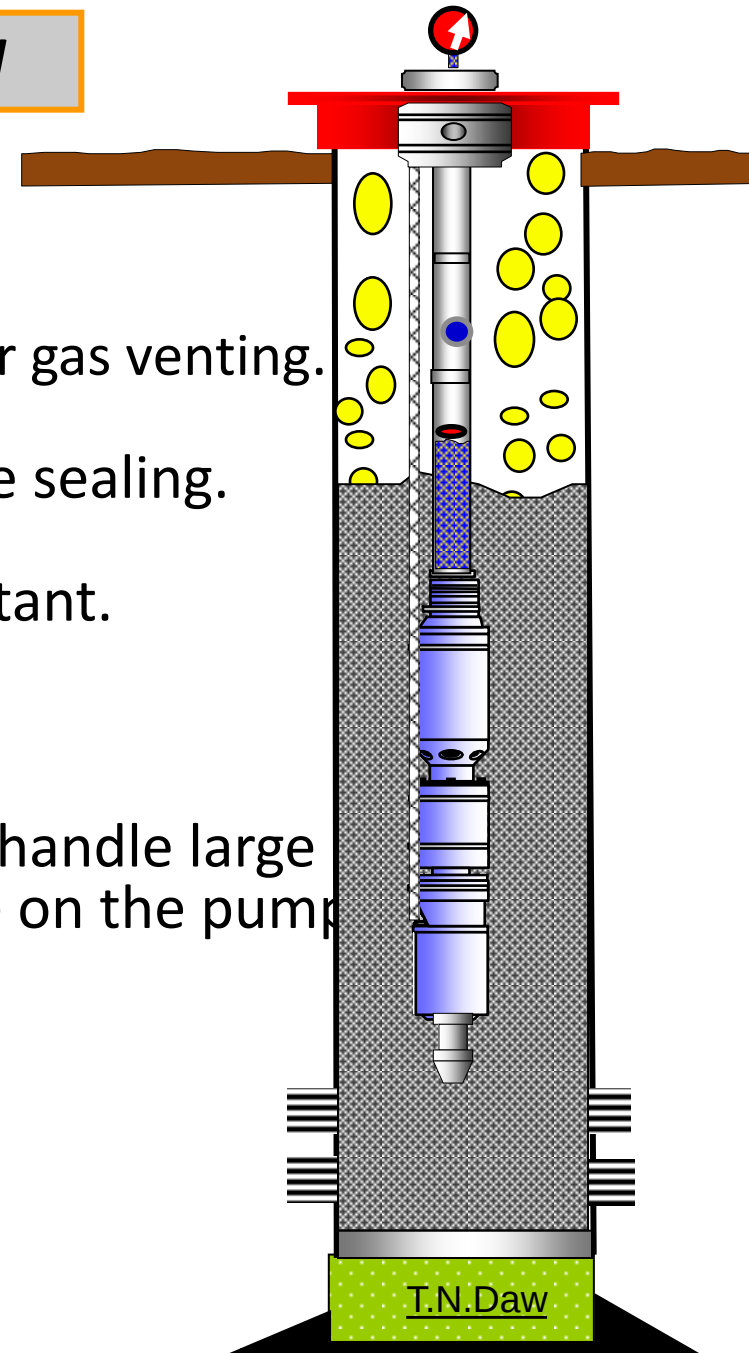
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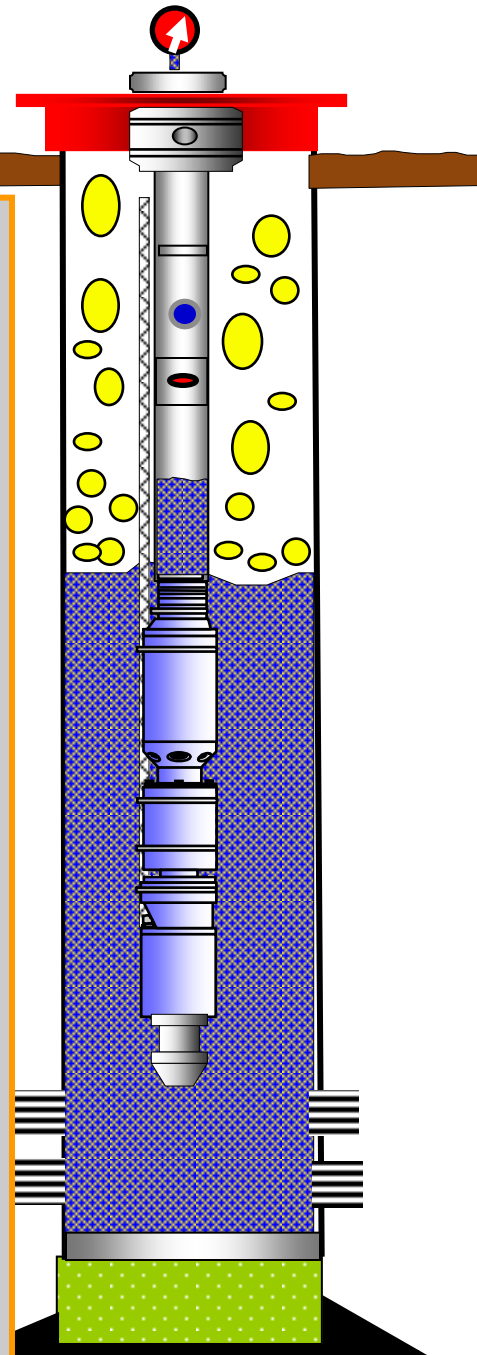
Attentions for ESP Operated

- The annulus is open to atmosphere for gas venting.
- Special well head is required for cable sealing.
- Power cable protection is very important.
- Motor cooling must be adequate.
- The tubing size must be adequate to handle large volumes with minimum backpressure on the pump

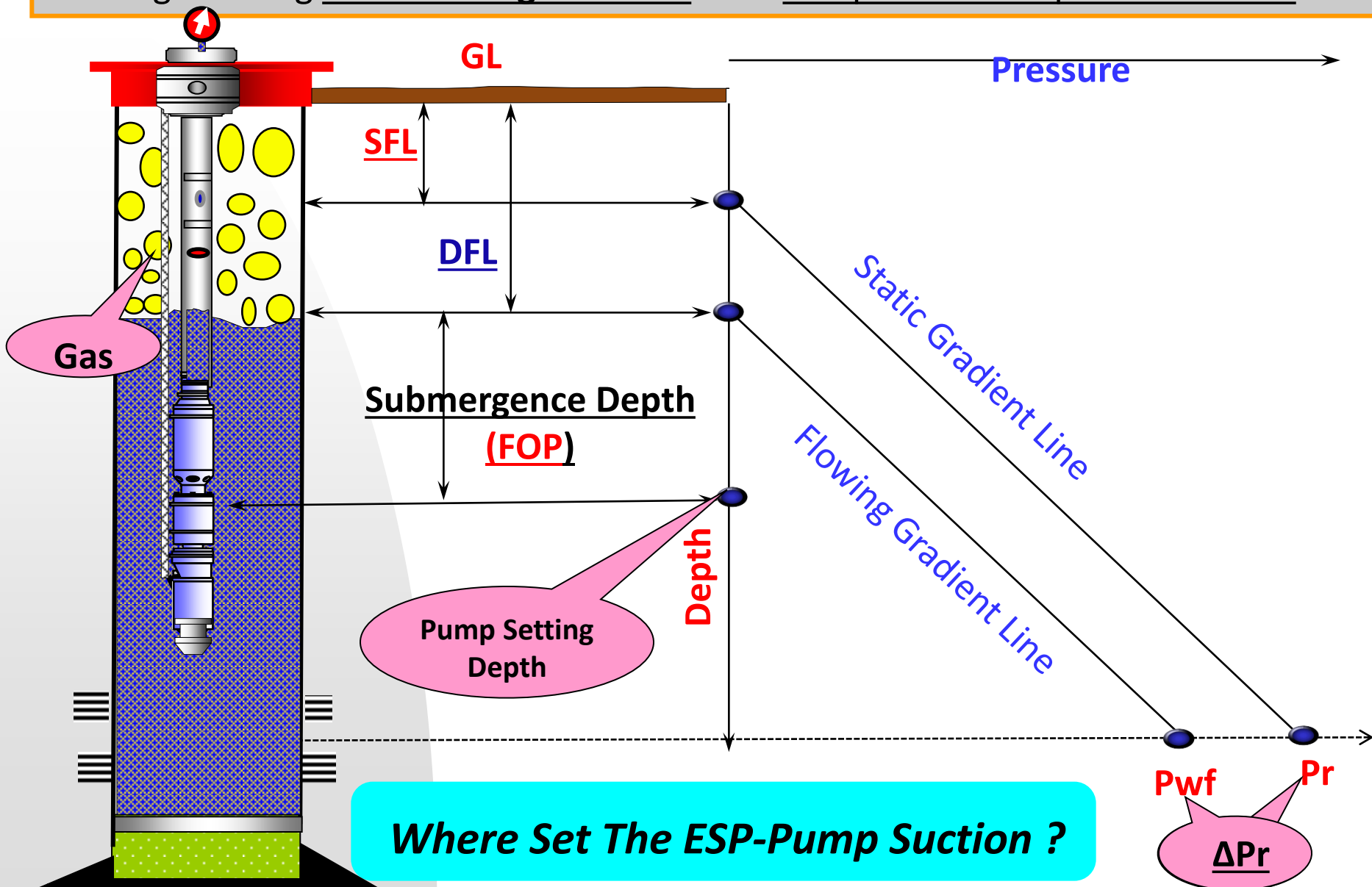


How the ESP Unit works ?

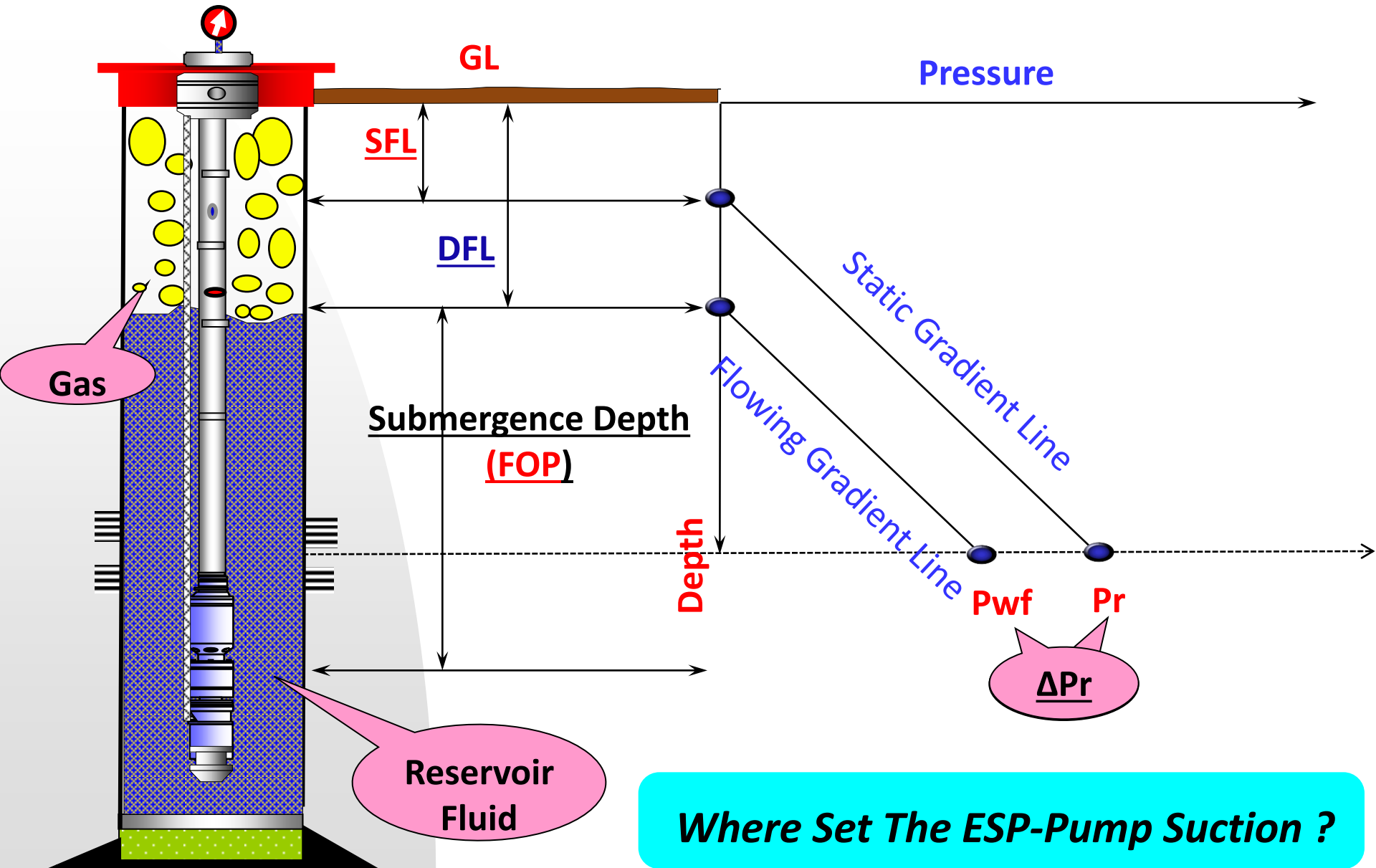
- When the ESP is started it draws the required voltage from the source.
- This powers generates pump and motor thus rotation.
 - As the fluid comes into the well through the perforations it passes by the motor and into the pump. This fluid flowing past the motor brings about a natural cooling process, critical for long and efficient motor life. The fluid then enters the intake and is taken into the centrifugal pump.
 - Each stage (impeller/diffuser combination) adds pressure or head to the fluid at a given rate.
 - The fluid will build up enough pressure as it reaches the top of the pump to lift it to the surface and into the separator .



The "standard" ESP-Unit configuration requires motor to be set above the perforations, in such a way that the well fluid moves up-ward around the motor housing allowing excess heat generated to be dissipated in the produced fluid.



Set the pump below perforation maximises potential draw down (increase production). And provide max. area for gas separation.



Where Set The ESP-Pump Suction ?

ESP System Application

Constraints	Typical Range	Maximum
Operating Depth	1,000' - 10,000' TVD	15,000' TVD
Operating Volume	200 - 20,000 BPD	60,000 BPD
Operating Temperature	100° - 275° F	400° F
Wellbore Deviation	10°	0 - 90°
Corrosion Handling	Good	
Gas Handling	Poor to Fair	
Solids Handling	Poor to Fair	
Fluid Gravity	>10° API	
Servicing	Workover Rig	
Prime Mover Type	Electric Submersible Motor	
Offshore Application	Excellent	
System Efficiency	35%-60% depending on pump type, fluid volume	

Advantages Of ESP Pump

- Suitable (Adaptable) for all Wells With above 5" Casing and high water cut.
- High Volume and Depth Capability.
- Minor Surface Equipment Needs.
- Simple to design, operate and low maintenance.
- Can be used in horizontal and deviated wells, but must be set in straight section .

Dis-advantages Of ESP Pump

- Only applicable with high voltage electric power.
- Repair of ESP equipment in oilfield conditions is difficult.
Faulty equipment must be sent to the manufacturer's repair shop.
- Expensive to change equipment to match declining well productivity.

The use of VSD can eliminate most of these problems but at an extra cost

- Gas, sand and solids production are restrict capacity and operating efficiency.
- Not suitable to deep and high temperature oil reservoirs, any type of survey test can be conducted only through (Y-Tool).
- Costly to install ESP systems have higher horsepower, tubing has to be pulled to replace pump (need Workover rig, cost of rig time).

ESP System Limitations

(Temperature at pump seating depth)

- Cable insulation weakens and will limit the life of the thrust bearing.

(Production casing / liners)

- Are required fit with pumping unit to avoid damage cable and good result for fluid velocity (Motor cooling).

(Free gas in oil)

- Excessive free gas results in pump cavitation that leads to motor fluctuations and reduces run life and reliability.

(Viscosity of oil)

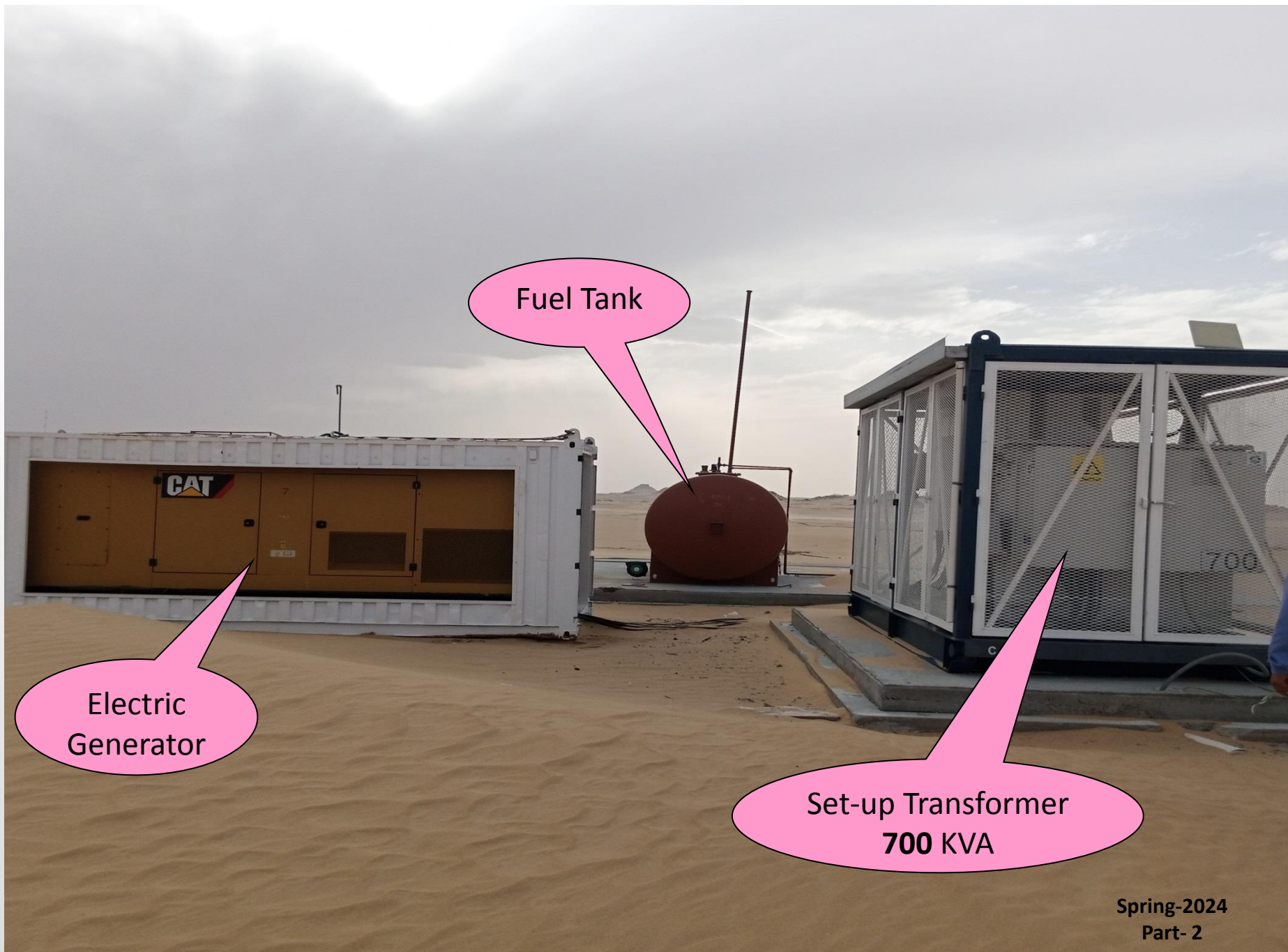
- Reduces the total head that the pump system can generate, which leads to an increased number of pump stages and horsepower requirements (**high cost**).

ESP Surface Components

- *The Electric Power Supply (High Line or Generator)*
- *The Electric Transformer.*
- *The Control Switchboard or VSD (*Motor controller*).*
- *The Junction (Vent) Box.*
- *The Well Head Assembly.*

Electric Power Supply

*(**High Line 11 kv**)*



Fuel Tank

Electric
Generator

Set-up Transformer
700 KVA

Electric Transformer

- The transformer sends the correct desired surface operating voltage to the switch board.
- The propose of the transformer is to provide correct surface voltage to proper submersible motor operation by changing electric power from a high voltage to low (Down-stepping).
- Or from a low voltage to high (Up-step).
- This based on power rating in KVA.



Control Switch Board (Fixed Speed Drive)

Switch Board Selected by voltage & HP values for the motor pump.

What does a Switch board Do ??

- **Power Supply Of ESP Unit.**

(provide accurate current & voltage for the motor).

- **Protection Of ESP Unit (monitors) :-**

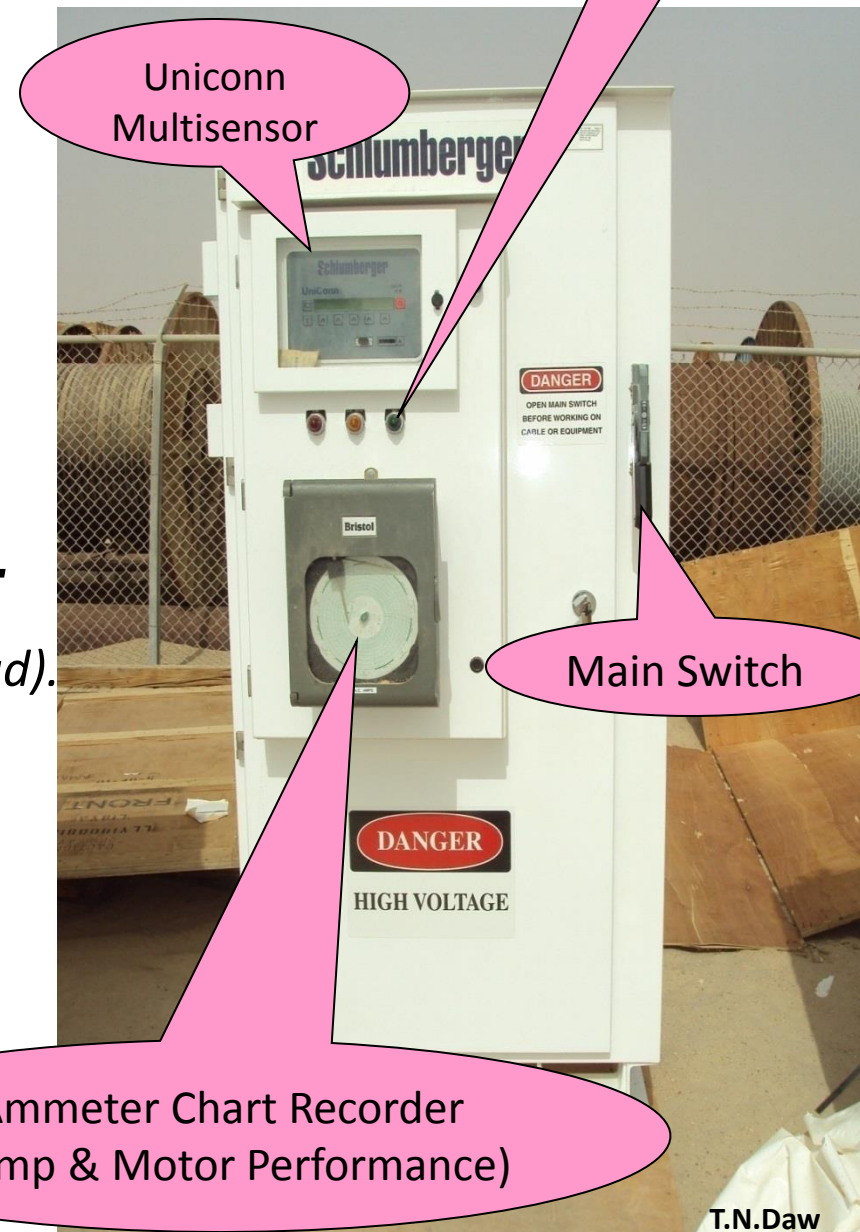
a- The running current (Overload & Under load).

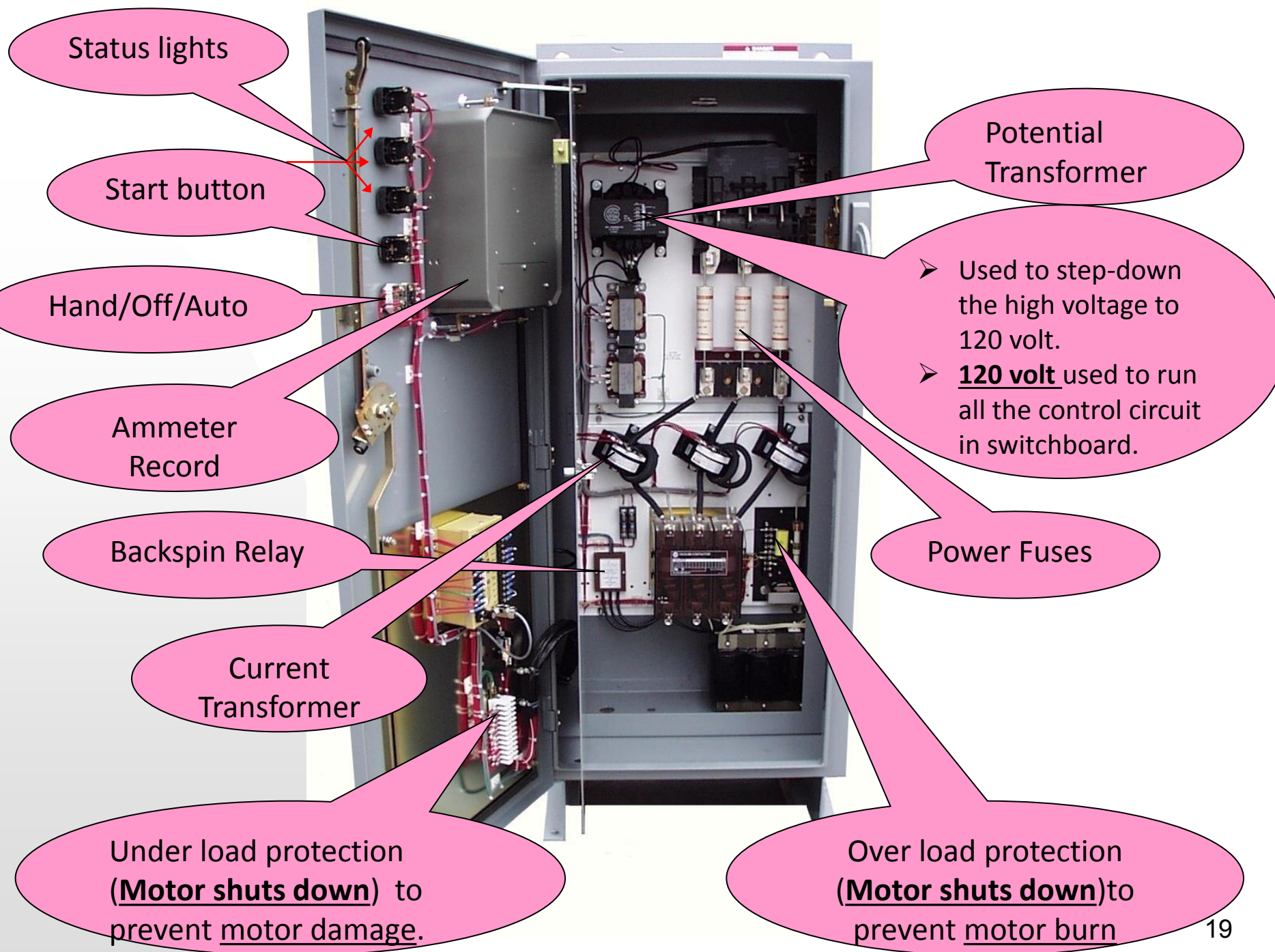
b- The voltage(Over volt & Under volt).

c- The frequency.

- **Control of ESP Unit (Motor starter).**

- **Diagnostics Of ESP Unit.**

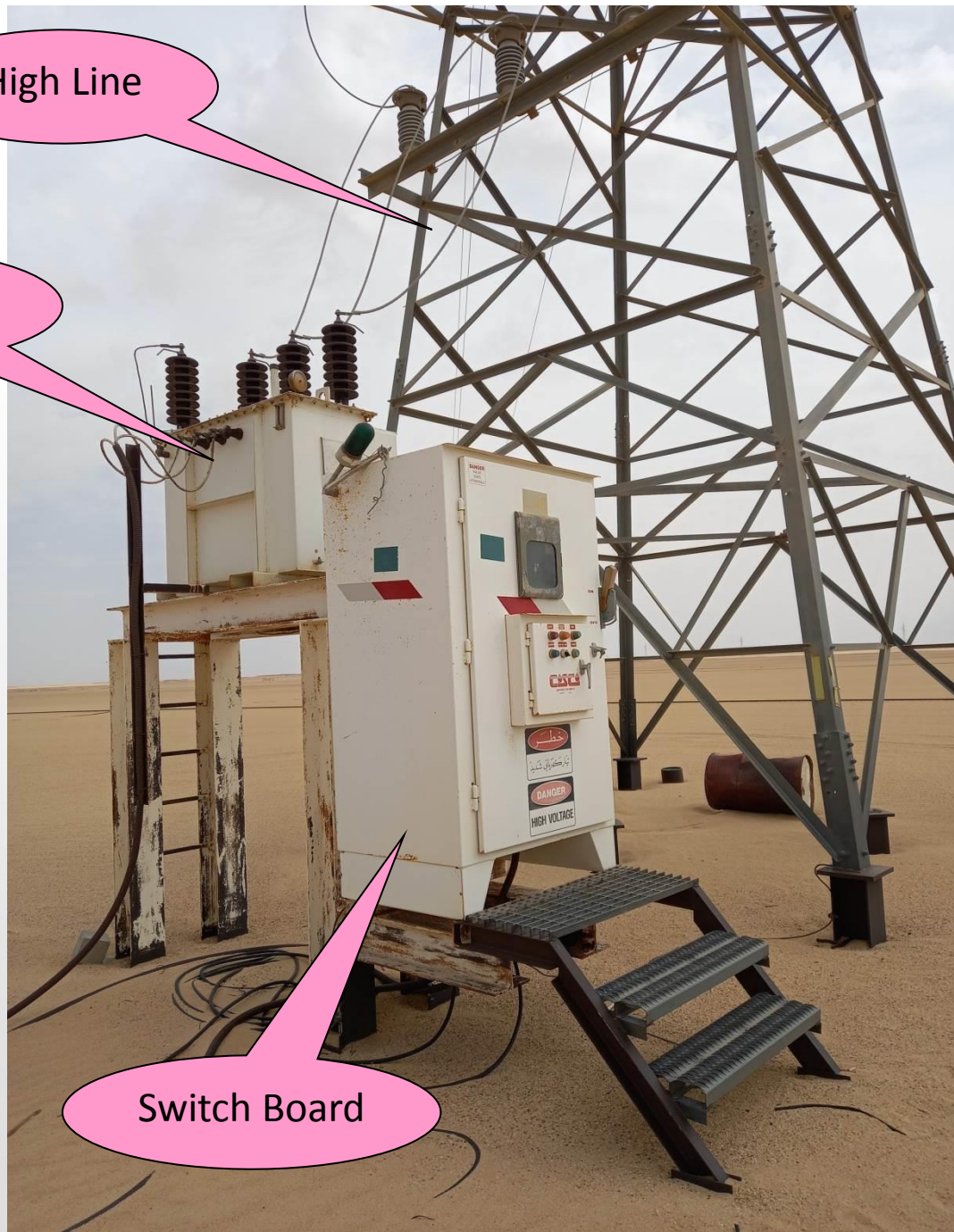




High Line

Transformer

Switch Board



Control Power Skid (Switch Board)



Generator

Switch Board

Transformer

Variable Speed Drive (VSD)

Introduction:

An ESP-system can be operated by using power supply with fixed (**speed**) frequency **50** or **60** Hz or using a **VSD** which is used to change the fixed frequency to other frequencies (record by **SCADA**).

*What does a **VSD** Do ??*

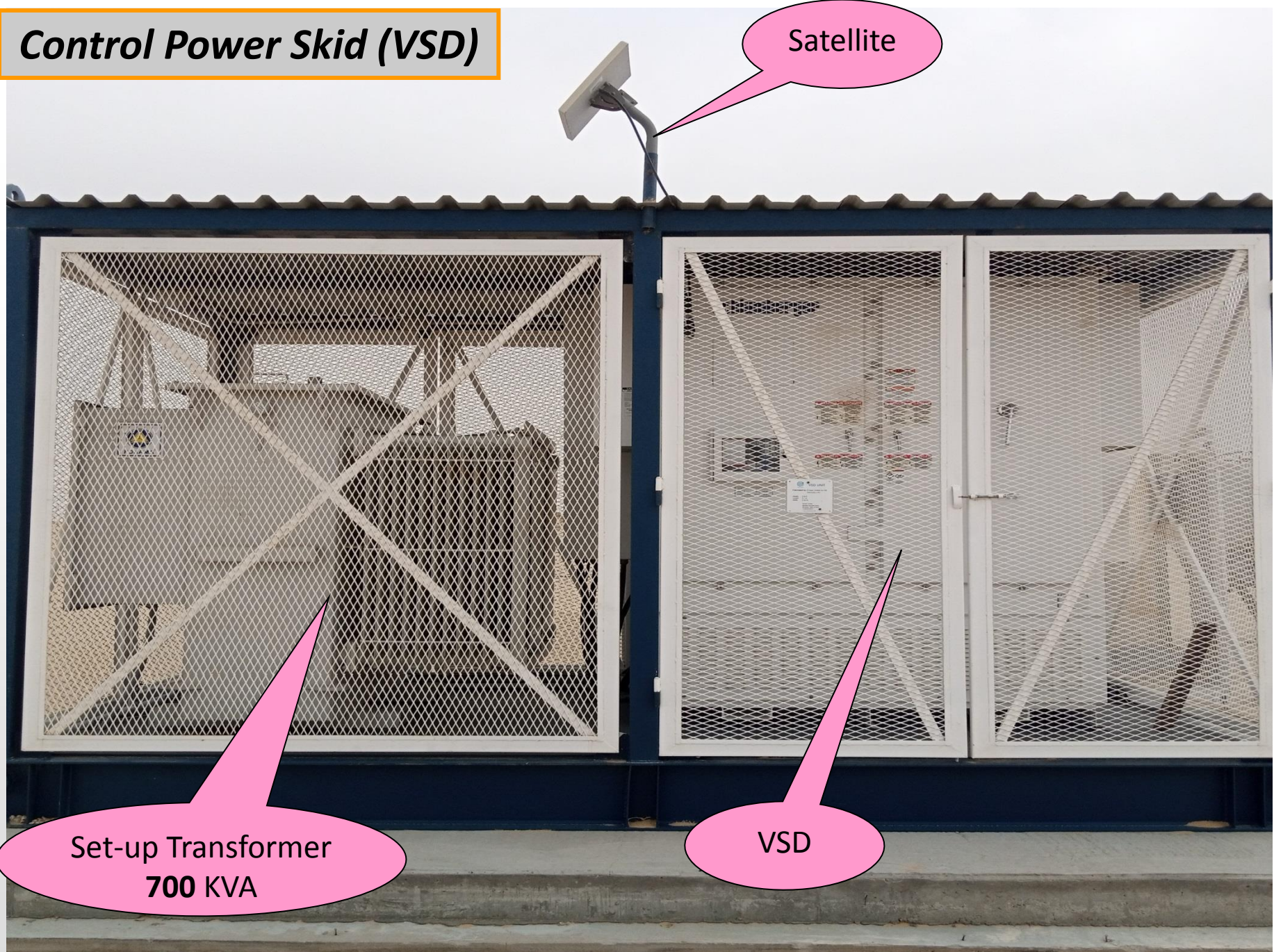
- It allow to adjustments the pump capacity (**BEP**)by changeable the motor speed (RPM) to achieve best run life.
- As the pump **wears**, the speed can be adjusted to maintain production (WOR, GOR, Pr,...etc) are not stabilized.
- To minimize operational costs and maximize run life.

Control Power Skid (VSD)

Satellite

Set-up Transformer
700 KVA

VSD



Variable Speed Drive Operation

ESPs may also be run at variable speeds.
Changing the speed or frequency of an ESP system.

“Affinity Laws”

$$New.flow = Old.flow \left[\frac{New.Hz}{Old.Hz} \right]$$

$$New.head = Old.head \left[\frac{New.Hz}{Old.Hz} \right]^2$$

$$New.Break.HP = Old.Break.HP \left[\frac{New.Hz}{Old.Hz} \right]^3$$



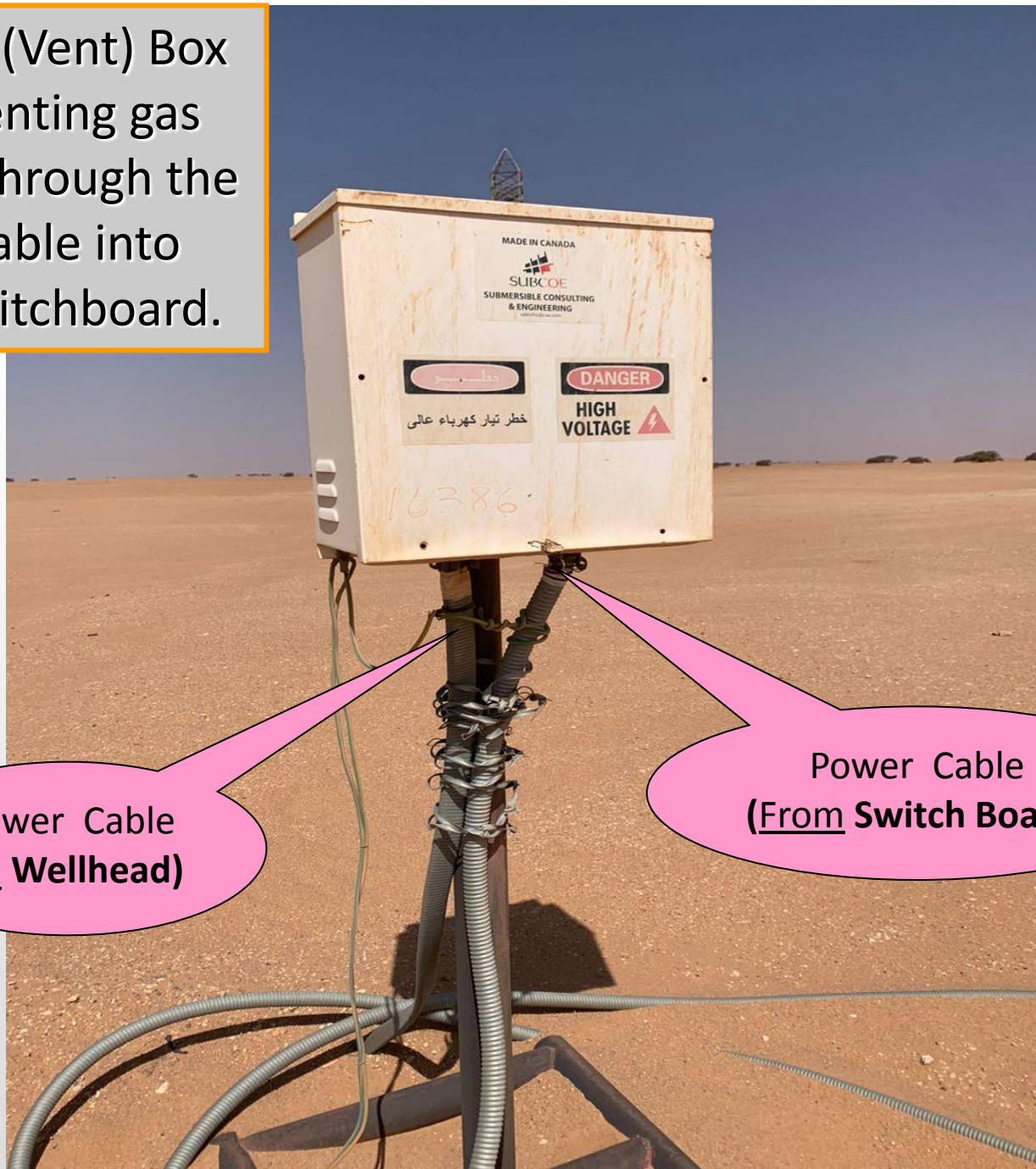
Junction (Vent) Box

Located between the well head & switch board for safety reasons.

What does a Junction Box Do ??

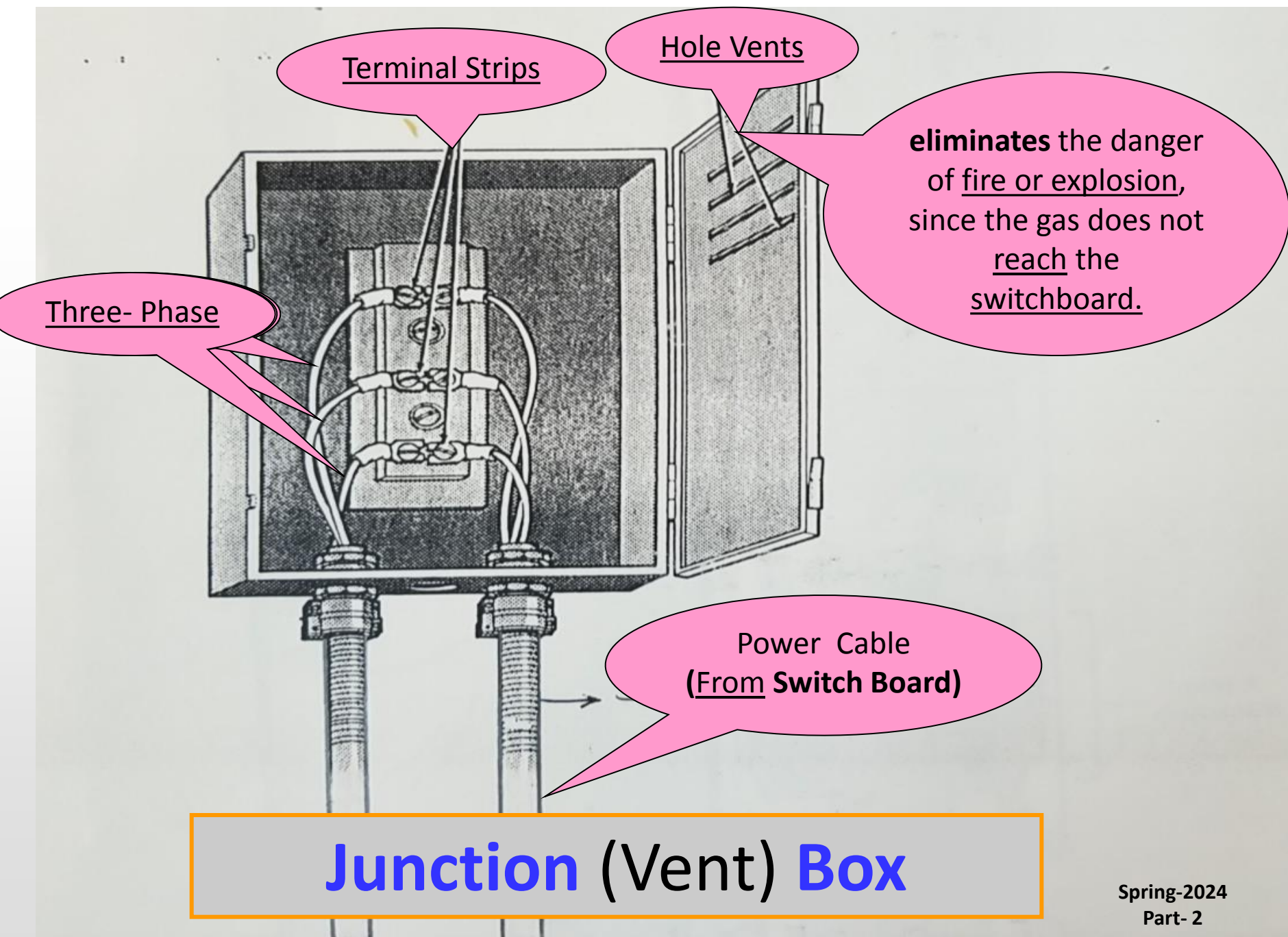
- 1-To provide an electrical connection between surface and down hole cables.
- 2- Provides a vent to the atmosphere for any Free gas that might migrate through the power cable into switchboard to avoid sparks (separates the surface cable from the down-hole cable).
- 3- Test point to change the motor rotation.
- 4-Test point for electrically checking the ESP – down-hole equipment.

A junction (Vent) Box will prevent gas migration through the power cable into control switchboard.



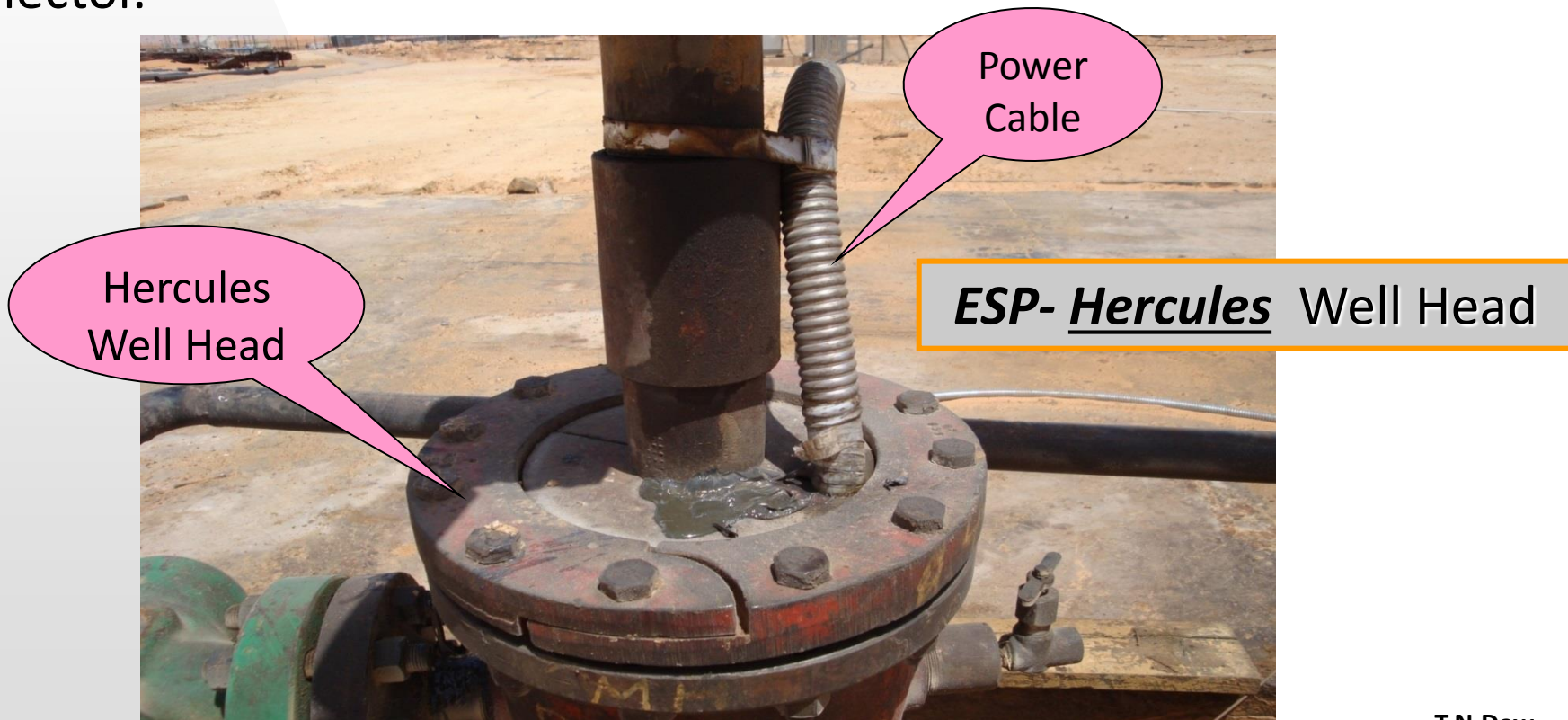
Power Cable
(To Wellhead)

Power Cable
(From Switch Board)

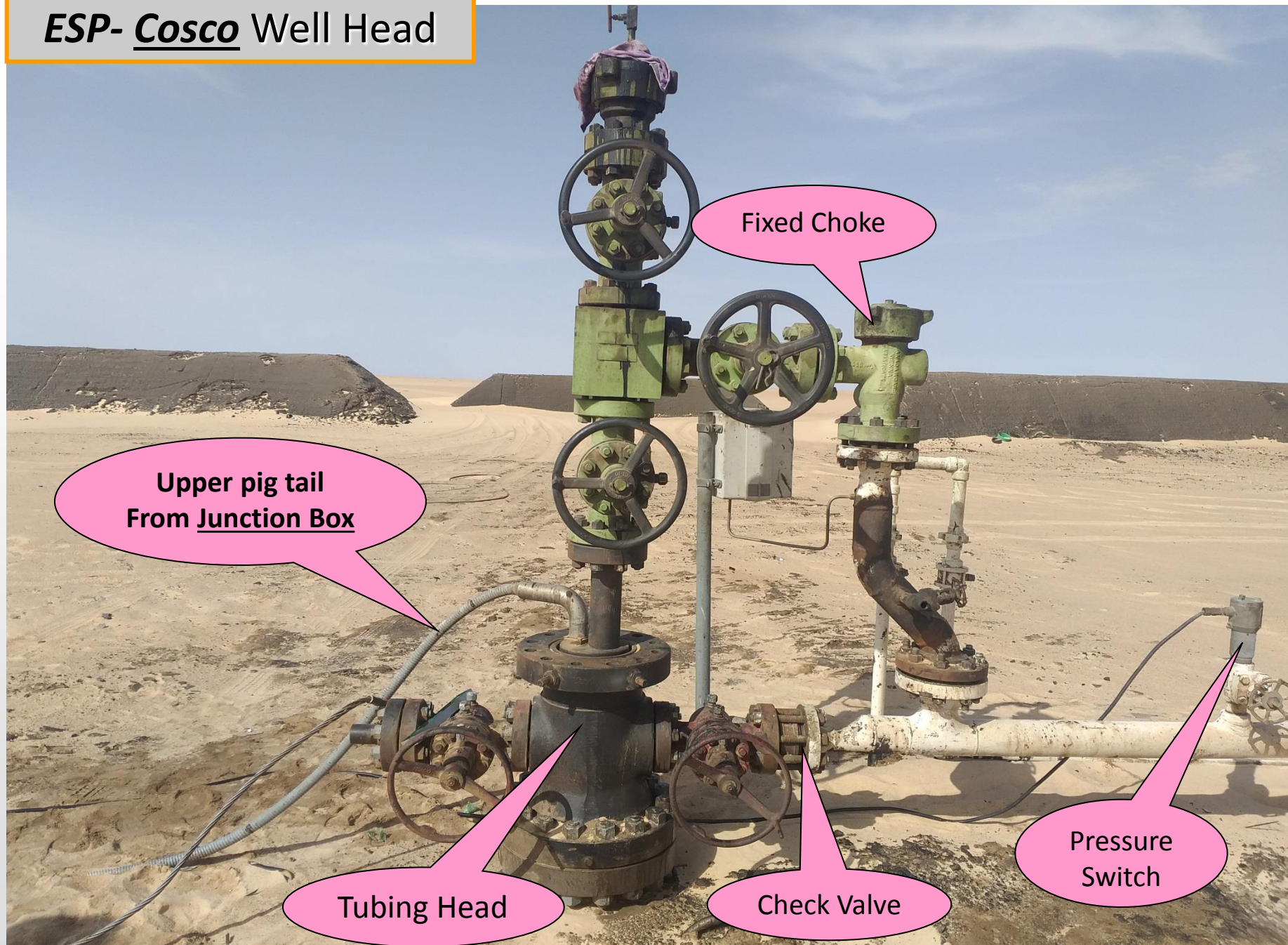


ESP- Well Head

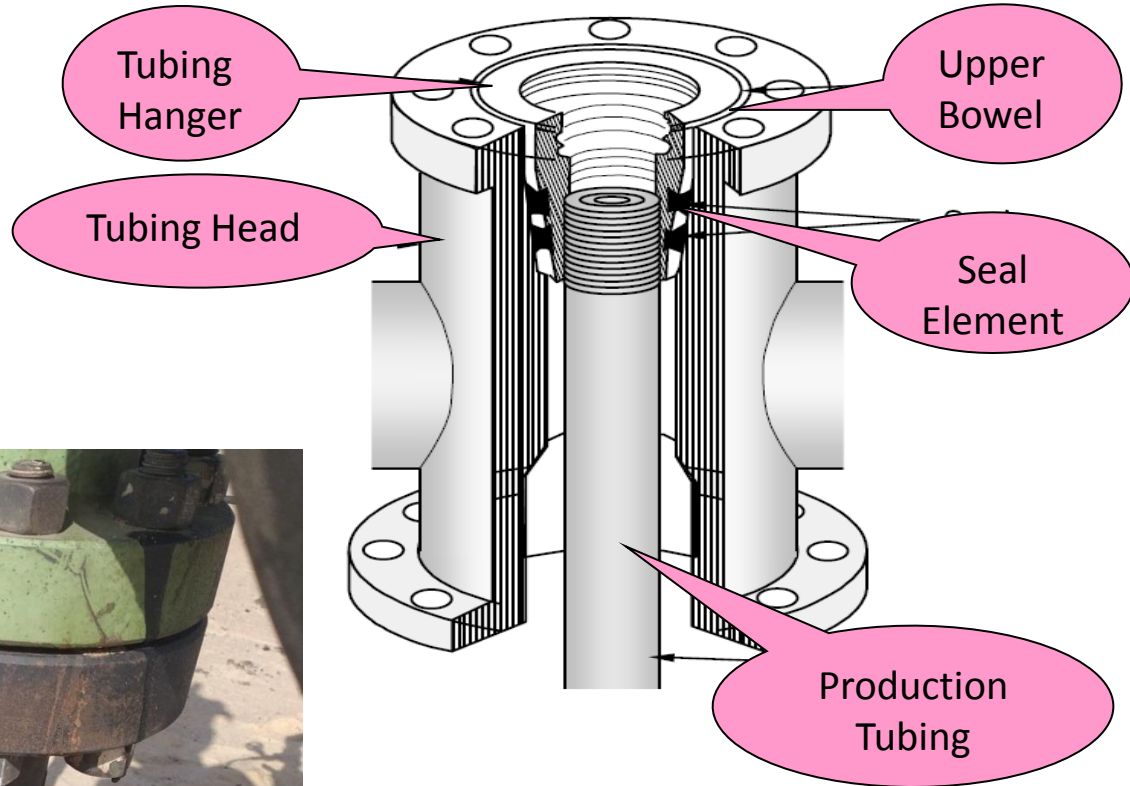
The **ESP- wellhead** is to maintain tubular control, as well as providing a positive seal around the cable and the tubing. There exist many different wellheads, with different types. The **Hercules wellhead** feeds the down-hole power cable directly through the wellhead. In other wellheads, the power cable is cut and its end is equipped with a power connector.



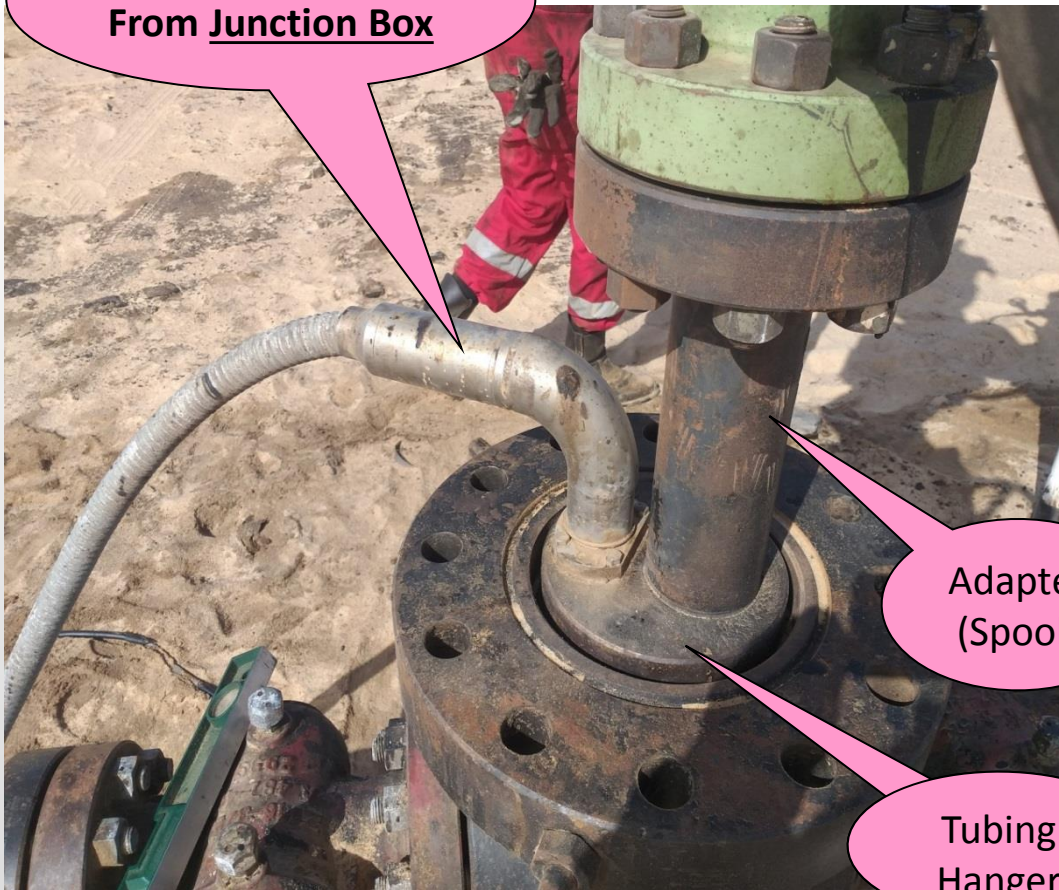
ESP- Cosco Well Head



ESP- Cosco Well Head



Upper pig tail
From Junction Box

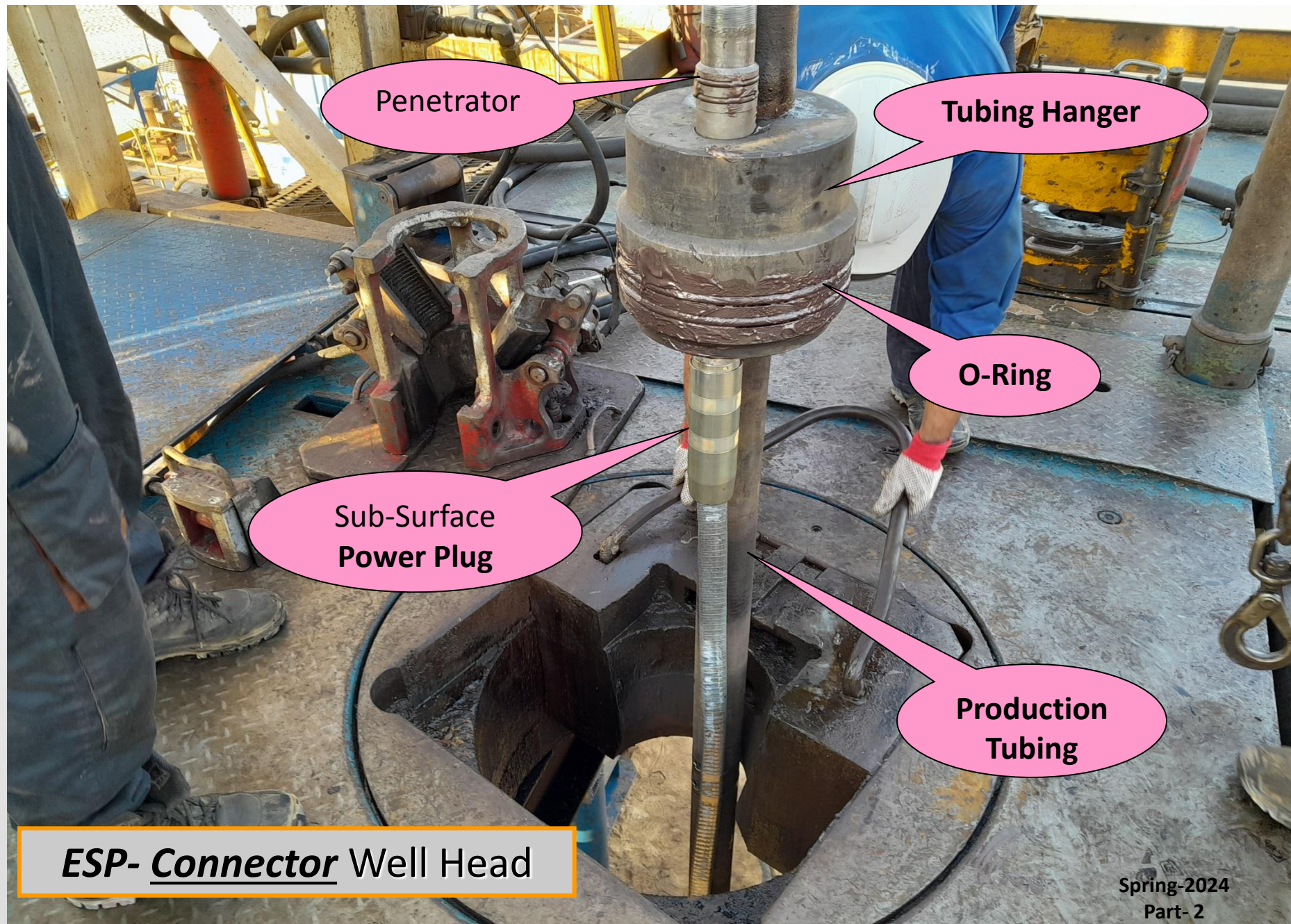


Adapter
(Spool)

Tubing
Hanger



Seal
Element



Penetrator

Tubing Hanger

O-Ring

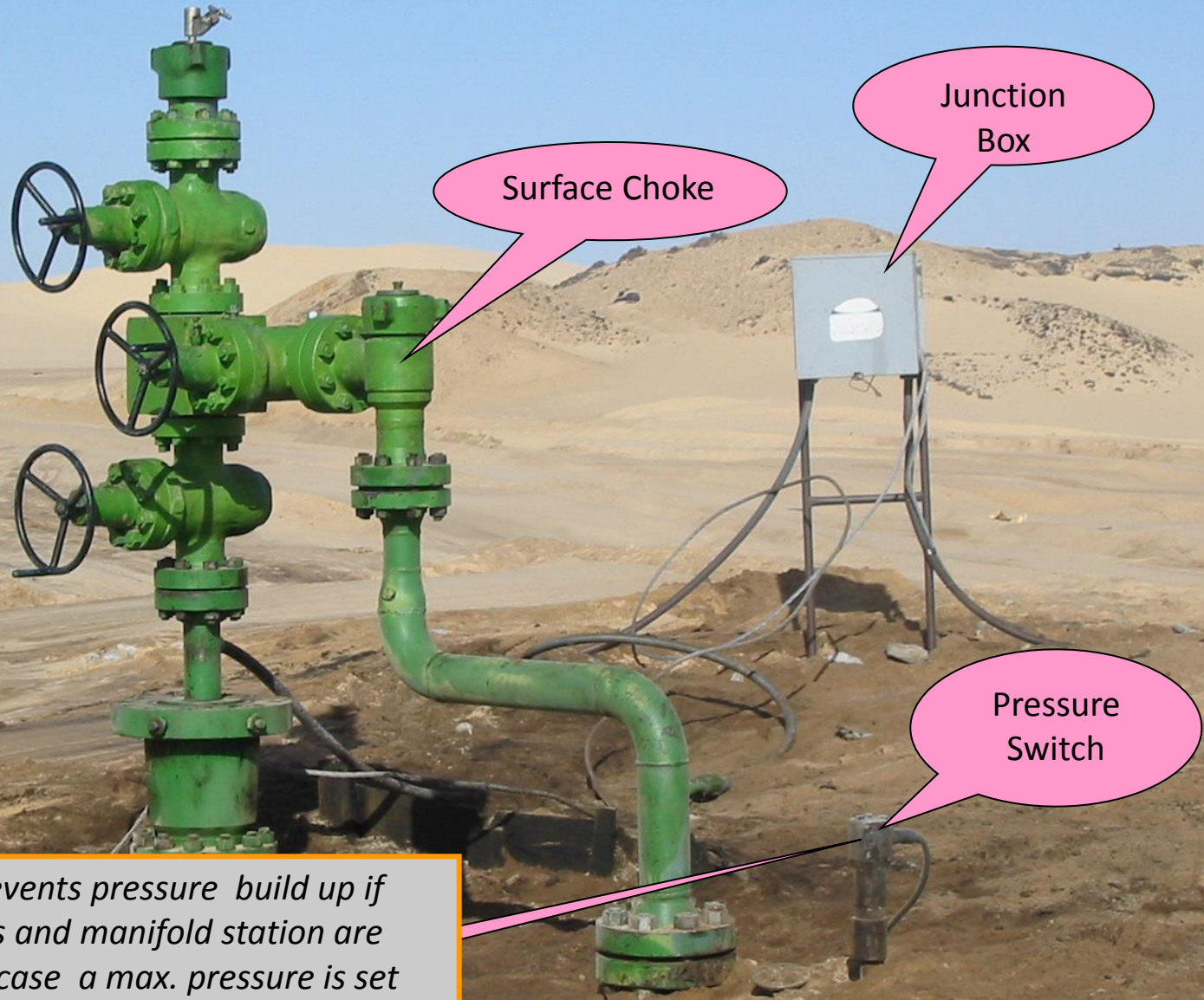
Sub-Surface
Power Plug

Production
Tubing

ESP- Connector Well Head

Spring-2024
Part- 2





High pressure occurs and prevents pressure build up if production flow line, valves and manifold station are closed or plugged. In this case a max. pressure is set in the switch at for example (+ or -300 psi) well head pressure which reading open the switch removing power from the control switch board.

ESP- Well (Christmas Tree)

Pressure
Gauge

Kill Valve

Production
Loop

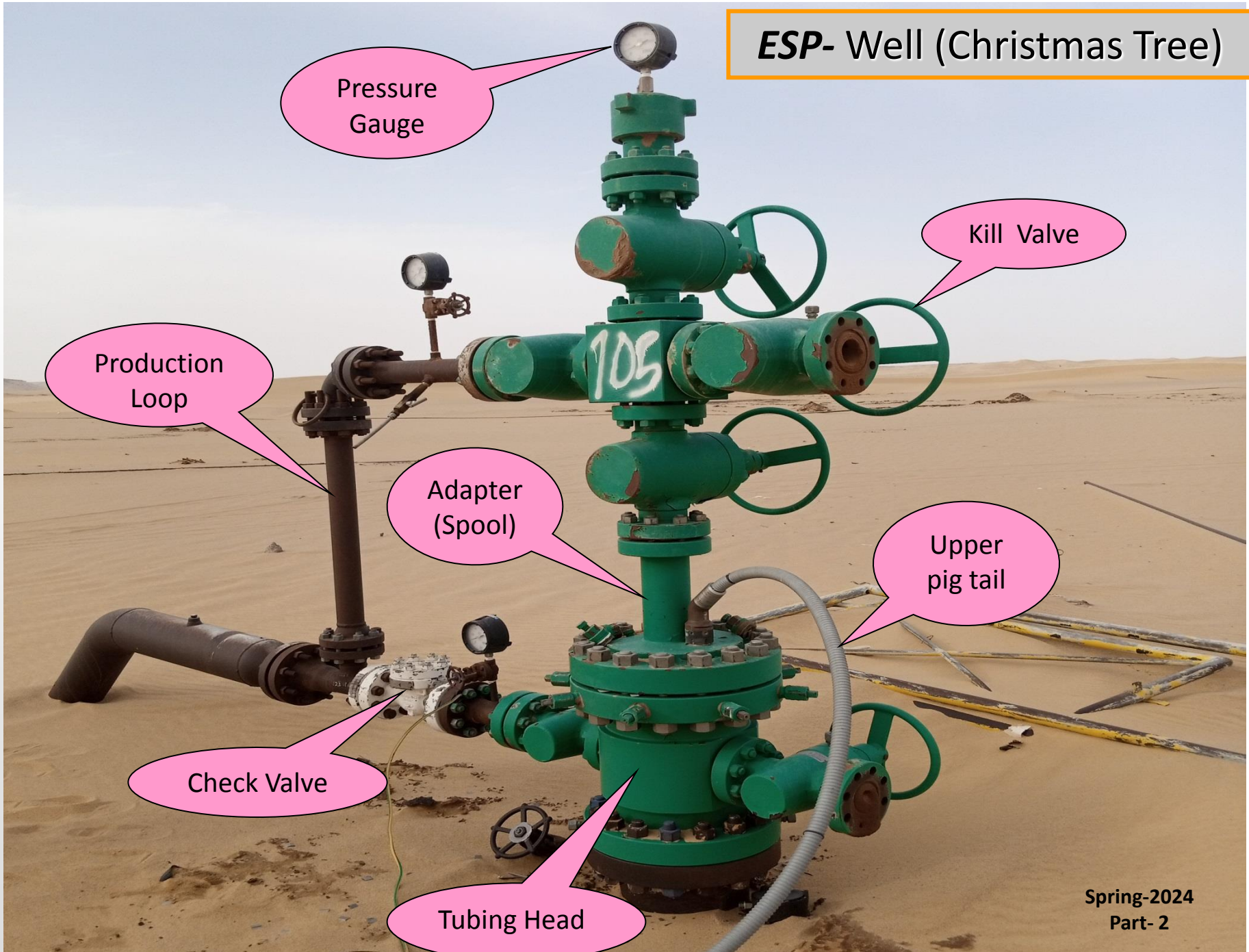
Adapter
(Spool)

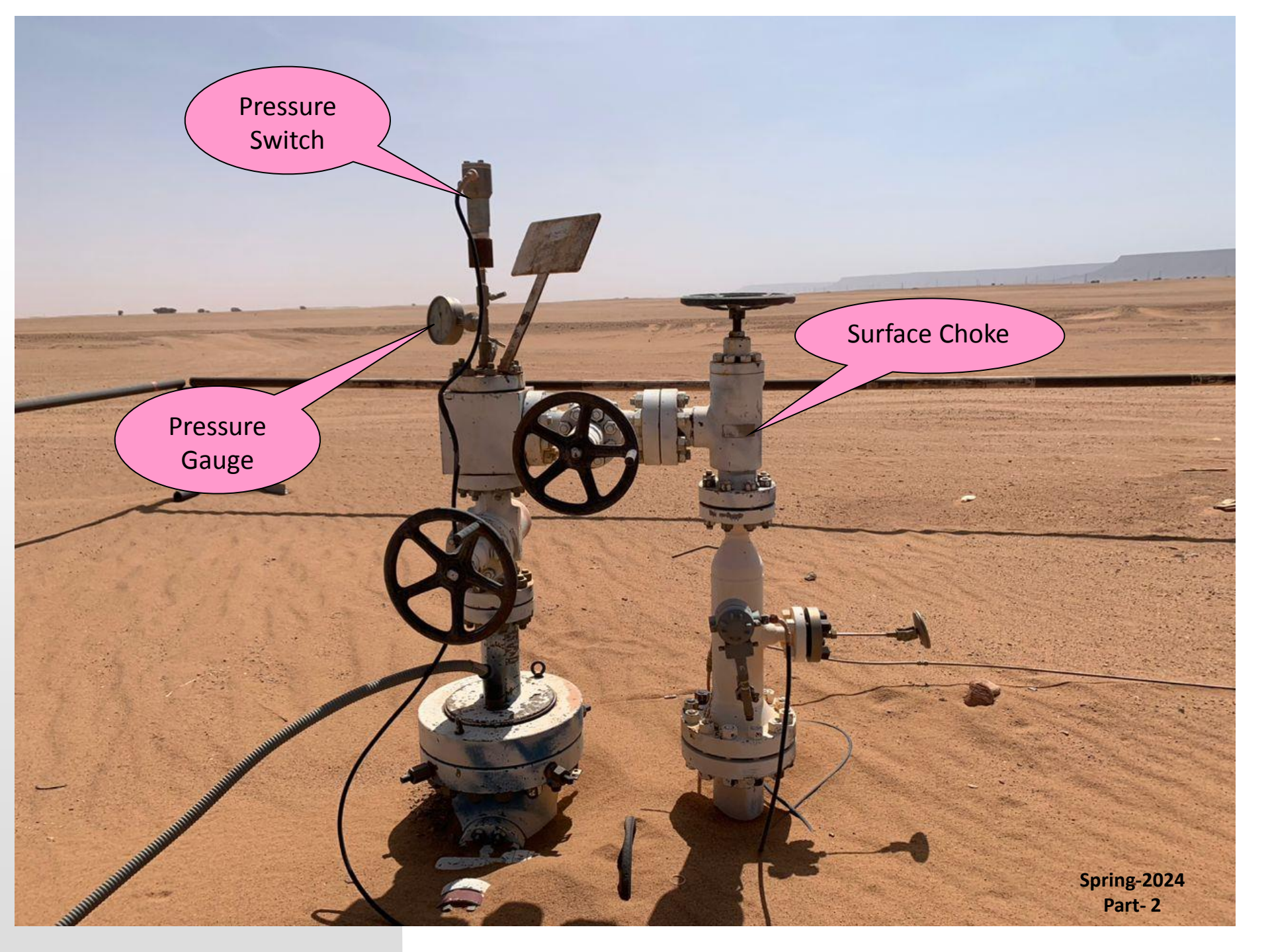
Upper
pig tail

Check Valve

Tubing Head

Spring-2024
Part- 2





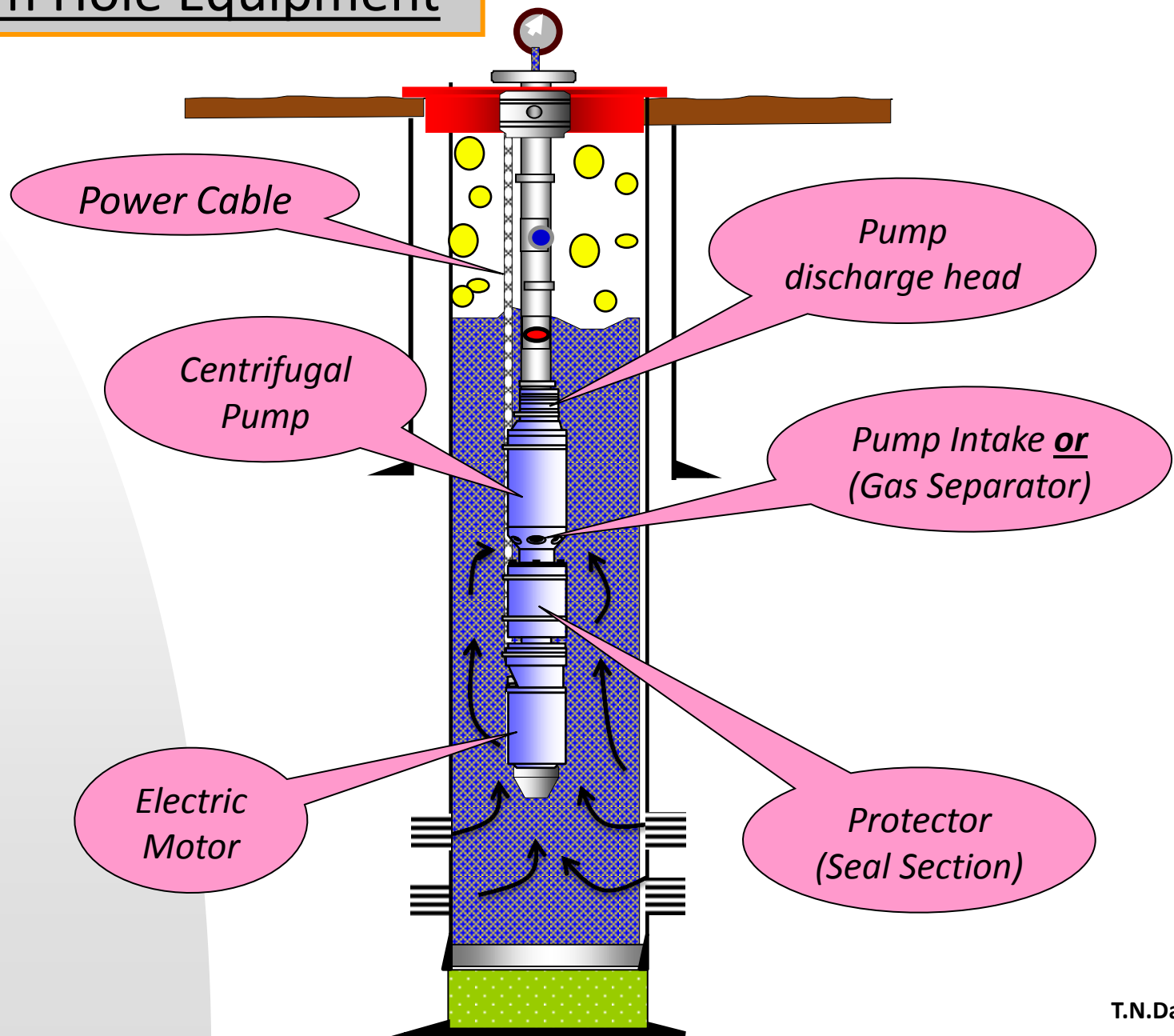
Pressure
Switch

The image shows two vertical wellhead assemblies in a desert. The left assembly has a pressure gauge and a pressure switch at the top. The right assembly has a surface choke. Both have large handwheels for valves. A corrugated hose is connected to the base of the left assembly. The background is a flat, sandy landscape under a clear sky.

Pressure
Gauge

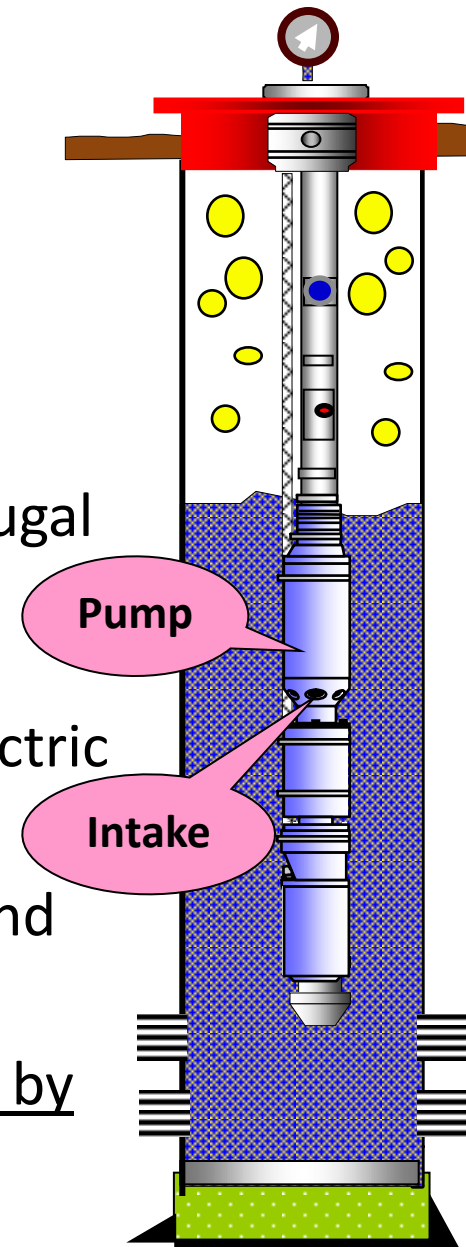
Surface Choke

ESP- Down Hole Equipment



ESP Centrifugal Pump

- Hangs from the production tubing.
- Lifts the fluid through the tubing to the surface.
- Is a multi-stage centrifugal or dynamic pump type, operating in a vertical position driven by motor.
- The electric motor connects directly to the centrifugal pump. This means that the electric motor shaft connects directly to the pump shaft by protector. So, the pump rotates at the same speed as the electric motor.
- The stage is constructed from rotating impellers and stationary diffusers.
- To supplies energy (Kinetic) in the form of velocity by rotation the fluid passing through it (**impellers**).

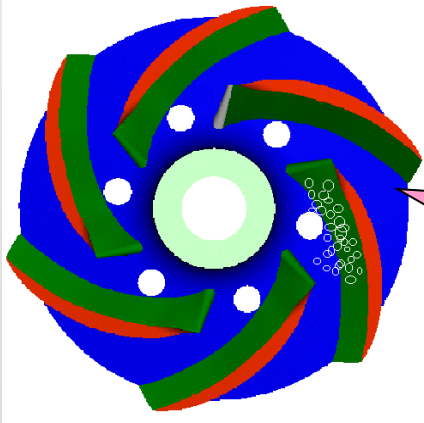




Rotating
Impeller



Stationary
Diffuser



Impeller

The pump capacity (volume) **increases** as the **diameter** of the impeller **increases**.

- Convert the kinetic energy from the rotating shaft into centrifugal forces to pressure (potential) energy (change the high velocity energy to low velocity energy to direction the flow to next impeller) that lift well fluids to surface and raising discharge pressure (**diffusers**).
- There are two types of impeller designs.
- The type or Geometry of the stage **controls** the volume of fluid production that can go through it.
- Number of stage **determine** the total dynamic head (***TDH***) generated and the motor horsepower required.
- The pumps can be connected together to create more lifting pressure. (Tandem Pump)
- The range of pump stages between 10 and more than 100 (depending on the pressure increase required).



Stationary
Diffuser

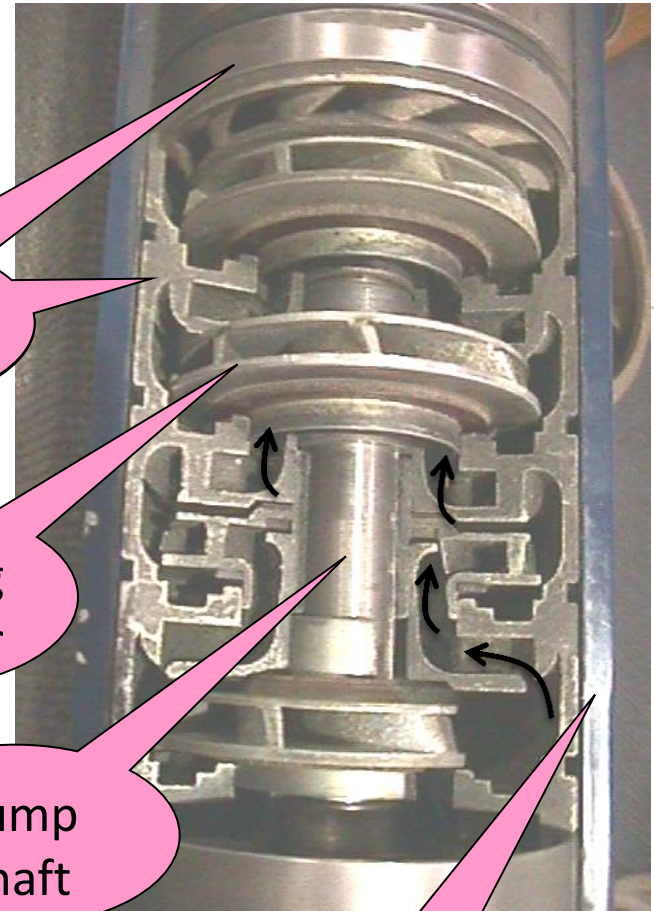
Rotating
Impeller

Pump
Shaft

Pump
Housing

Pump Stage

Stages are set in series along an axial shaft and compressed in a housing, increase the pressure of produced fluid and drive by the submersible electric motor.

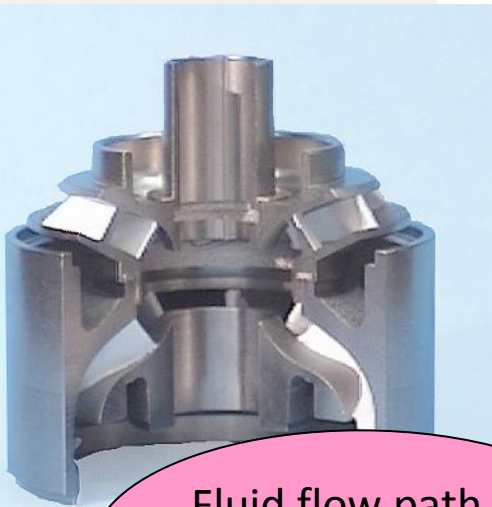


Impeller Geometries

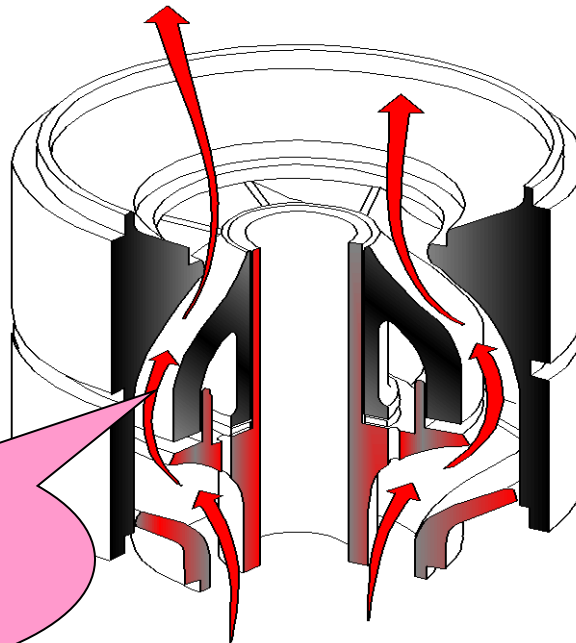
Mixed flow Impeller

A mixed flow impeller (The flow going from impeller and go **45 degree** to the shaft) and give more head.

Usually found in pumps designed for **higher flow rates** (>1,700 bpd).



Fluid flow path in a "mixed flow impeller"



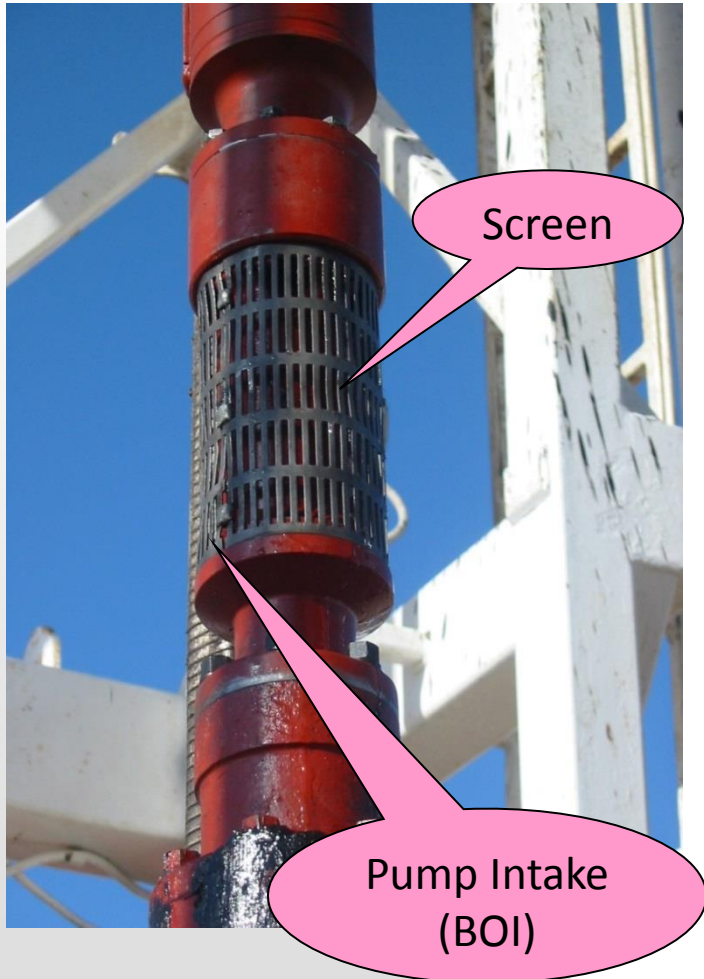
Radial flow Impeller

A radial flow impeller (The flow going from impeller and go **90 degree** to the shaft) and give less head.

Usually found in pumps designed for **lower flow rates** (<1,200 bpd).



ESP Standard Pump Intake (BOI)



- Where the well fluid enters the pump and is directed into the impellers.
- Different types of intakes are used depending on fluid properties.
- BOI is simply a perforated tube with a shaft connection between pump and protector
- Is screened to block large particles that would damage internal pump components.
- Designs do not separate gas.
- Using in wells that produce low GLR.
- Transmits torque from seal section (**Protector**) to pump.

ESP Gas Separator

Introduction:

- When free gas (Slug Flow) enters the pump section, the pumps performance is reduced. This is caused by the big difference in the specific gravities of liquids and gases. The existence of free gas at pump suction conditions, pressure and temperature affects the ESP pump in several ways:
- The head developed by the pump decreases.
- When producing gassy fluids, cavitation can occur at higher flow rates causing mechanical damage of the pump stages.
- When producing with very high gas rates, gas locking may occur when no pumping is taking place.

- In case the GLR high and low BHP, the well fluid contain amount of **free gas**. This effect on pump performance, the intake replaced by a gas separator.
- Gas separator designed to separate any free gas in liquid before enter to the pump.
- This minimizing the potential of gas locking.

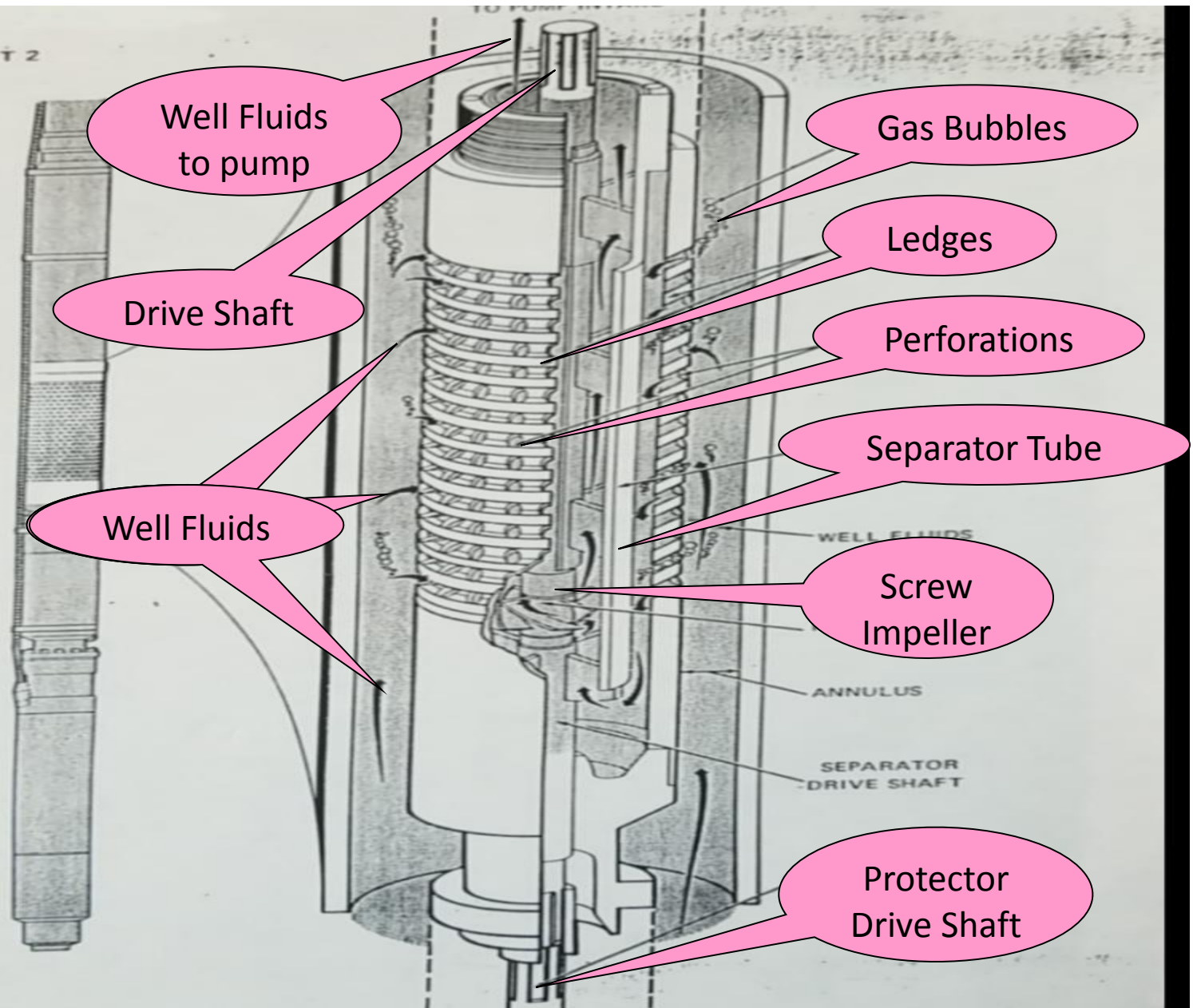
Note

Gas separator **do not** used when install Hydo-Paker.
(Where is the gas will be go ??)

Gas separator types include:

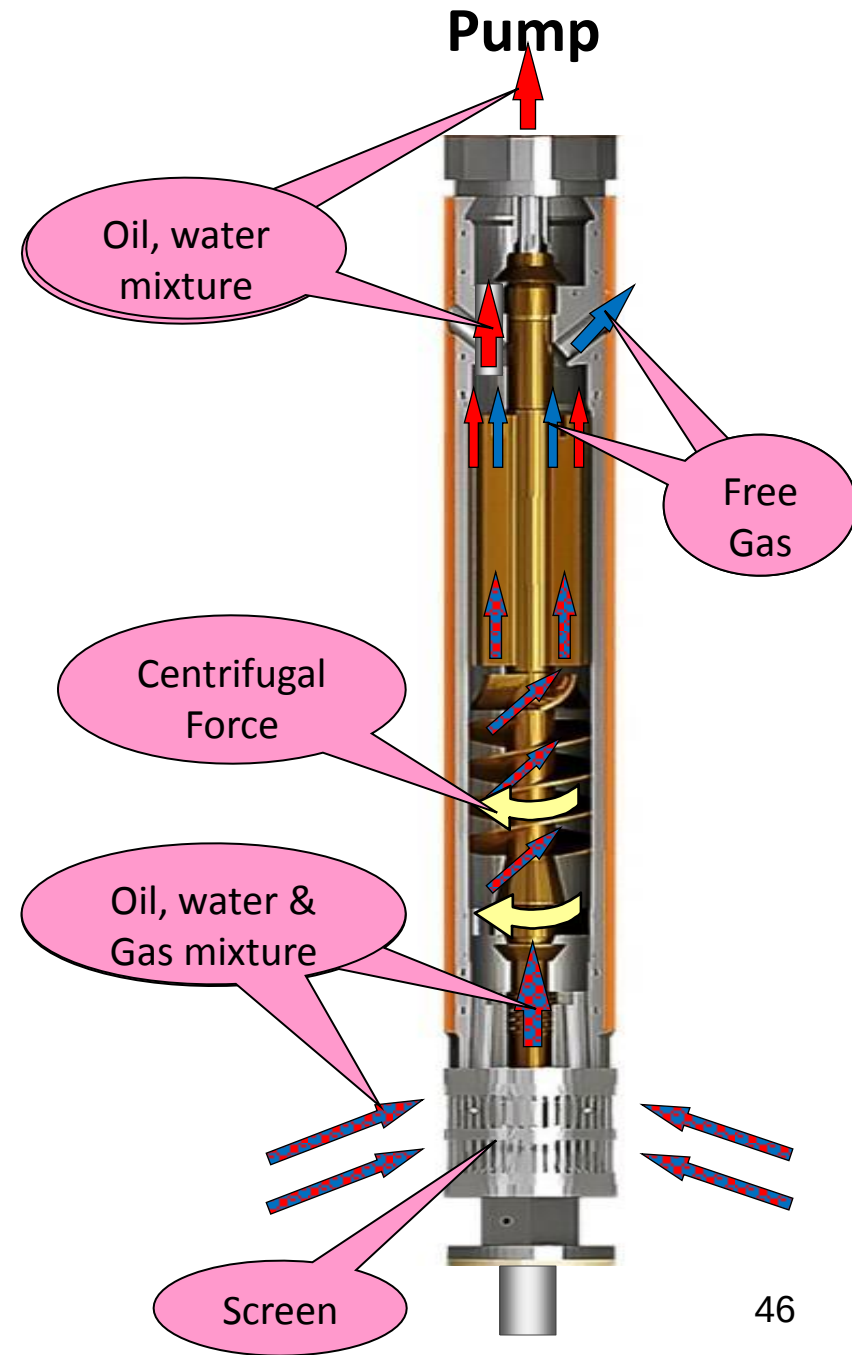
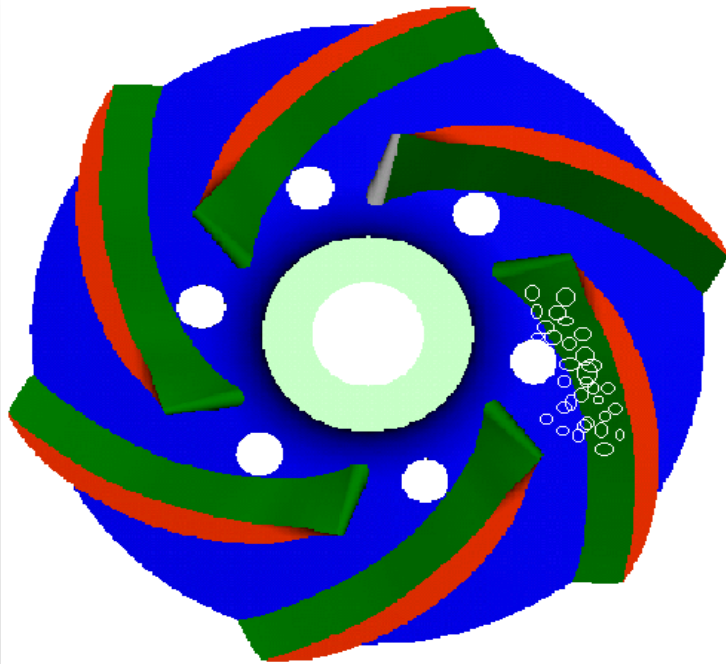
- ◆ Reverse Flow (**Vortex**).
- ◆ Rotary (**Dynamic**).
- ◆ Advanced Gas Handle (**AGH**).
- ◆ Poseidon Multi-Phase Pump (**MPP**).

EXHIBIT 2



Reverse Flow - Gas Separator

Rotary Gas Separator



What Can We Do About Gas ?

Options Available

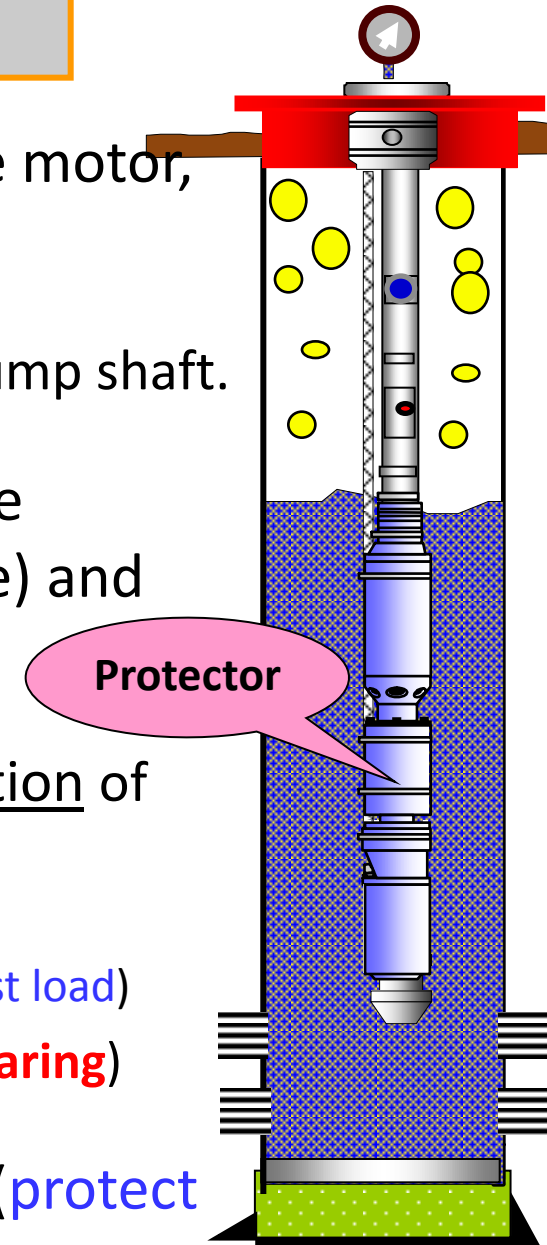
- Avoid It →
 - Increase Pump Seating Depth.
 - Used Mixed Flow Impeller.
- Separate It →
 - Install Gas Separator.
(Rotary or Reverse)
- Handle It →
 - Multi-Van Pump (**MVP**)
 - Advanced Gas Handle (**AGH**)

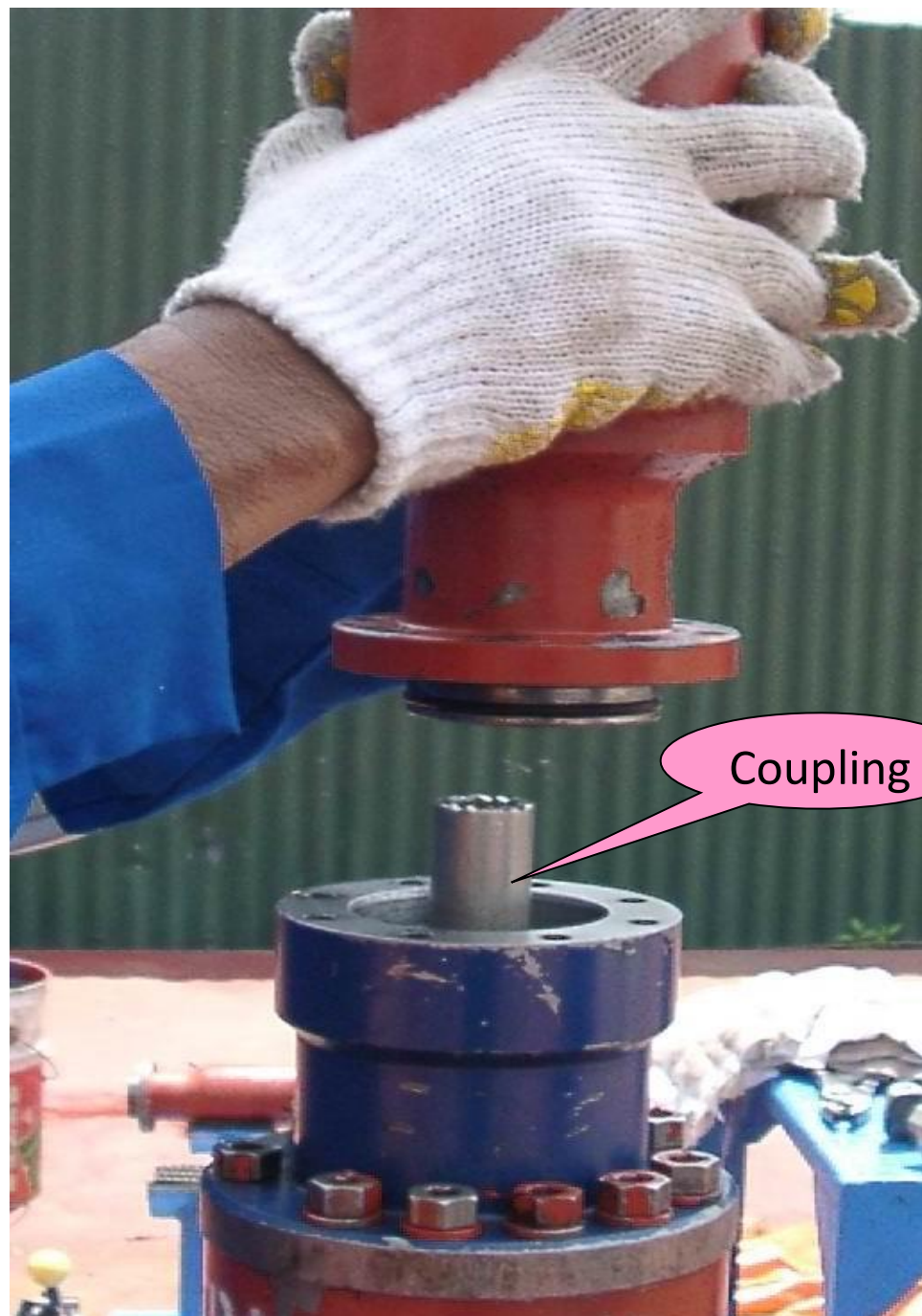
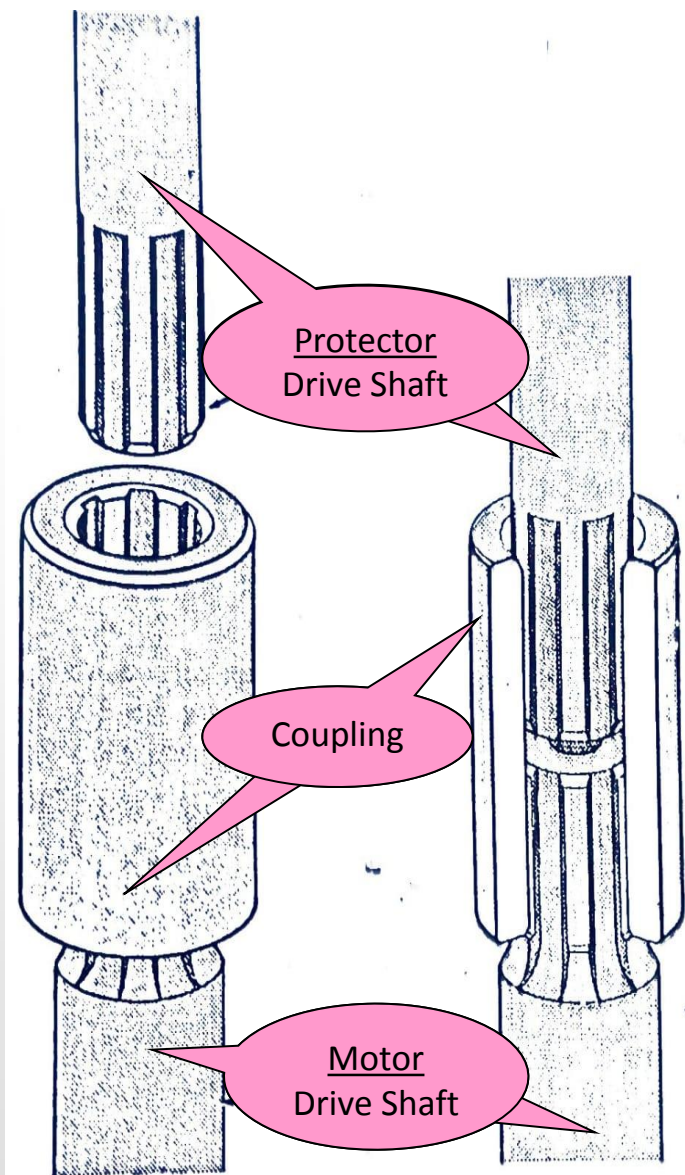
Note

The **AGH** does not separate gas - but conditions it by compressing the total fluid, So it can be produced by the pump.

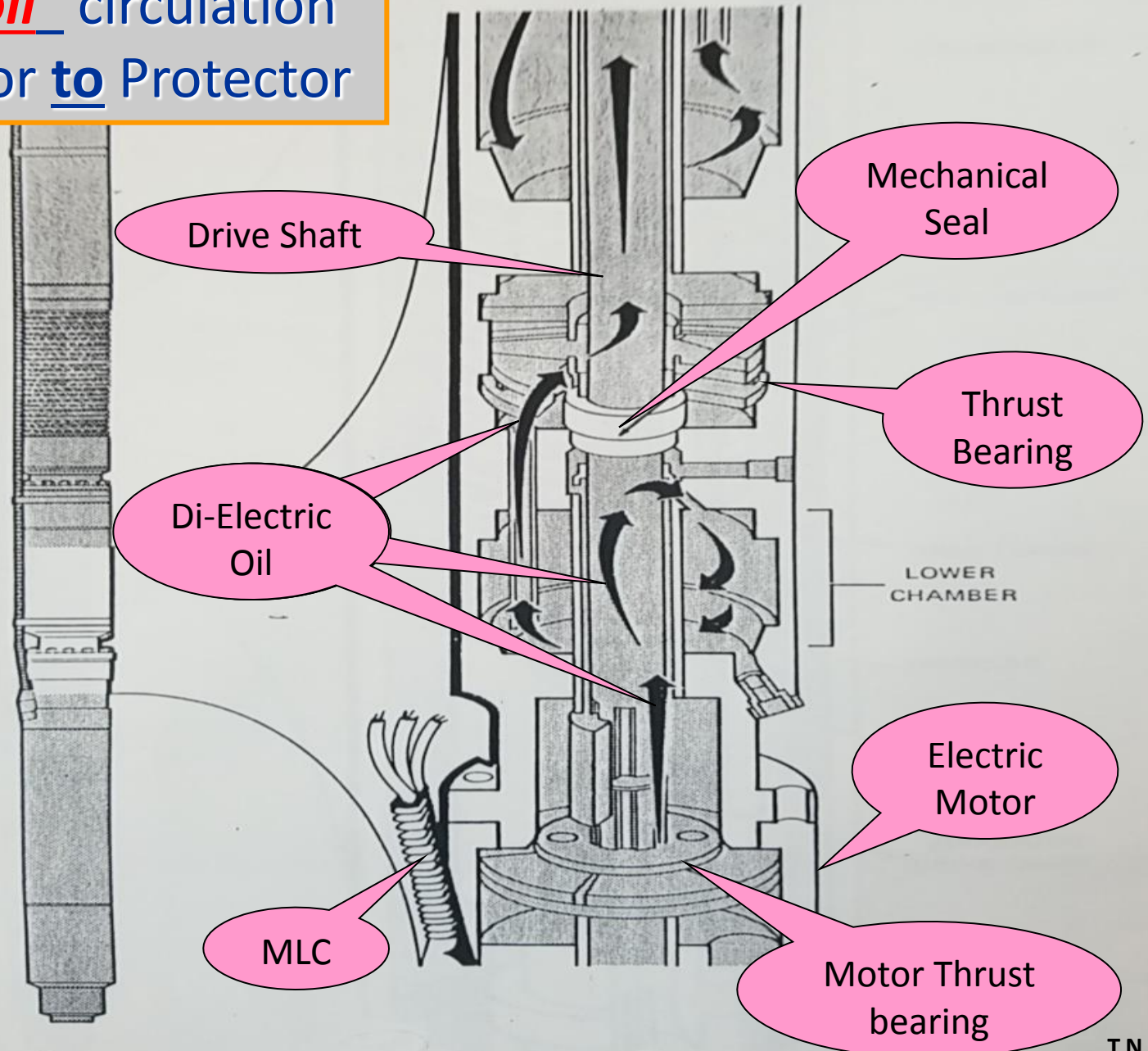
ESP Protector (Seal Section)

- The main purpose is to protect the submersible motor, is located between the pump and motor.
- Transfers the motor torque (Rotary Motion) to the pump shaft.
- Filled with (Di-electric oil) that create a pressure equalizing between motor oil (internal pressure) and well bore pressure.
- Provides area for thermal expansion and contraction of the motor oil.
- Provides the absorbs for any (Up or Down axial thrust load) that transfers through pump shaft. (**Pump thrust bearing**)
- Isolates the well fluid from the clean motor oil (protect the electric motor from damage).

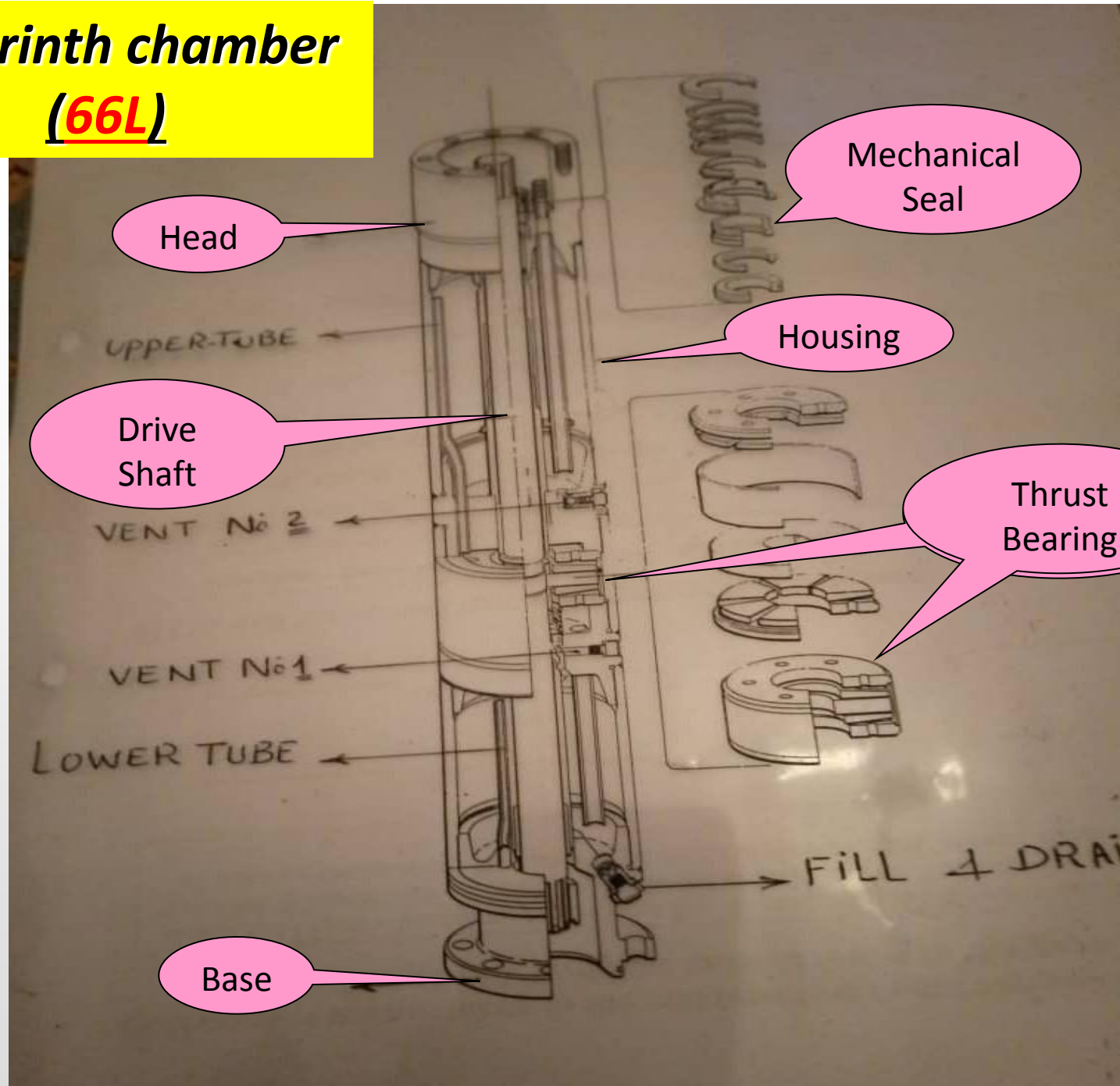




Mineral oil circulation
from Motor to Protector



Labyrinth chamber (66L)



Protector Types

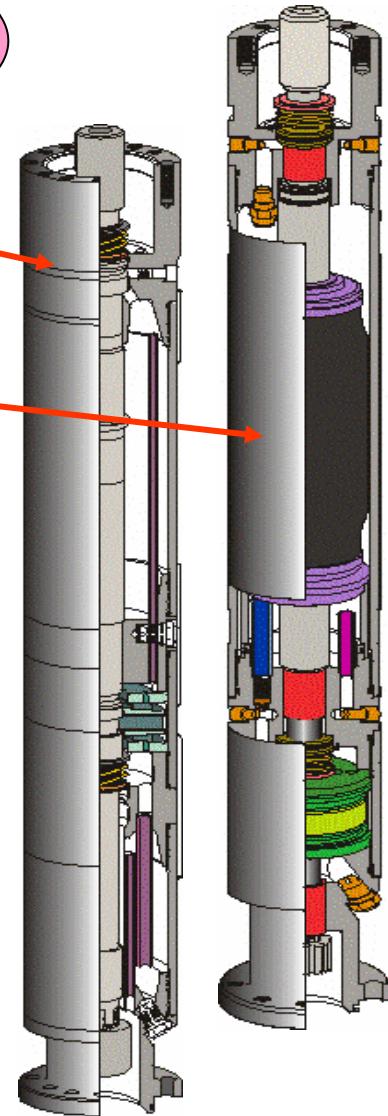
Motor
Drive Shaft

▮ *Labyrinth chamber (66L)*

▮ *Positive Seal chamber*
(Elastomer *Bag*).

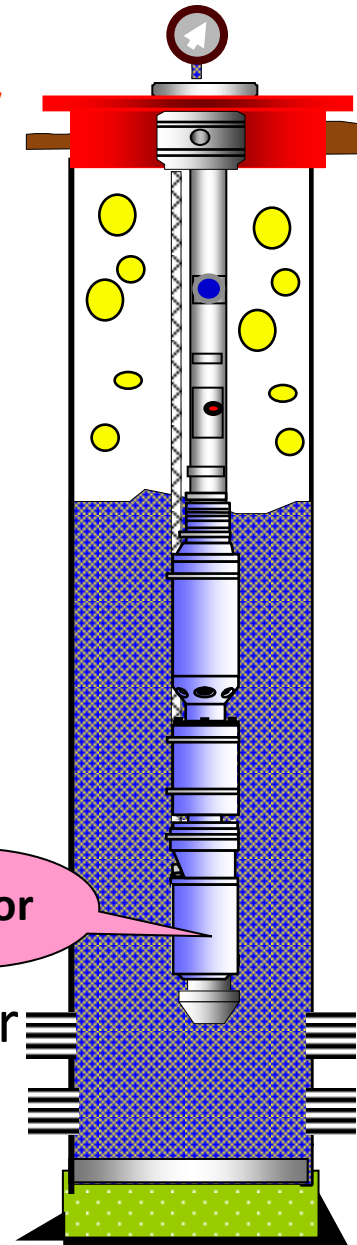
✓ The advantages of a bag type over the Labyrinth type is that can be used in highly deviated wells application.

Motor Thrust Bearing
is installed at the top of the
rotor string and carry the
load of the motors rotors.



ESP Submersible Electric Motor

- Its providing the **HP** (10 -750) & 10 -100 **A** , 230-5000 **V** necessary to turn the pump.
- The motors can be run tandem to create more HP.
- 3400 to 3500 RPM @ 60 HZ (North America)
2850 to 2950 RPM @ 50 HZ (Europe).
- Is made-up of rotating part (**Rotor**) and stationary part (**Stator**).
- The motor is filled with a mineral (Di-electric) oil that has a specific gravity between **0.8** and **0.83** for lubrication, insulation and cooling.
- The minimum fluid velocity of **1.0 ft/sec** is recommended for cooling the motor.
- The motor is manufactured in four different diameters **3.75**, **4.56**, **5.40** and **7.38** inches.



Rotor

(rotating part)

Usually about 12 – 18" long
located in an electric field
generated by the stator

Housing

Cover for the motor and
is threaded at both ends
for head & base

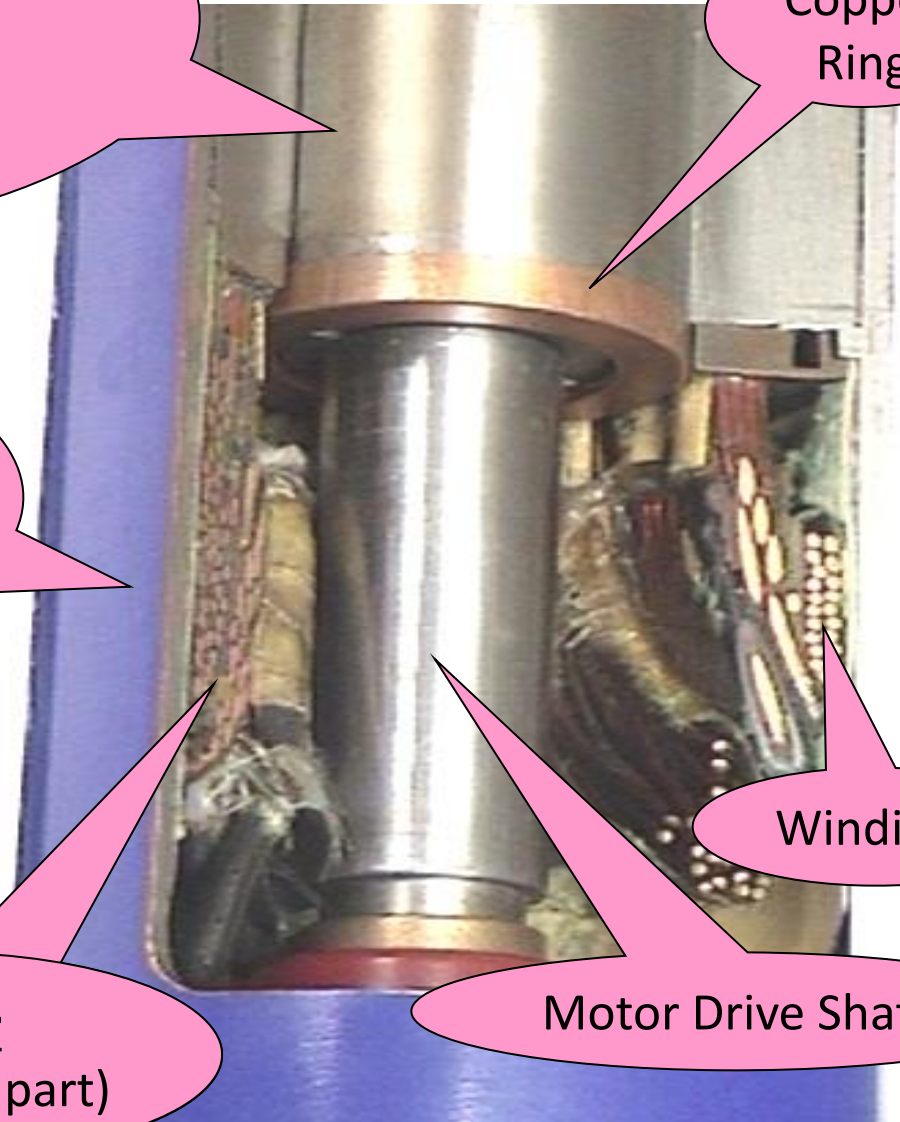
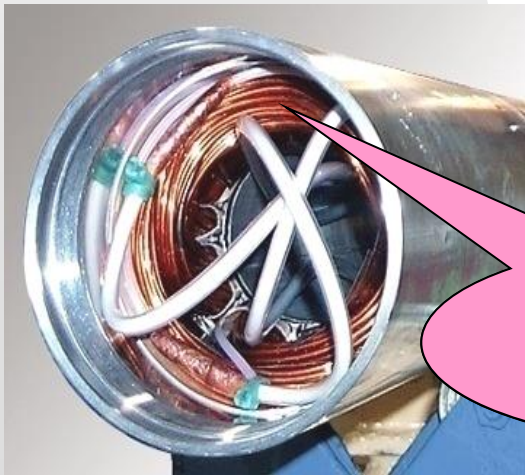
Copper
Ring

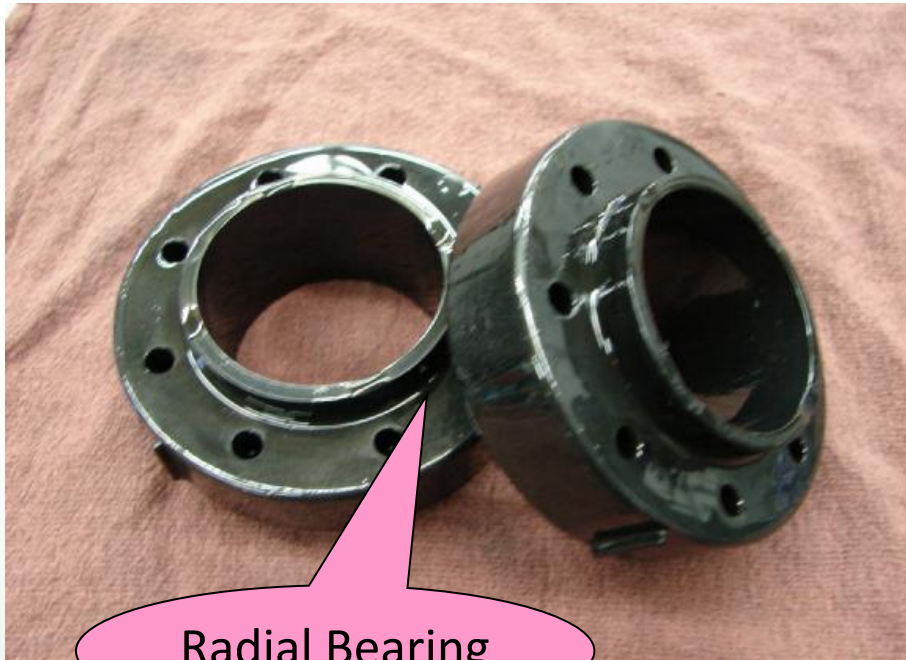
Windings

Stator

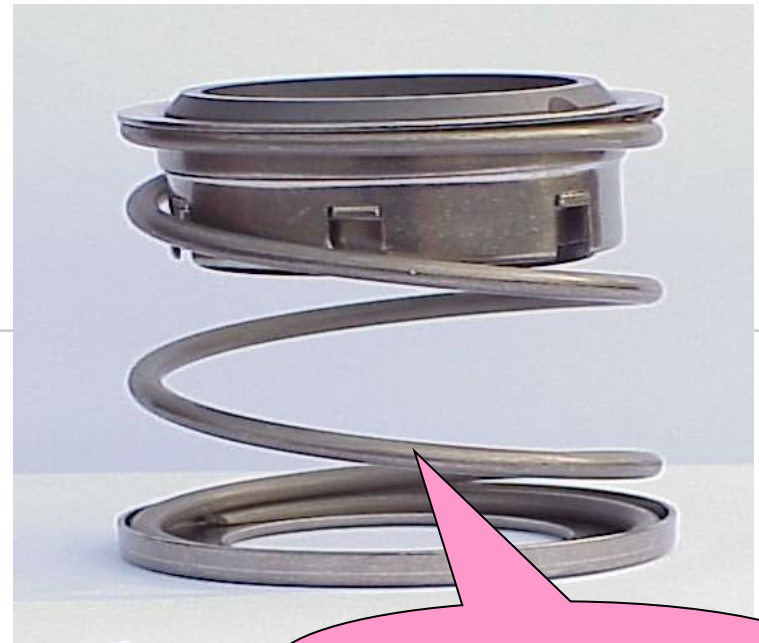
(stationary part)

Motor Drive Shaft

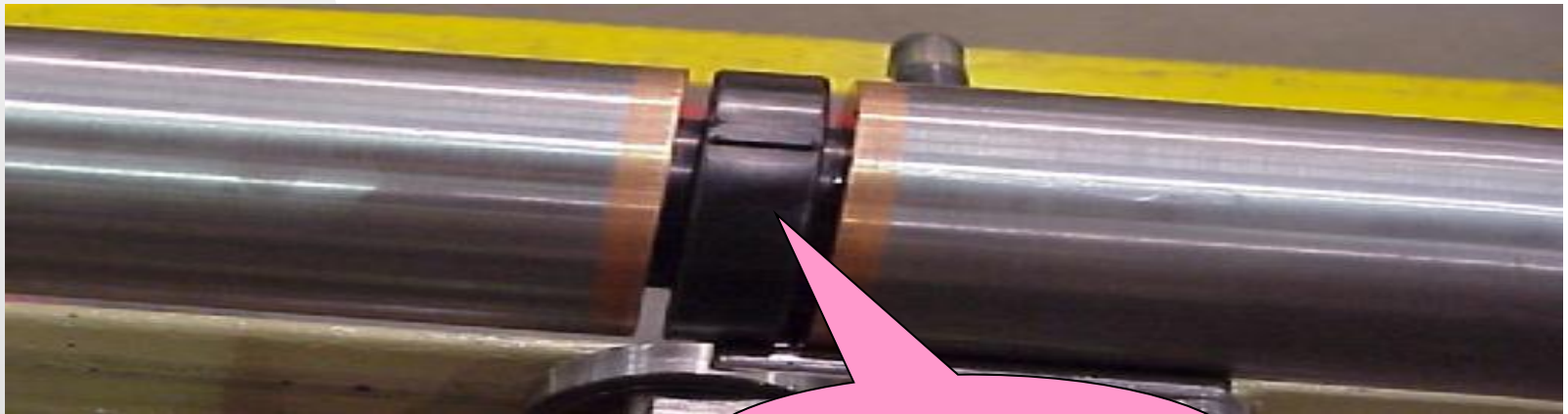




Radial Bearing



Mechanical Seal



Bearing
between Rotors

Electric Motor Cooling & Lubrication

Cooling:

- Electric motor require fluid flowing through the surface to reduce the operating temperature (prevent motor from overheating).
- The motor is cooled by heat transfer from the motor housing to the well-bore fluid.
- To accomplish adequate cooling, a fluid velocity past the motor of one ft/sec.
- For this reason, submersible pumping units should be set above the perforations zone, So all reservoir fluid must pass by the motor.

Lubrication:

- The motor oil circulates up the hollow motor shaft, out on the rotor bearing, down the rotor stator gap and through a filter at eye base of the shaft.

Di-Electrical Oil:

- 1- To prevent short circuits between motor parts (**isolate motor electrically**).
- 2- Proper lubrication for the bearings & drive shaft.
- 3- To transferring the heat generated from motor to the outside of the housing in the wellbore.
- 4- Dissipates the heat that generated from bearings.

Motor Size

The motor size depends on casing size & fluid velocity.

Motor Cooling

Reservoir fluid flowing past the motor provides cooling essential for submersible operation.

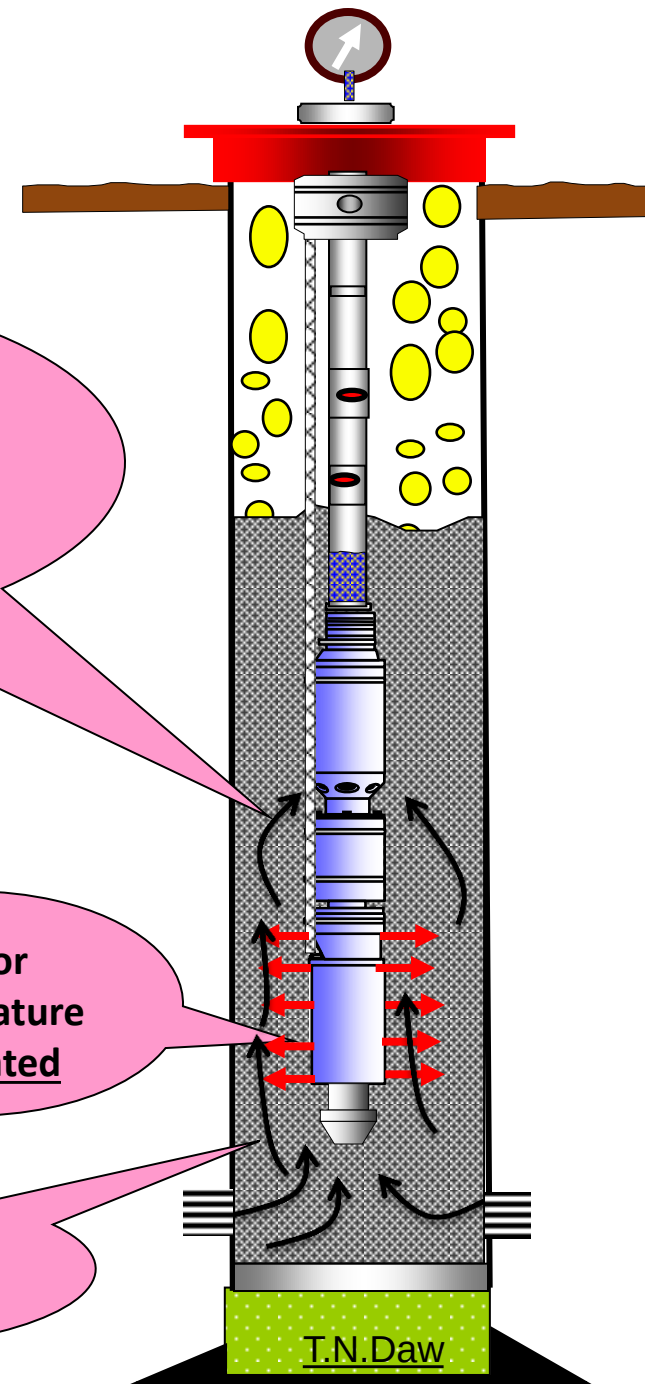
✓ In order to have some cooling, I need motor temperature is go-out from this area .

We need the fluid which pass around the motor to carry hot fluid temperature to go-up and compensate with another fluid from the formation.

Heat

Motor Temperature generated

Bottom hole Fluid Temperature



ESP Electric Motor Cooling

Annular velocity equation used to determine fluid velocity around the motor.

$$V_{\left(\frac{ft}{sec}\right)} = 0.0119 \times \left[\frac{Q}{\left(ID_C^2 - OD_M^2 \right)} \right]$$

Where;

V = annular velocity, ft/sec.

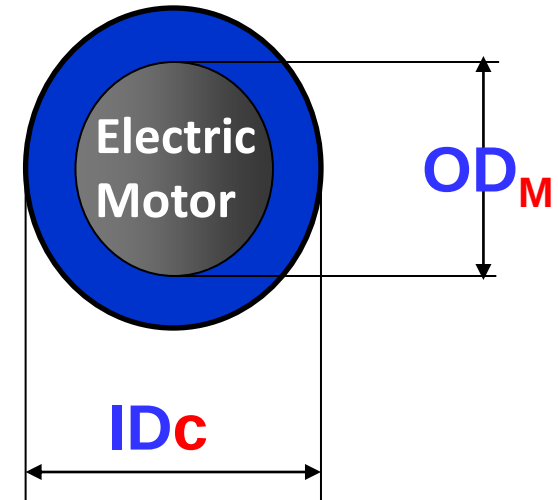
Q = well flow rate, BBL/Days.

IDc = casing inside diameter (inches).

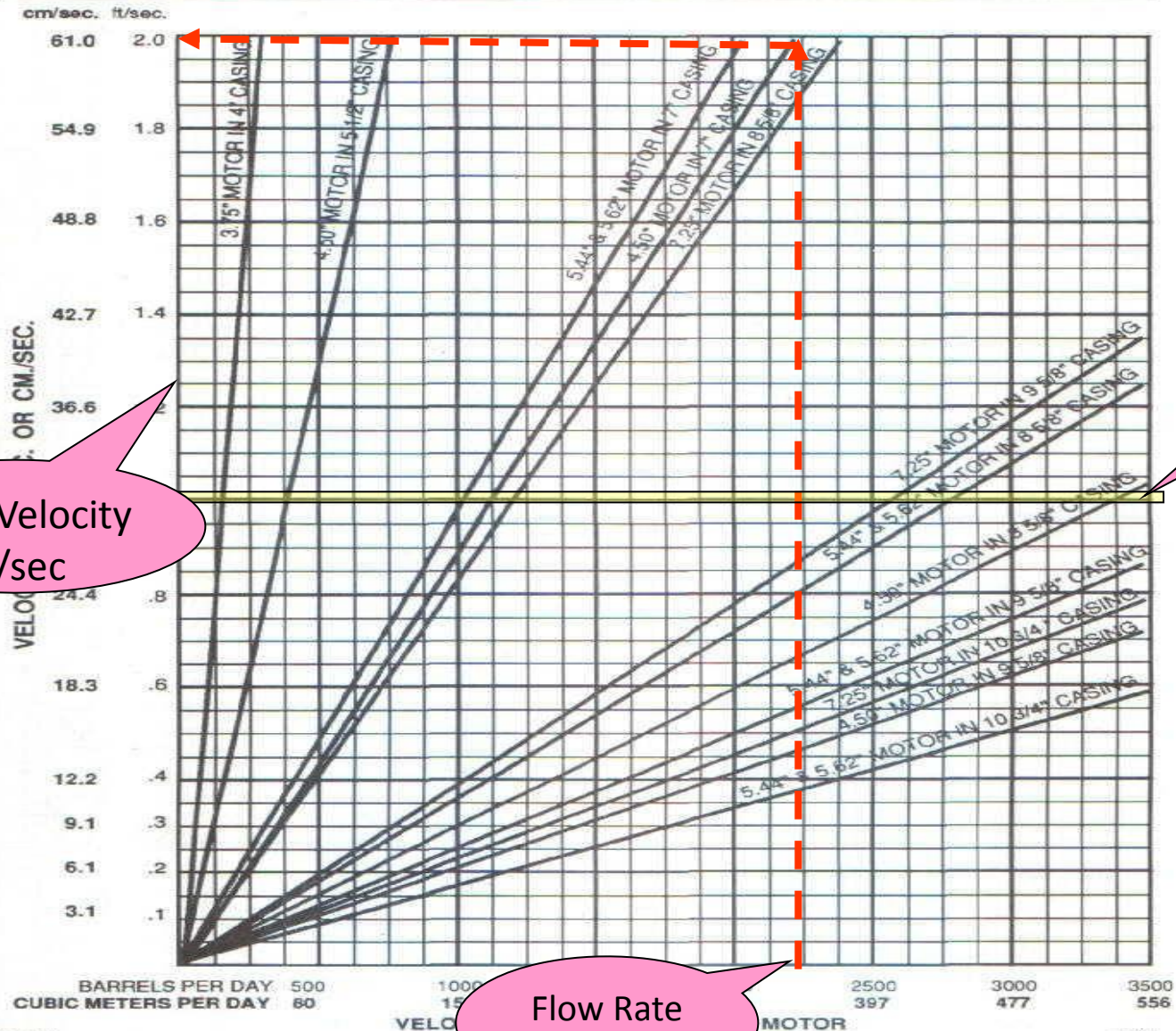
ODm = Motor Outside diameter (inches).

The above Equation From (Graft, Holden and Graves)

$$Cooling.Flow = \left[\frac{Quantity.Discharge}{Ring.Space} \right]$$



Fluid Velocity Past Motor-Chart



Fluid Velocity
ft/sec

1-0 ft/sec
ROT

Flow Rate
B/D

Example:-

CASING = 7" OD , 23 Lb/ft , 6.366" ID , N-80

MOTOR = 450 Series (OD) = 4.5"

Flow Rate = 2200 B/D

What is the fluid velocity passing to the motor?

Solution

$$V_{\left(\frac{ft}{sec}\right)} = 0.0119 \times \left[\frac{2200}{(6.366^2 - 4.5^2)} \right]$$

$$V = 1.3 \frac{ft}{sec}$$

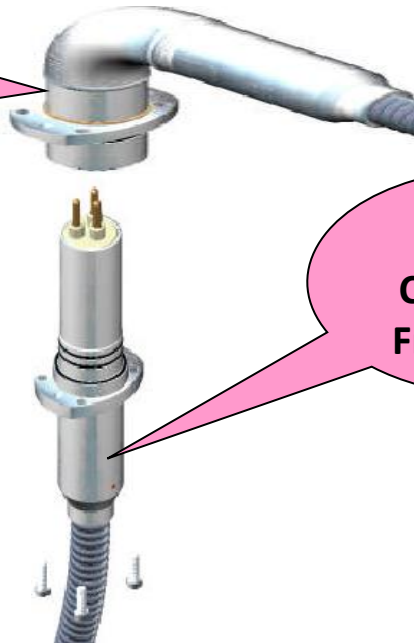
* This velocity adequate for cooling the motor

ESP Power Cable

- The electric power is transmitted current from the switch board at the surface down to electric motor housing through Three Conductor electric cable at the Pothed.
- Especially designed for oil field application.
- Power cable is a available in three voltage ratings 3000, 4000 and 5000 volt & different conductor sizes #1, #2, #4.
- Is constructed of three insulated , jacketed, conductor copper wires. contained by metal armor (outer material).
- Proper applications must address electrical (Volt & Amp), physical size, H₂s and operating temperature requirements.
- Cable is available in round and flat styles & must have small enough diameter to fit in the annulus along the well tubing.

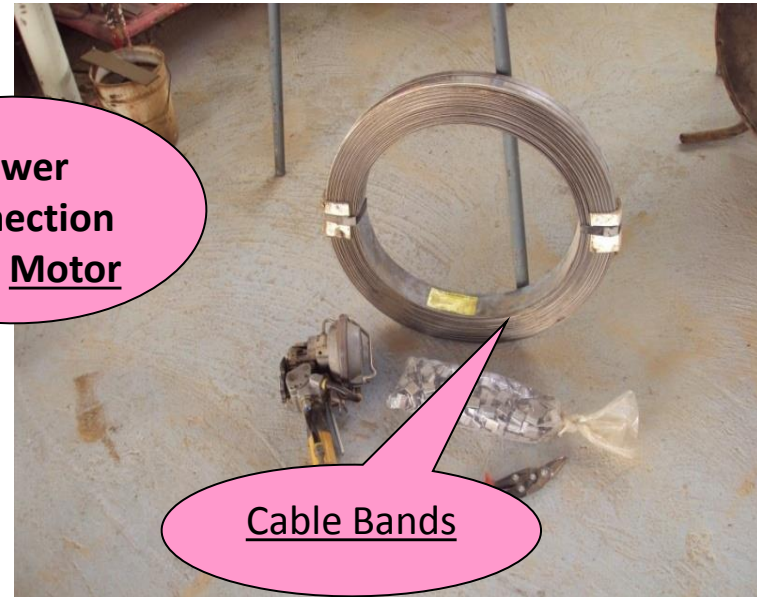
- Depending on the type of insulation used cables may be installed in oil wells with temperatures in excess of 300 F.
- Is strapped or banded to the out side of production tubing by cable bands at the middle of the joint and below the tubing collar at steady intervals (2 bands /Jts) to prevent mechanical damage during running & pull out of hole in workover.
- Penetrates the well head by special Tubing Hanger.

Upper Connection
From Junction Box

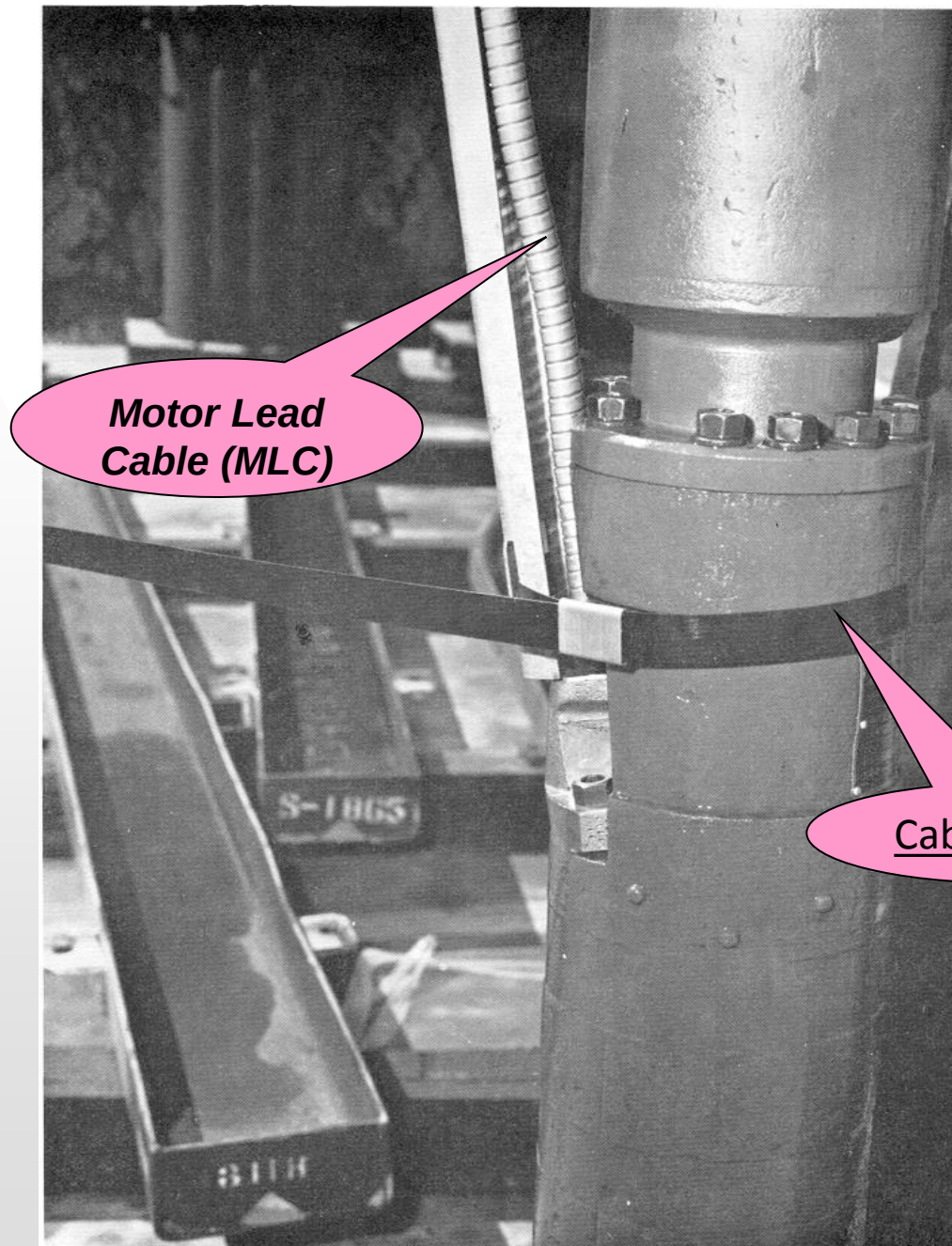


Lower
Connection
From Motor

Tubing
Hanger

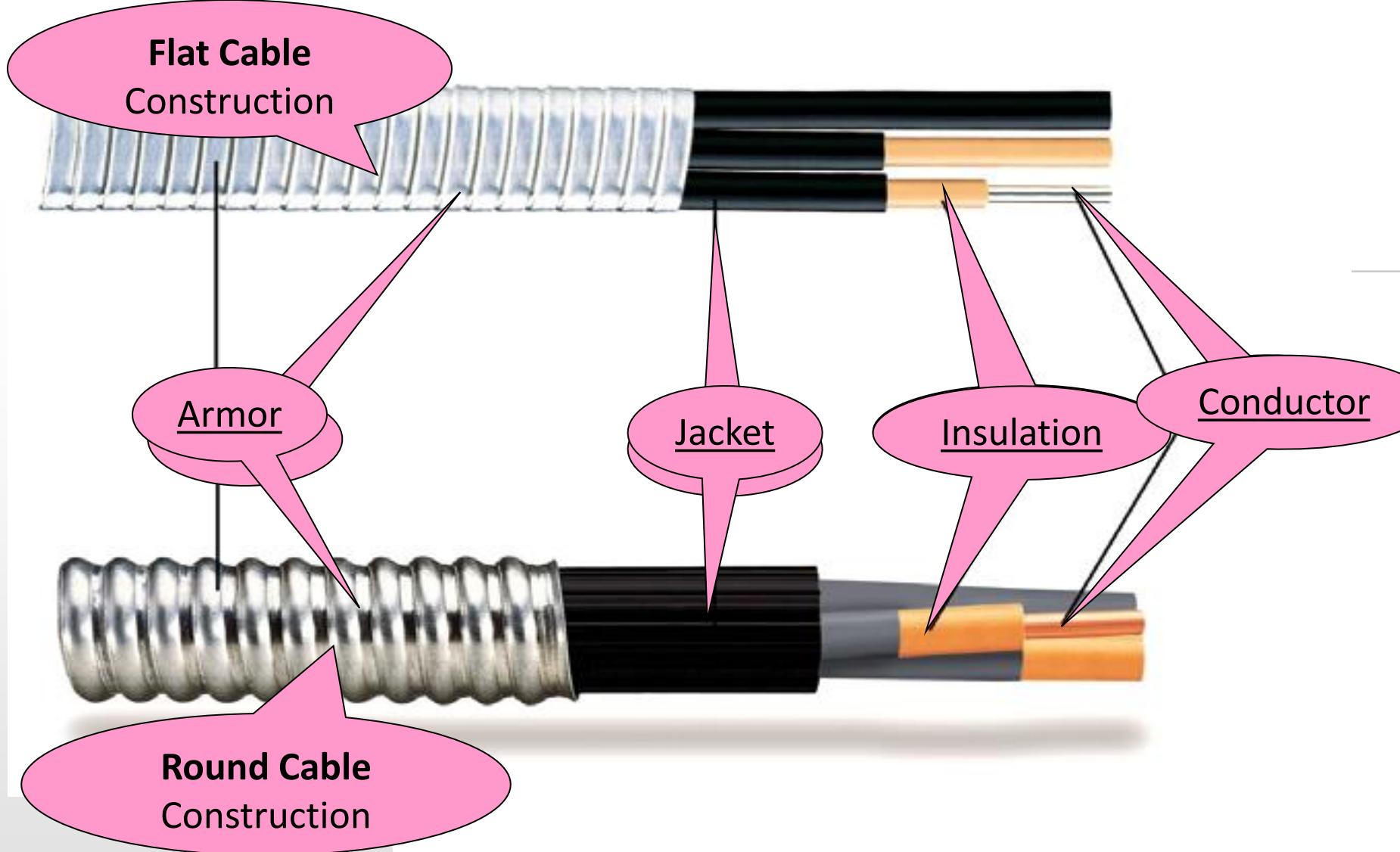


Cable Bands



***Motor Lead
Cable (MLC)***

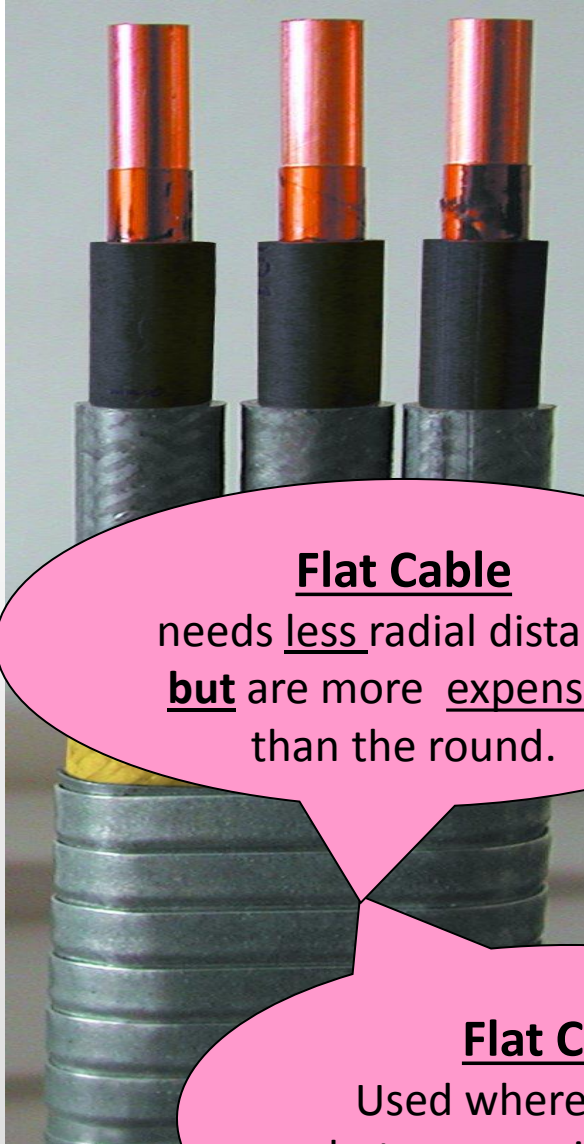
Cable Bands



✓ **Types:** Black Steel - Stainless Steel - Monel armor

✓ Cable is **selected** based on (**well fluid properties** & **BHT**),

✓ **Size** of cable depended on (A, V, HP & space between **tubing collar** and **casing**).



Flat Cable

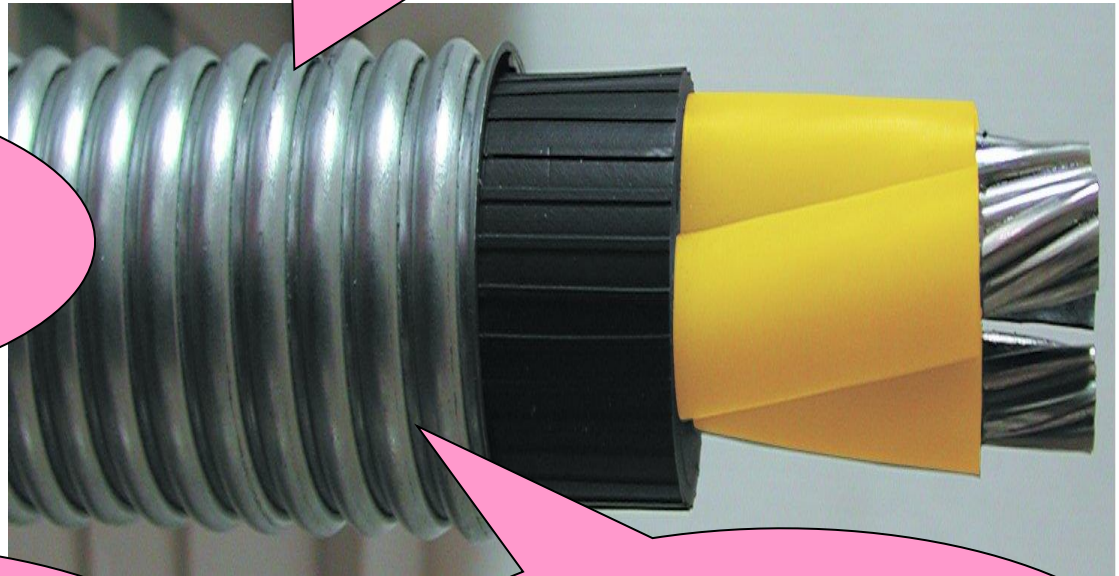
needs less radial distance
but are more expensive
than the round.

Flat Cable

Used where clearance
between casing & tubing
collars is a problem.

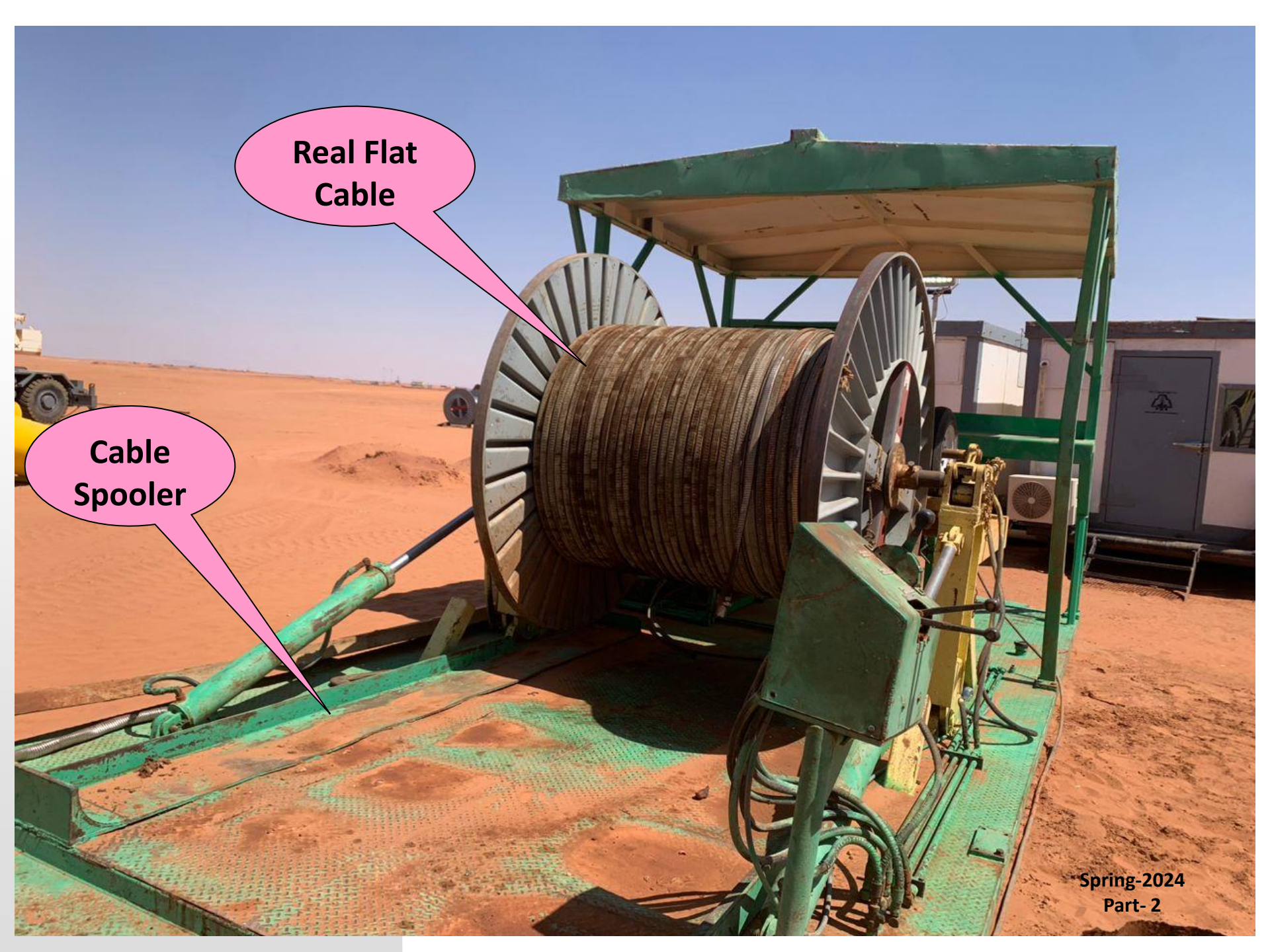
Cable Armor

- *Protect cable from mechanical damage.*
- *Protection for corrosive oil wells.*
- *Armor is available in different materials (**stainless steel, Galvanized and Monel**).*



Round Cable

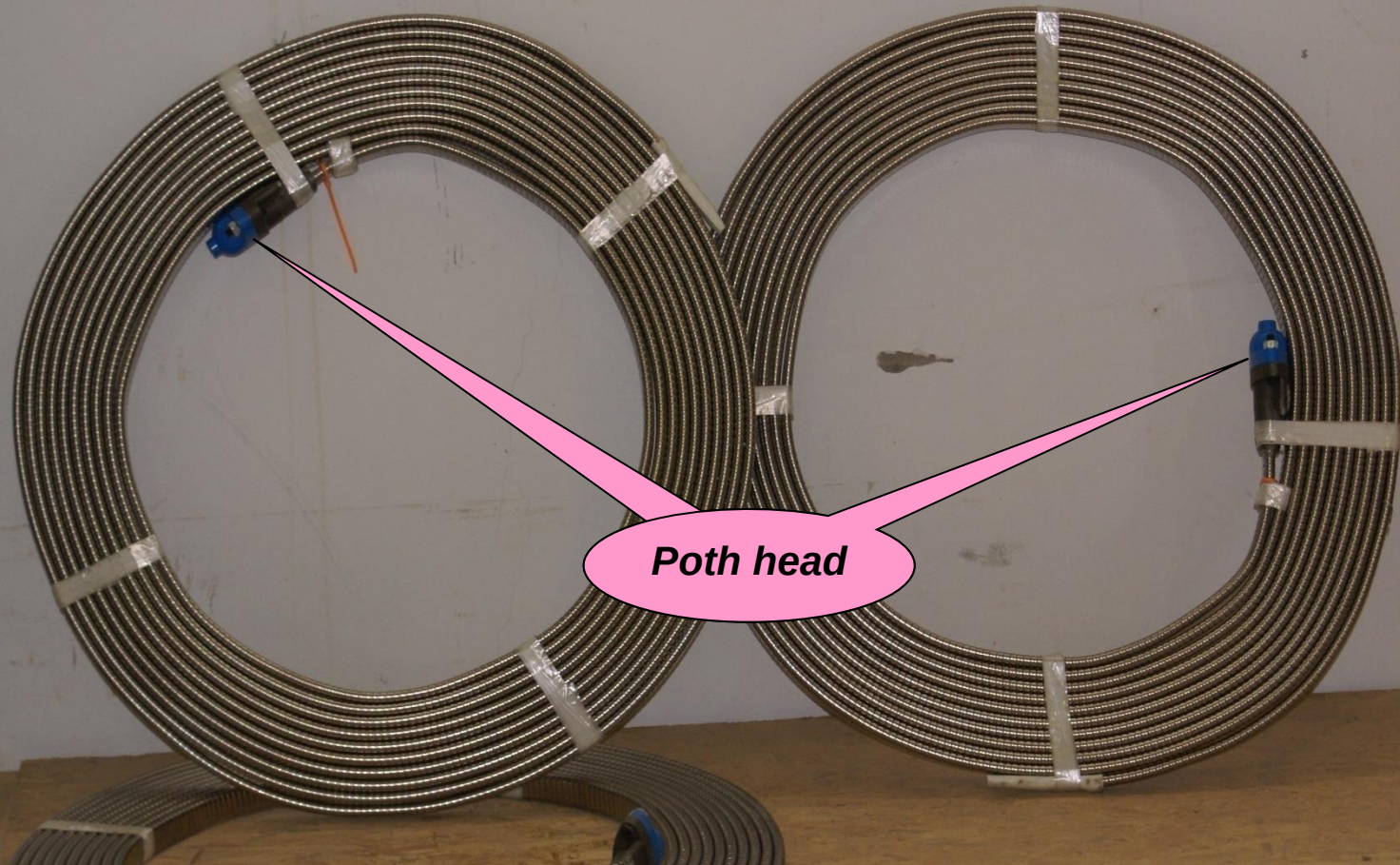
Used where clearance
between casing & tubing
collars is not problem.

A large industrial cable spooler is shown in a desert environment. The spooler is a green metal frame with a large spool of flat cable. A pink speech bubble points to the cable, and another points to the spooler. The background shows a sandy desert under a clear blue sky, with a small building and a vehicle in the distance.

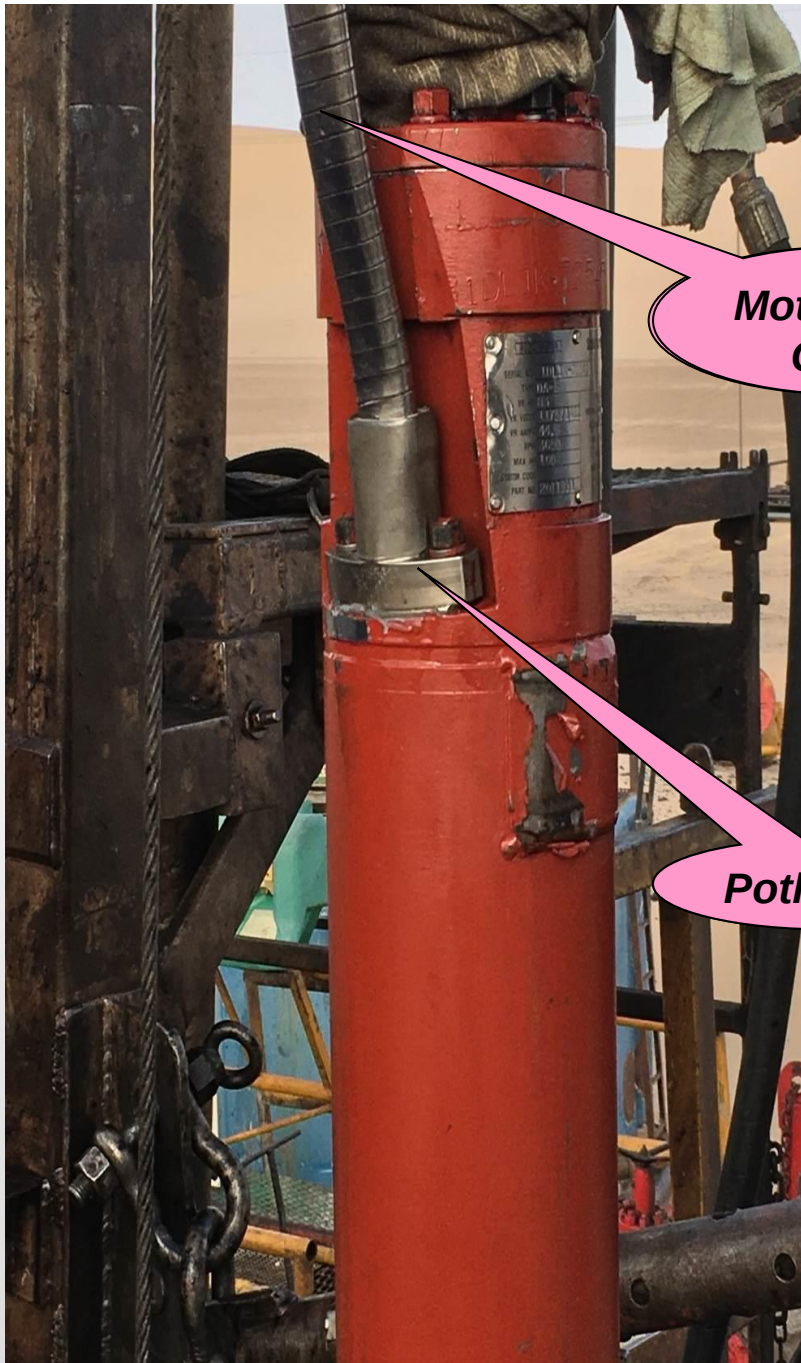
**Real Flat
Cable**

**Cable
Spooler**

ESP Motor Lead Cable (MLC)

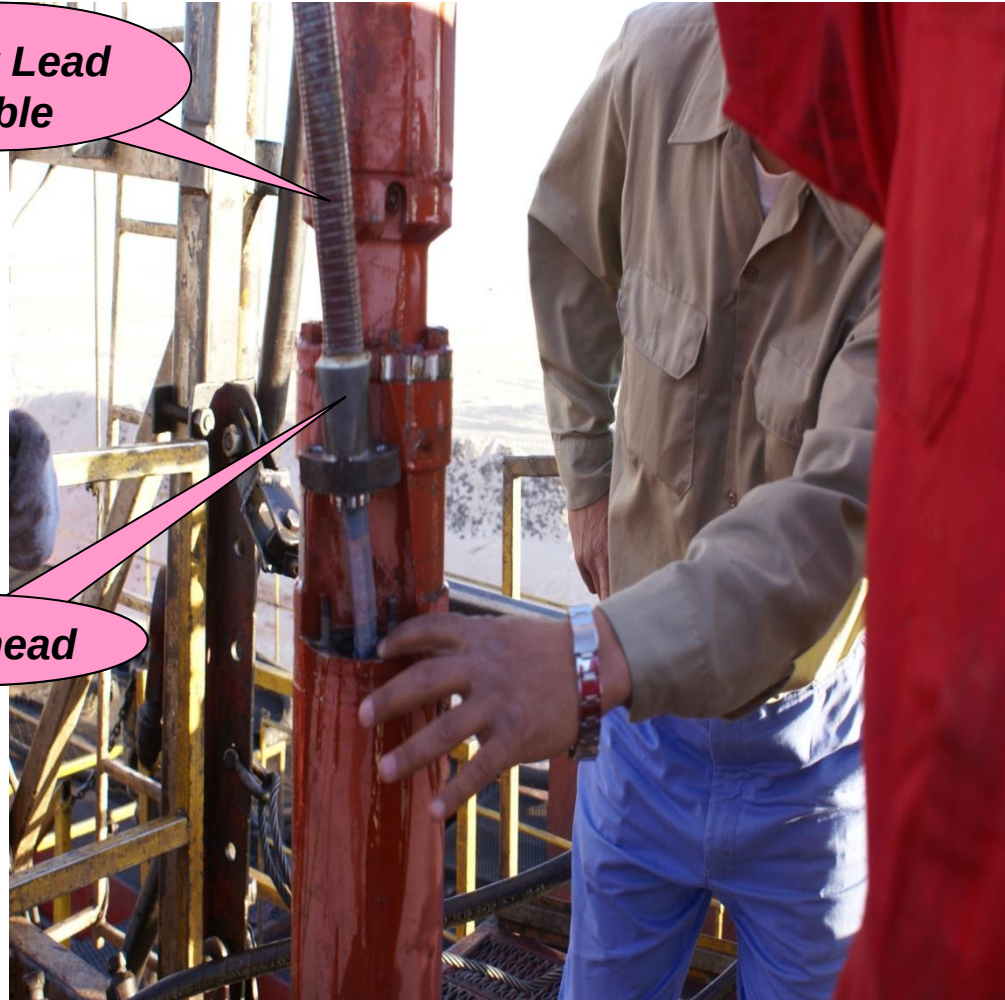


MLC (FCE) → Is the part of the ESP main cable running along the ESP unit and is directly connected to the motor pothead.

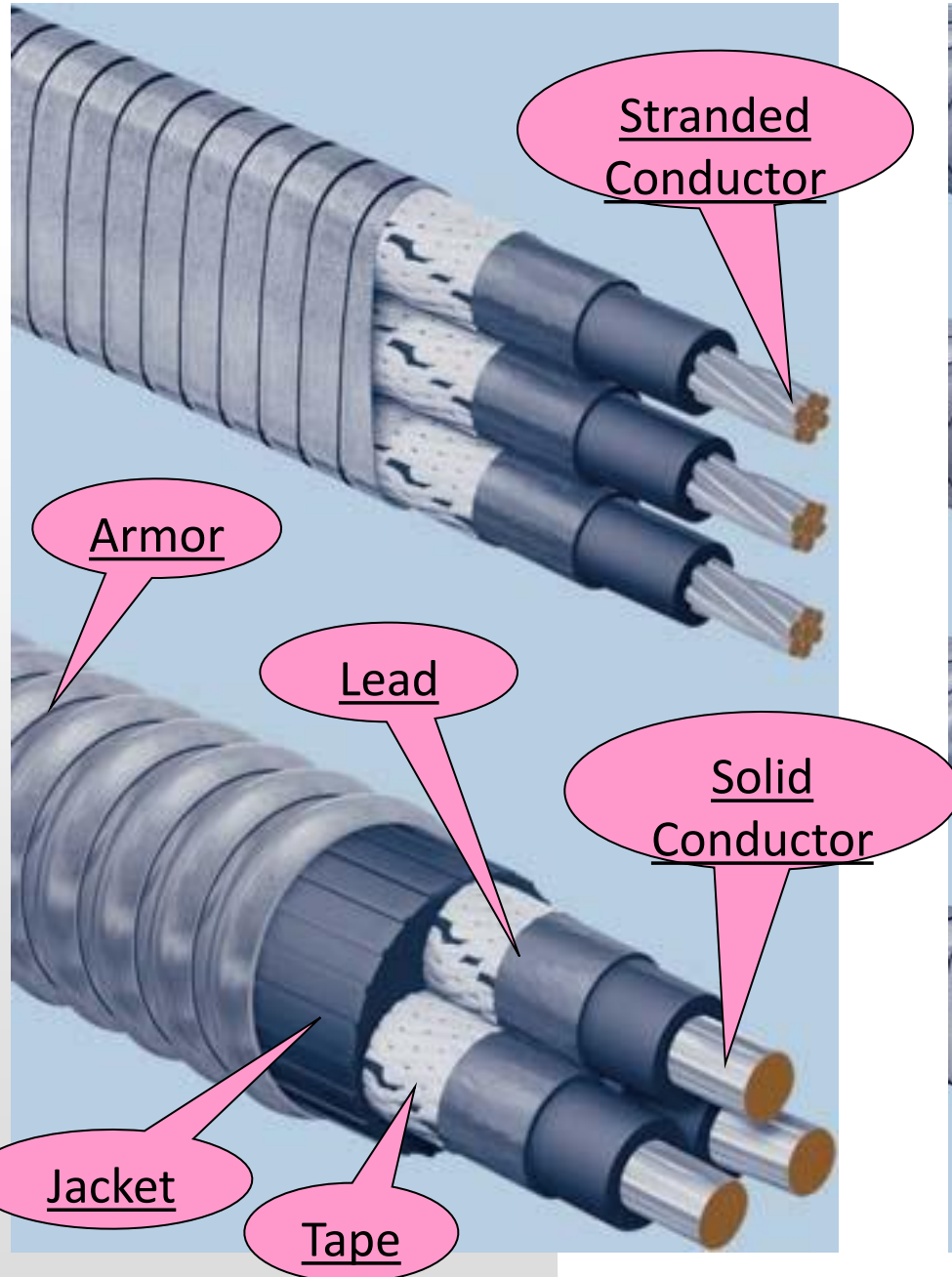


**Motor Lead
Cable**

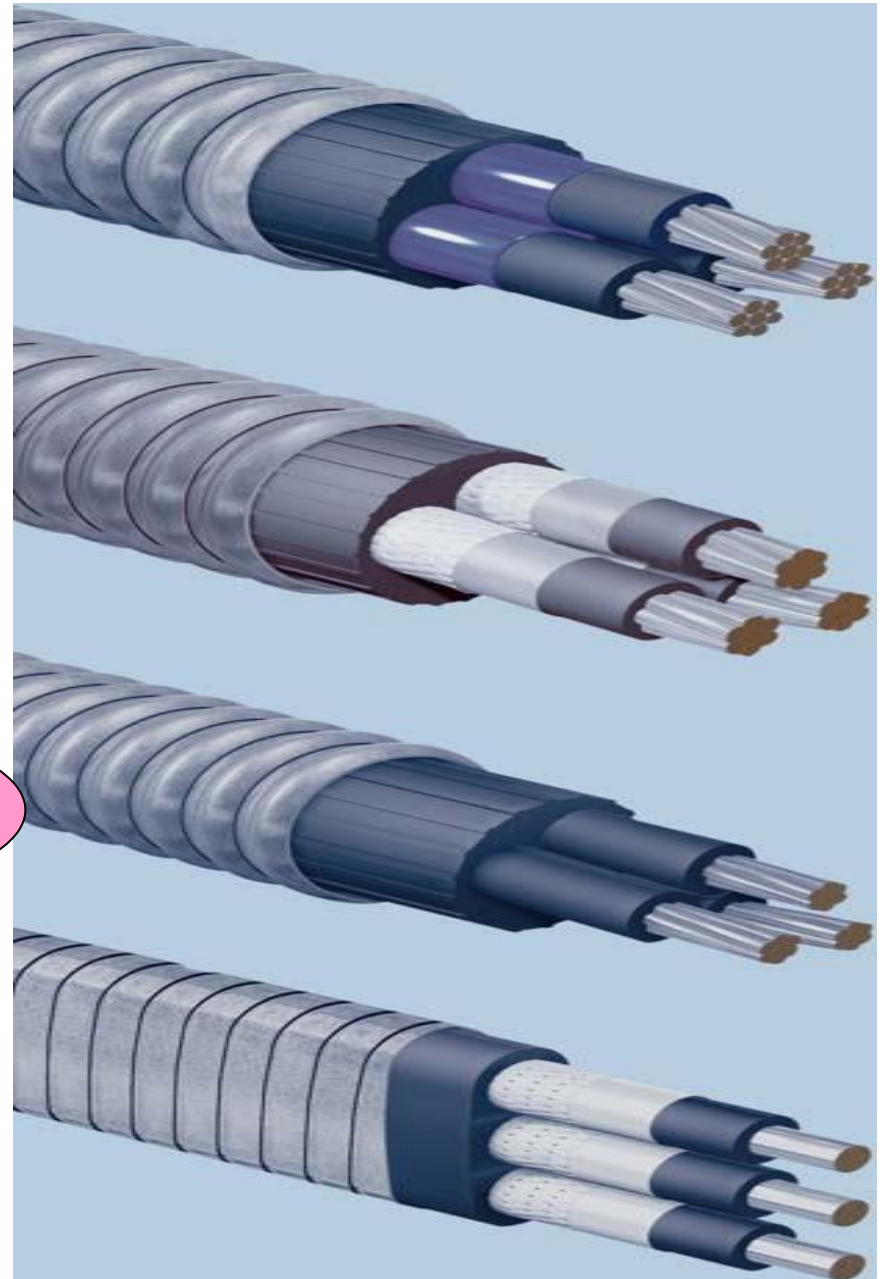
Poth head



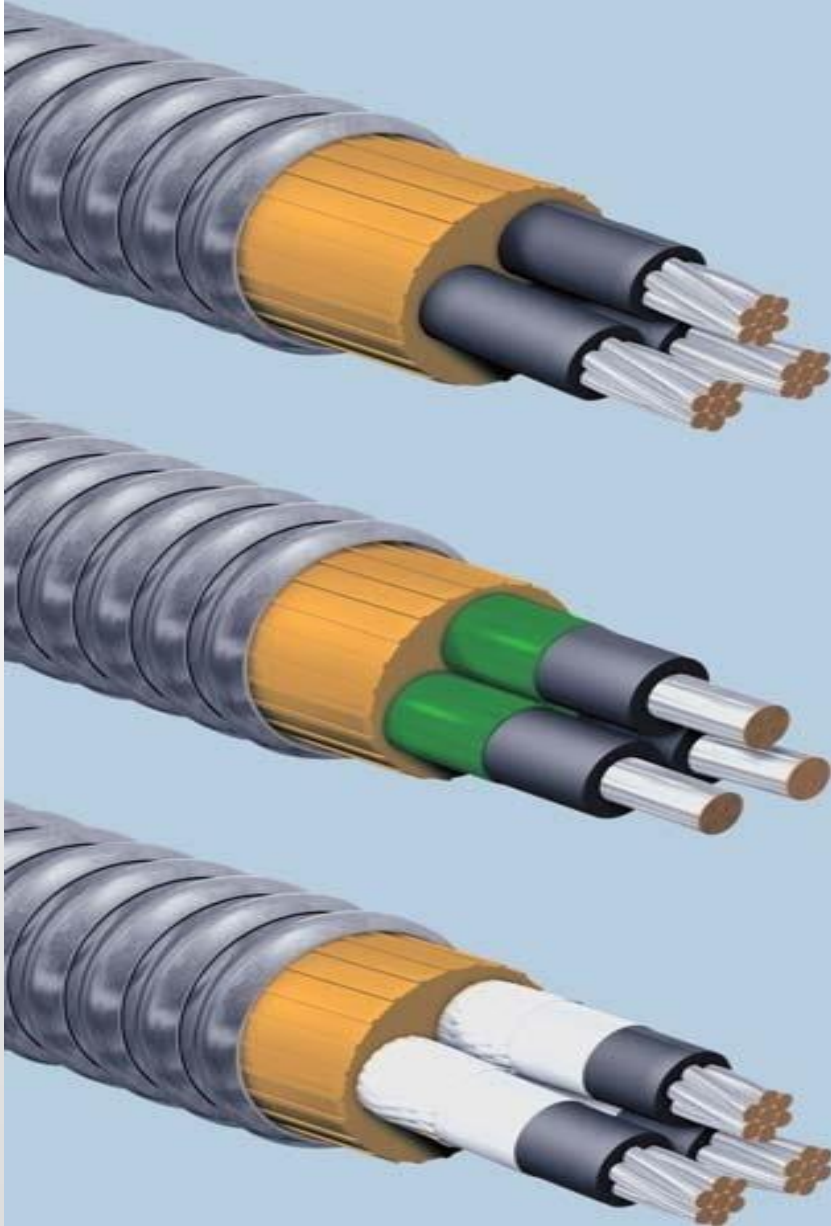
Reda Lead: 400-450°F (203-232°C)



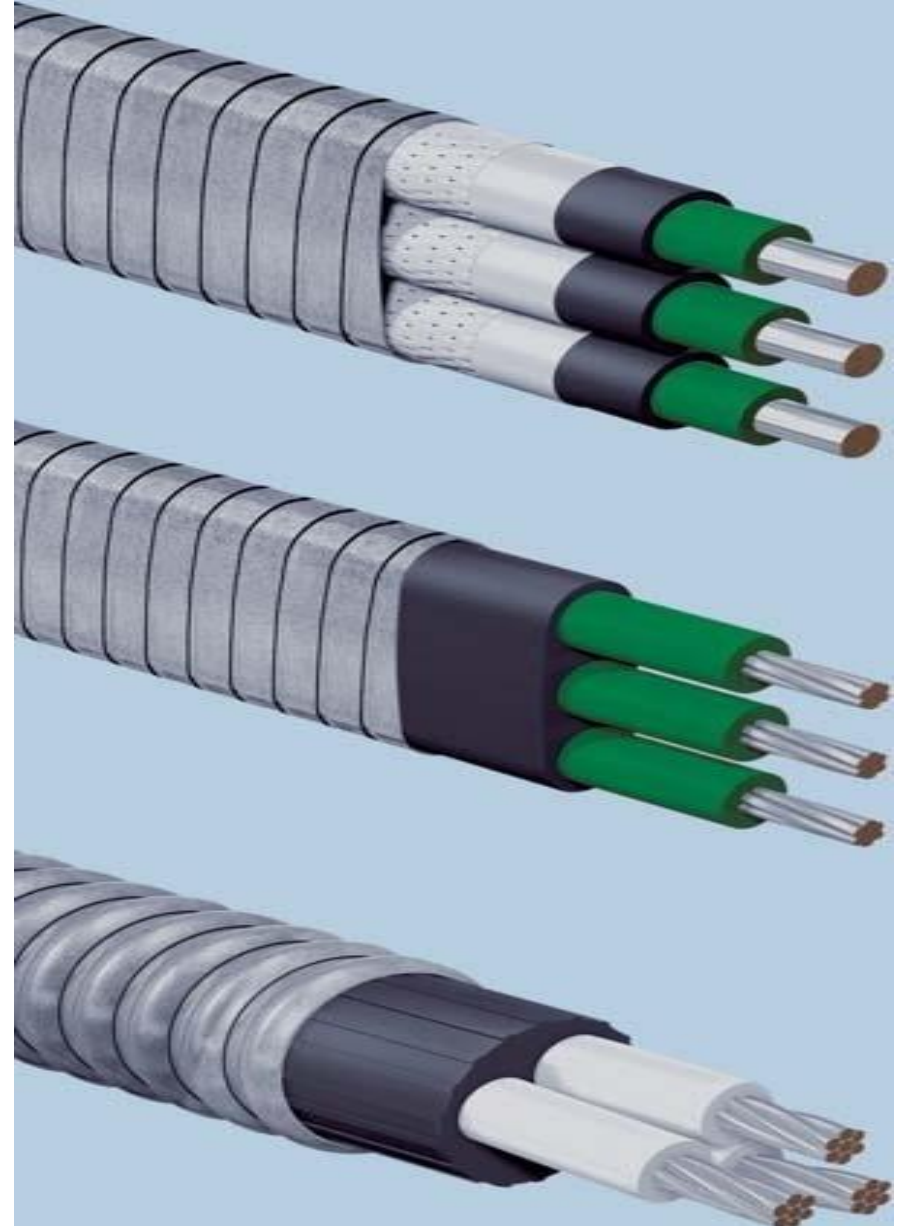
Reda Black: 300-400°F (149-204°C)



Reda Hot: 250-350°F (121-177°C)



Reda Lene: 205°F (96°C)





Sleeves

Lead Insulation

Rubber Insulation

Cable Splicing

*We performed splicing to connect two different type of cable
(Flat or round) by a pump company serviceman.*

ESP Standard & Optional Installations **(Miscellaneous Equipment)**

▶ **Check Valve.**

▶ **Bleeder (Drain) Valve.**

▶ **Y-Tool.**

▶ **Down Hole Sensor.**

🔥 **Motor Shrouded.**

🔥 **Centralizer.**

🔥 **Down Hole Desander.**

🔥 **Twin Seal Packer (Hydro II).**

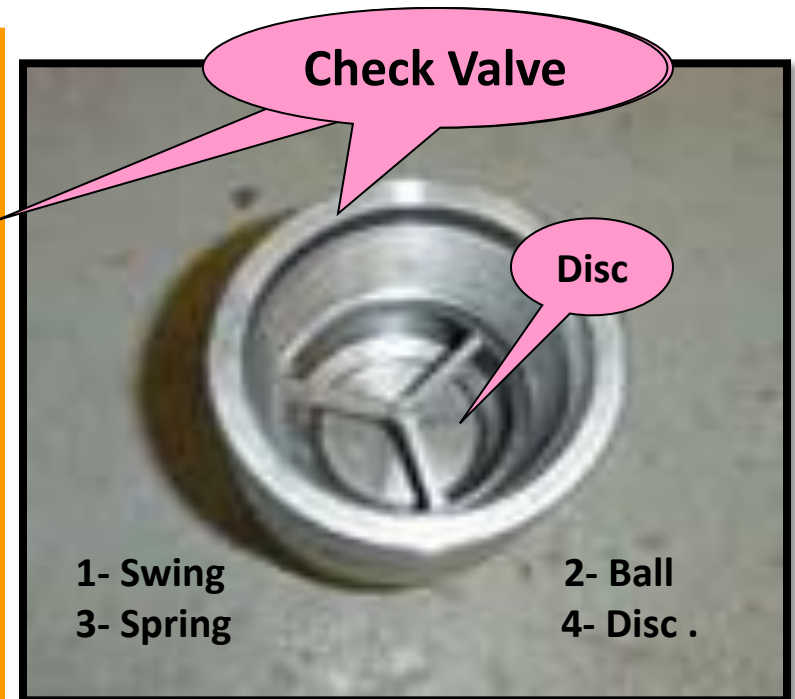
Introduction:

- When the motor is switched off.
- The head of fluid present in the tubing will reverse the flow direction through the pump as it flows back into the well bore .
- This will cause the motor to turn backwards.
- Trying to restart the motor while it is rotating backwards will lead to the motor burning out very quickly. **This can be avoided by installing a check valve in the tubing.**
- **Or** using **a sensor** to detect backspin and preventing motor restart.

✓ It is located **2-3** joints above the pump assembly. (**For Gassy oil wells**) The **C.V** should be installed from 7-8 Joints above the pump.

✓ To prevent the pumped fluid from flowing back down during shutdown. (**Reverse Rotation**).

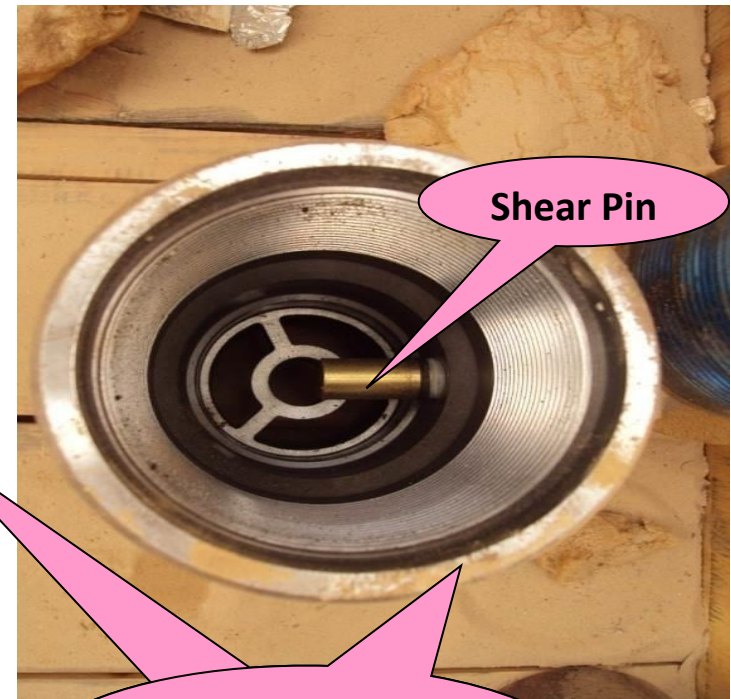
✓ prevents any solids in the produced fluid from setting back into the pump.



Introduction:

- When check valve is used in tubing string, it is necessary to install a Drain valve immediately above the C.V .
- If C.V is not run and the well has no Sand problem, then No reason to run B.V , as the produced fluid in the tubing will drain through the pump intake while pulling.

- ✓ It is located **2-3** joints above the check valve.
- ✓ To pull out the pumping unit with tubing empty.
- ✓ Needle sheers by a steel bar (Sinker bar) prior to pulling pumping during workover operations.



**Bleeder (Drain)
Valve**

Check & Drain Valve



Shear Pin



Bolt-on-head

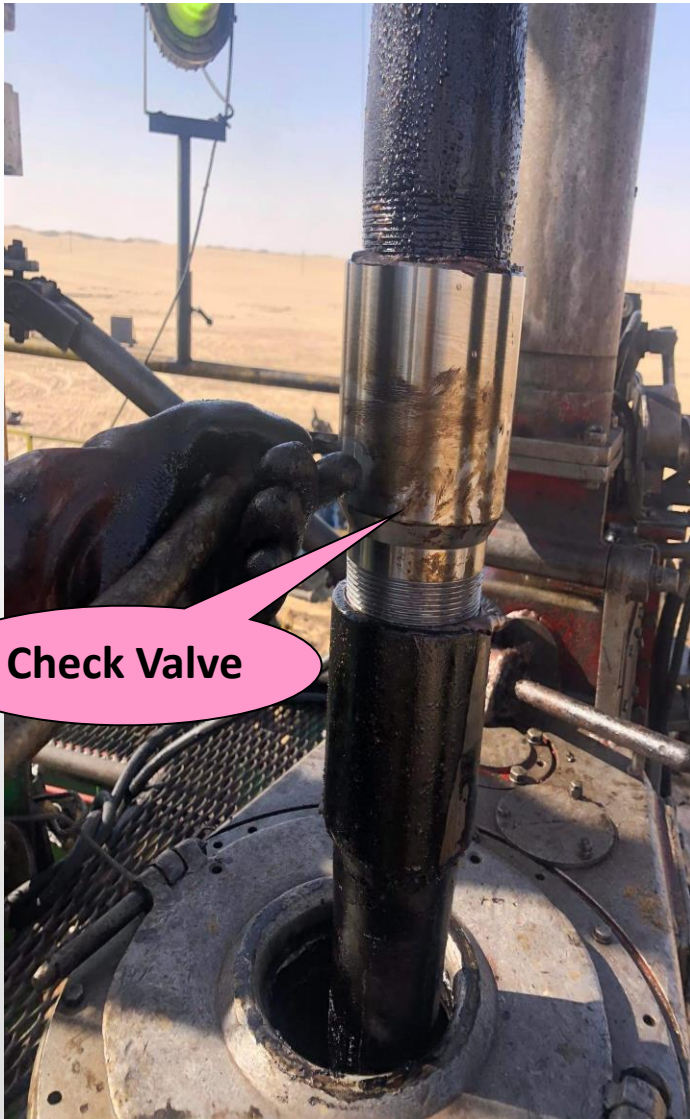


Check Valve



Drain Valve





Backspin Relay

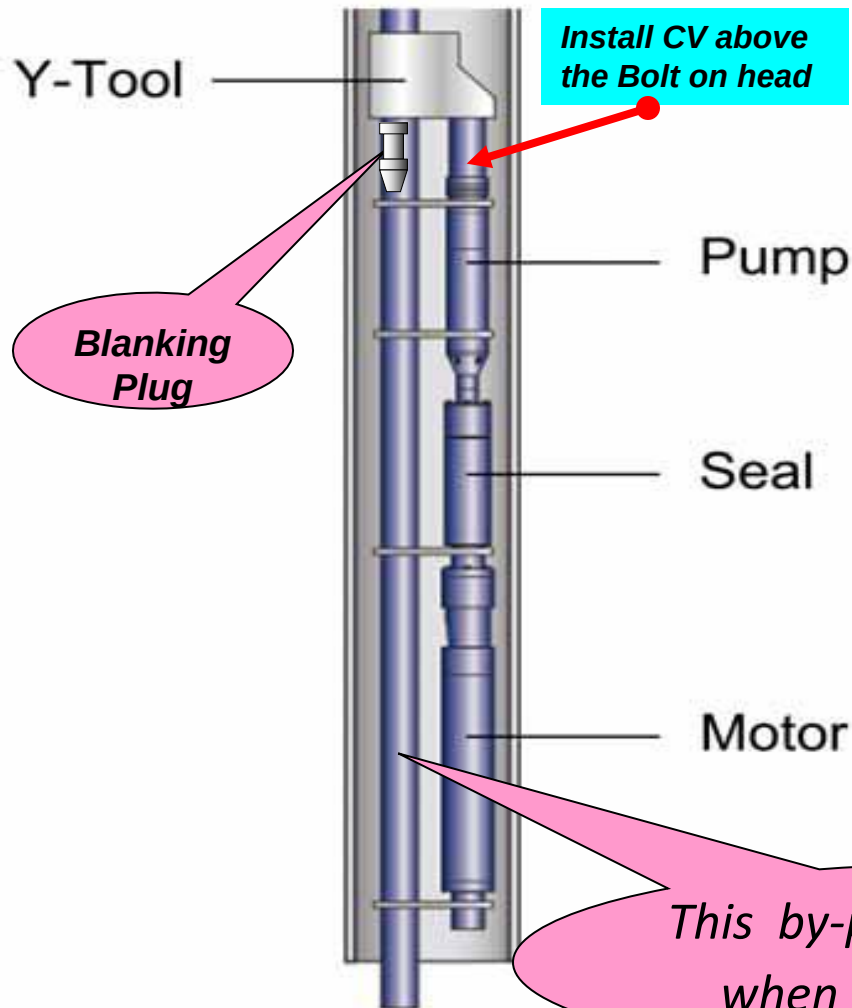
When plugging of ESP unit with scale, sand and we cleaning by pumping down chemical would be impossible because the check valve installed on the tubing string.

During the ESP is shutdown the fluid flowing turns the pump and motor in the wrong direction .

Power generation from motor is detected by the backspin relay and controller prevents restarting the ESP unit.

ESP- Y-Tool (By-pass)

Is a device to allow wire line or coiled tubing access below the ESP unit.



1- Running pressure and temperature survey tool by using wire line tool

2- Running well logging or coiled tubing tool.

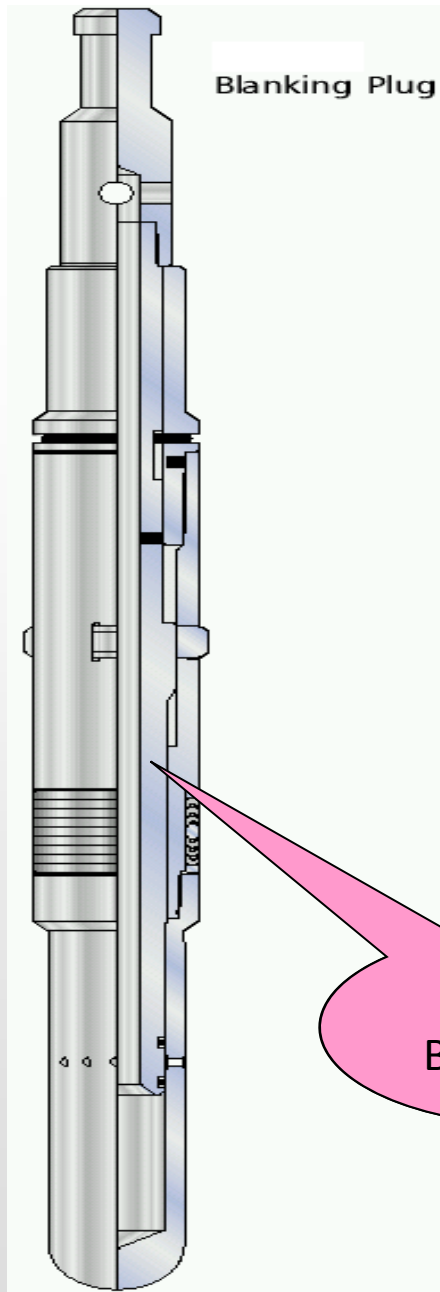
3- Can be also used for installing two parallel ESPs in the well.

4- Running down hole sampling.

This by-pass tubing plugged when the pump running.

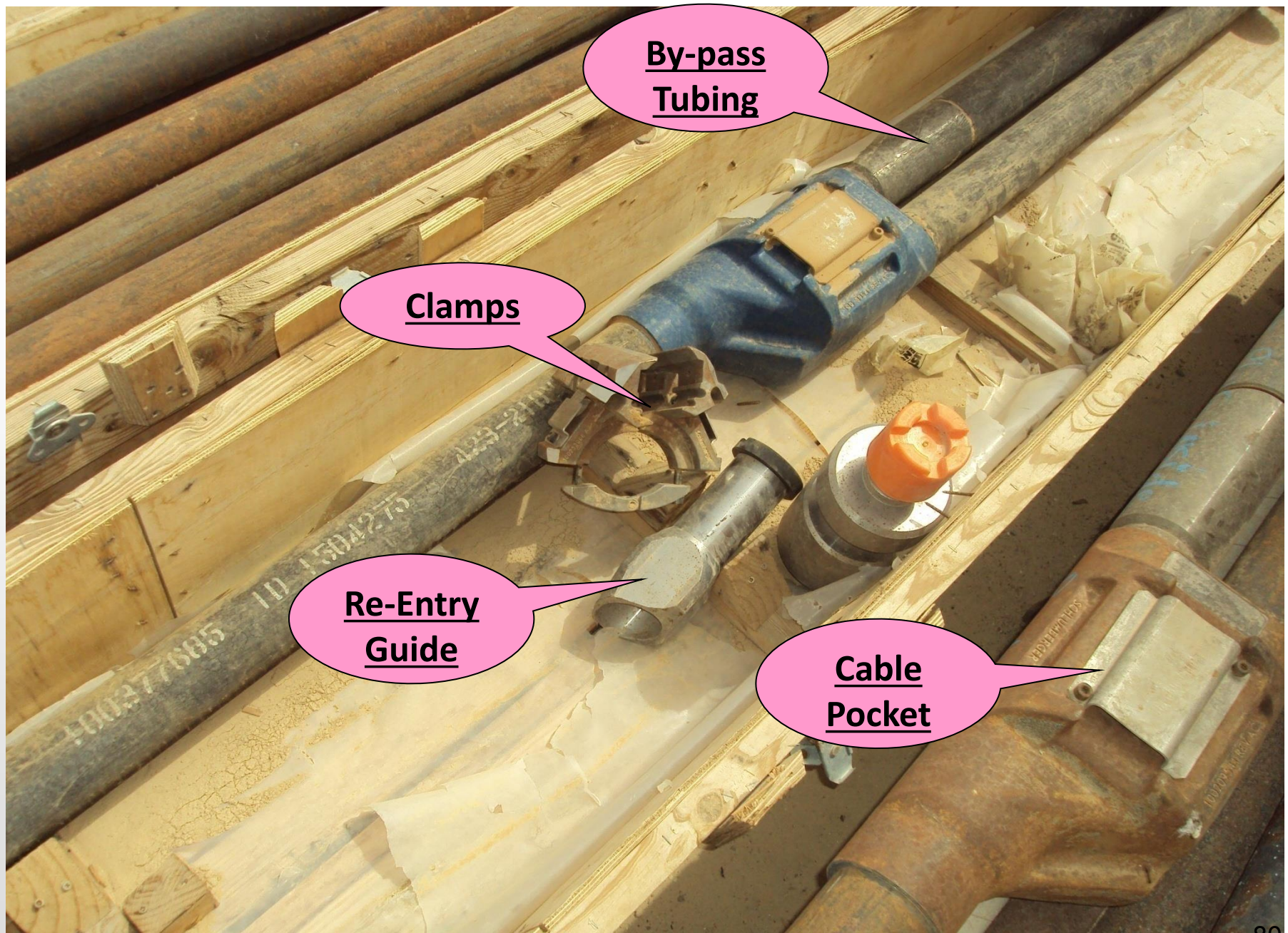


Pump
Head



Y- Tool
Balking Plug

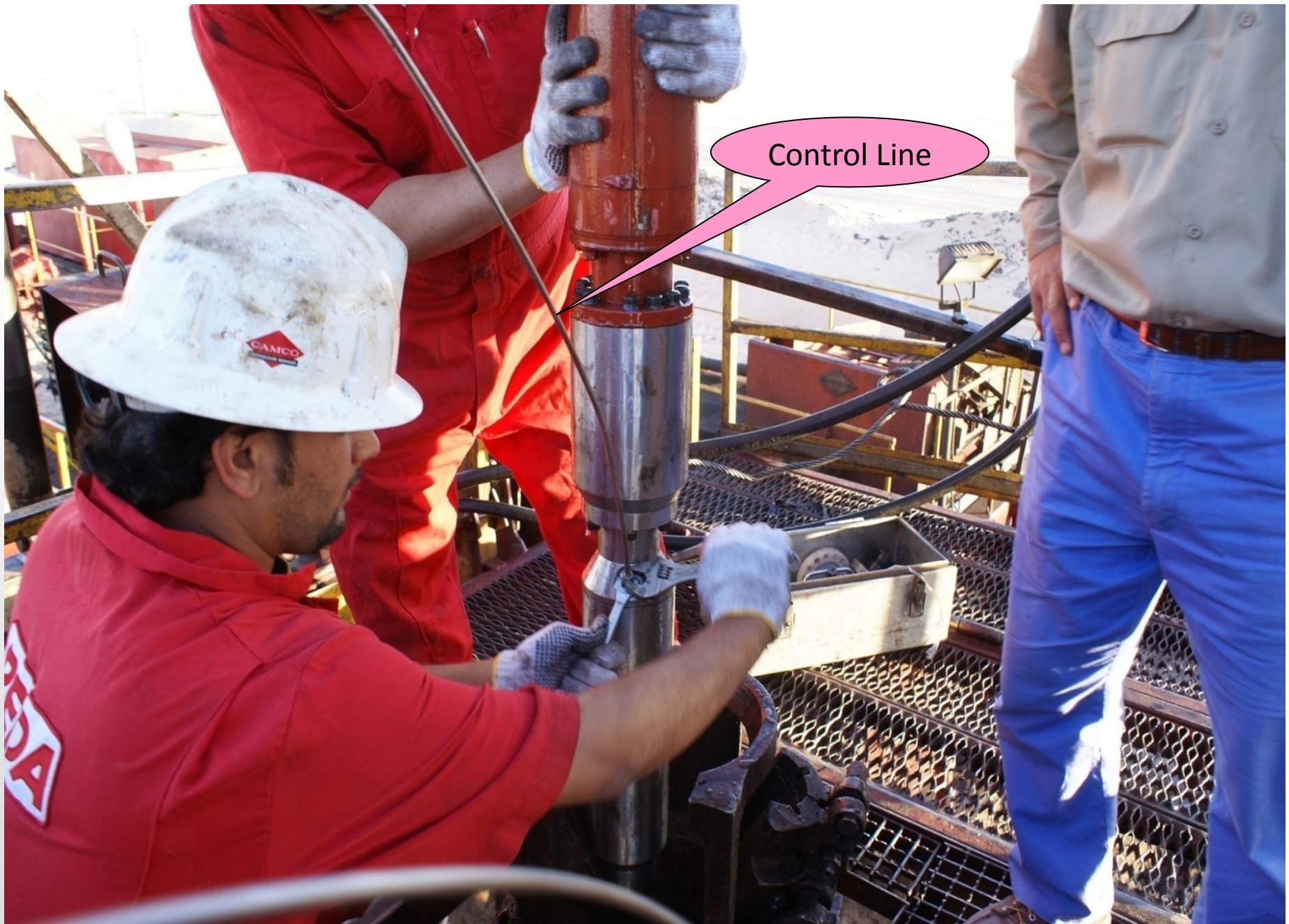




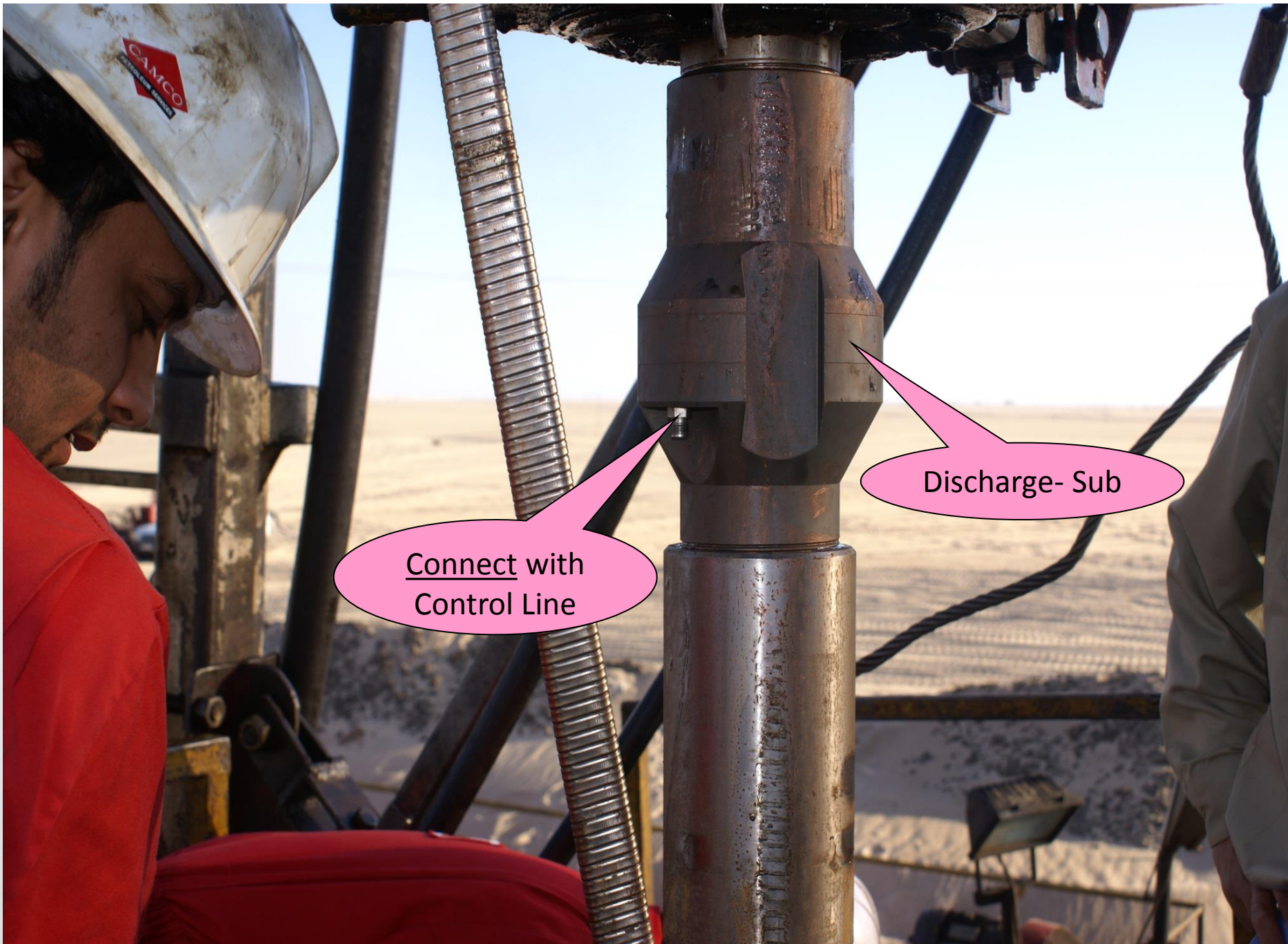
ESP Down hole Multi-Sensors

The Down-hole sensor is located below the motor and transmits important well and system data by the main power cable to the surface. Down hole sensors measure well parameters and provide critical data to increase ESP system efficiency and reliability.

Measures	<u>Type 0</u>	<u>Type 1</u>	<u>Type 2</u>
Pump Intake Pressure, PIP	Yes	Yes	Yes
Pump Discharge Pressure, PDP	<u>No</u>	Yes	Yes
Fluid Intake Temp, T_i	Yes	Yes	Yes
Electric Motor Temp, T_m	Yes	Yes	Yes
Flow Rate, q	<u>No</u>	<u>No</u>	Yes
Motor Vibration, Vib	Yes	Yes	Yes

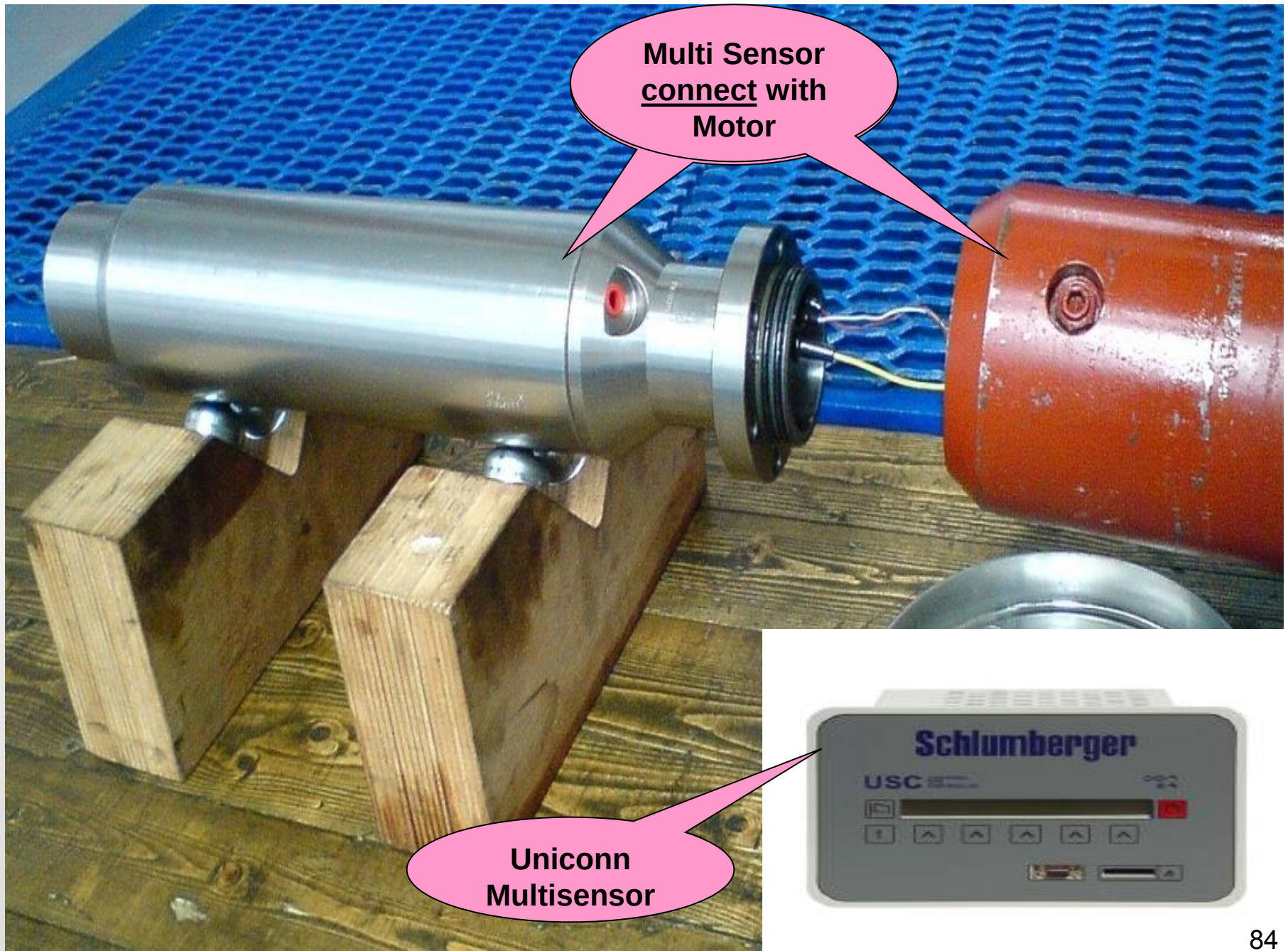


Control Line



Connect with
Control Line

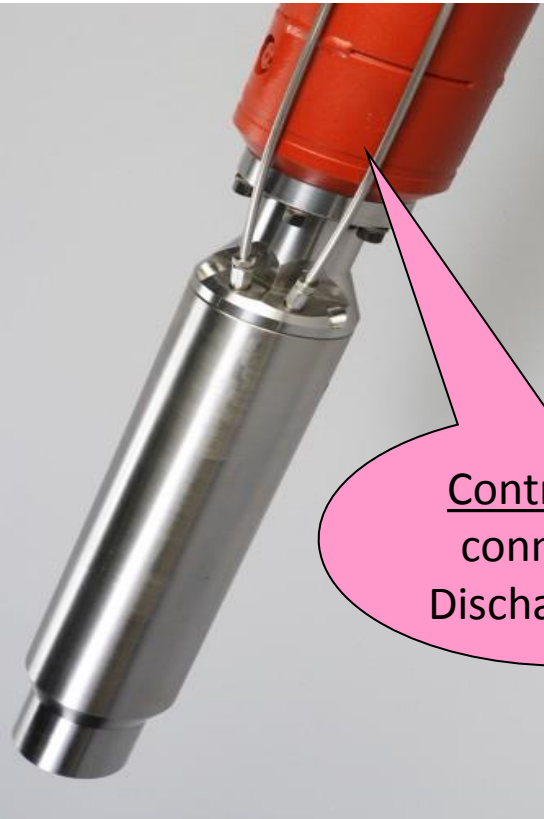
Discharge- Sub



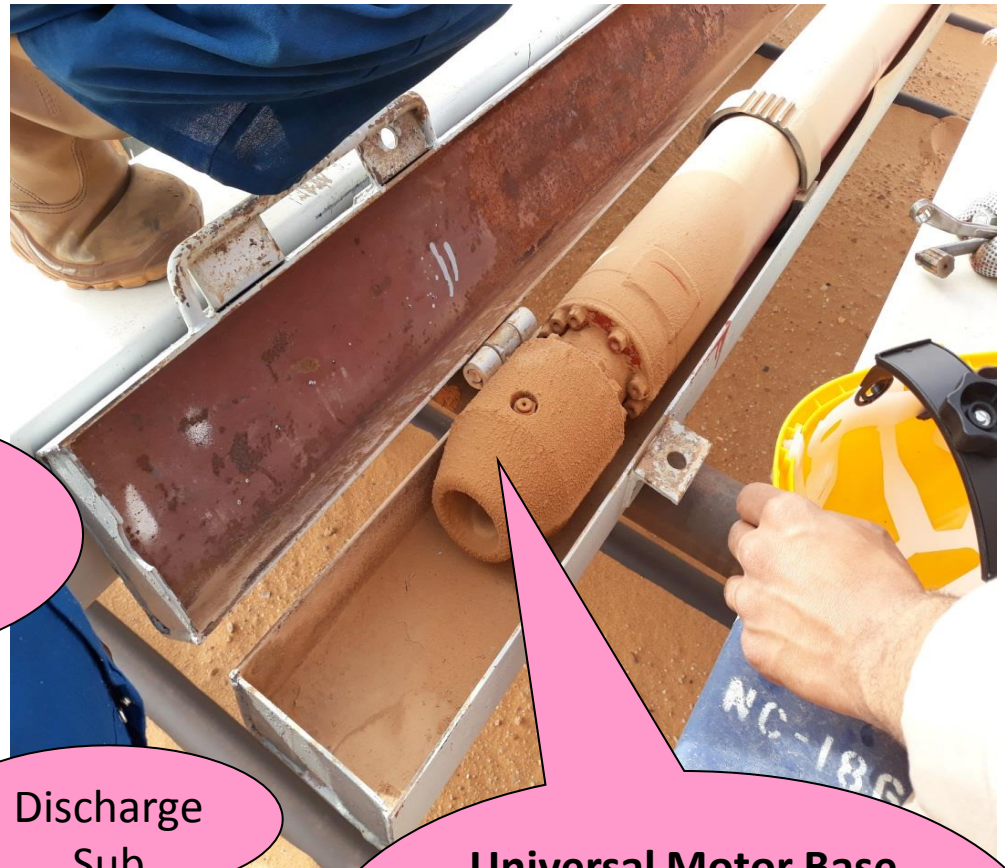
Multi Sensor
connect with
Motor

Uniconn
Multisensor





Control Line
connect to
Discharge Sub



Discharge
Sub

**Universal Motor Base
(UMB)**

It is connected at the
end of motor to close
the sensor connection

Down hole Multi-Sensors

What Can We Do From Multi-Sensor Data...?

- Increased Production. (**Monitoring**)
- To Optimized ESP Performance.
(**use the analysis result for optimization**)
- To Identification Of Pump Wear.
(**Protection**)
- To Increased Pump Run Life.
(**Result**)



The time between date put on production and date failure.

ESP- Multi-Sensors Output Data

<i>Time</i>	Pi	Pd	Ti	Tm	Vib	Cl
9:56:34	3029	4414	125.6	133.8	0.23	0.1
9:56:55	3027	4426	125.8	134	0.253	0.102
9:57:16	3026	4410	125.8	134.1	0.249	0.105
9:57:36	3031	4418	125.5	133.8	0.24	0.101
9:57:57	3024	4408	125.8	134	0.231	0.105
9:58:17	3023	4404	125.5	133.8	0.246	0.102
9:58:38	3033	4420	125.8	134.1	0.251	0.099
9:58:58	3029	4410	125.6	134	0.262	0.096

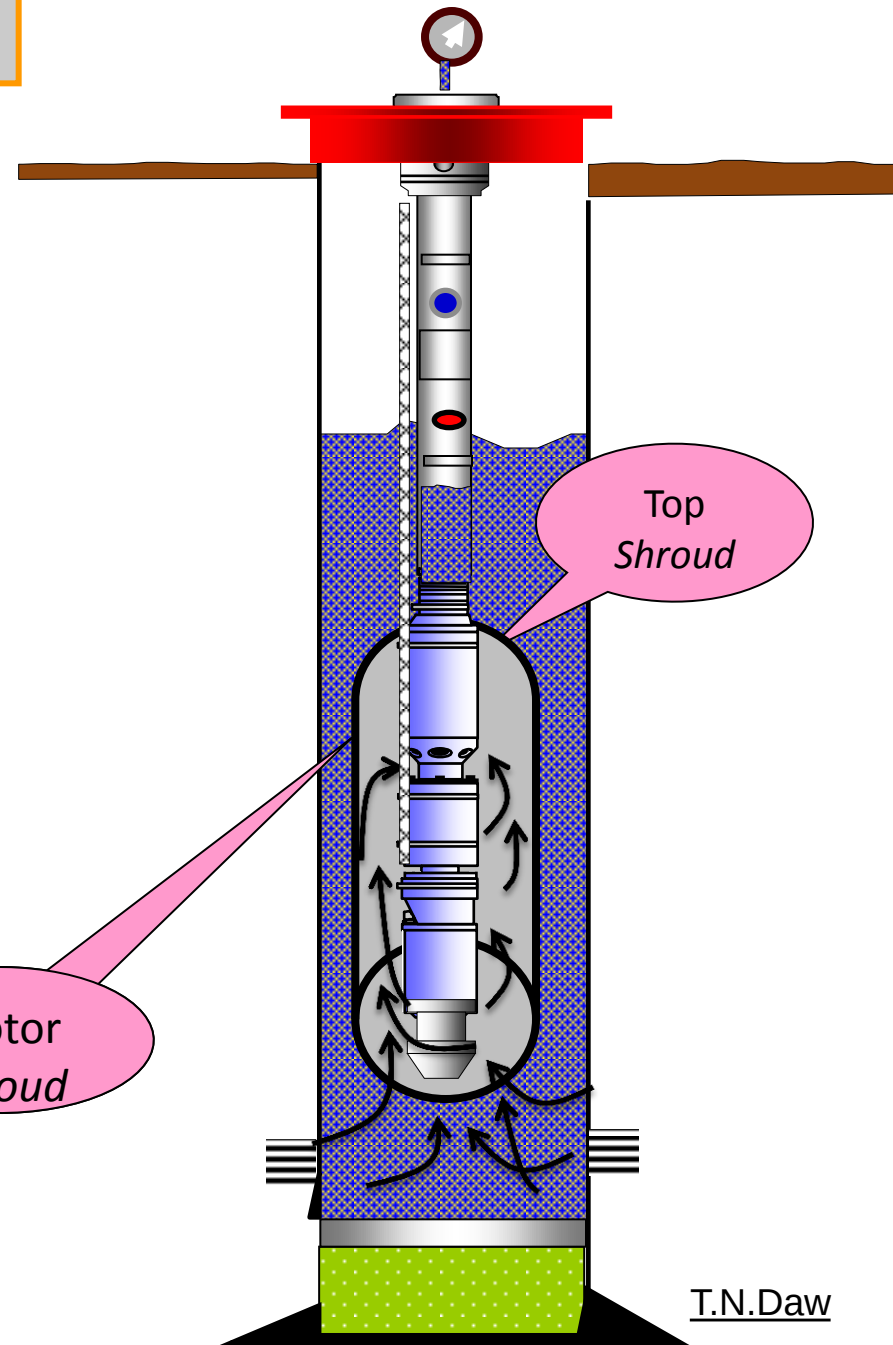
ESP- Motor Shrouded

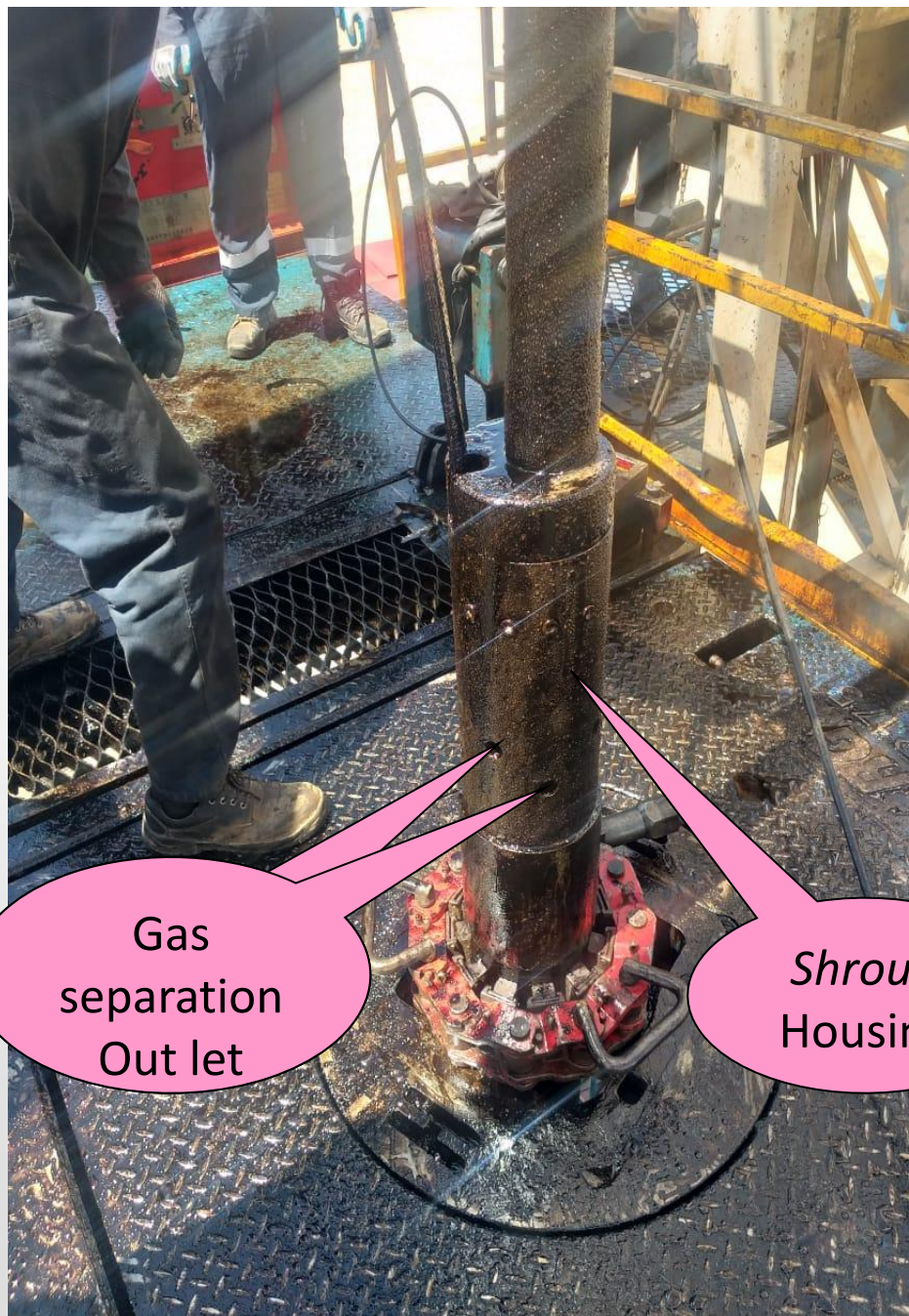
✓ Motor Shroud is a short sections of pipe around the length of the ESP unit (*that is bolted on the pump intake or gas separator and covers with protector and motor*).

✓ Motor Shroud Can not be used in casing smaller than 5 ½" OD.

- ❖ Directs fluid past the motor for cooling when fluid velocity is low (below minimum range).
- ❖ Allows pumps to be set below perforations .
- ❖ Reducing the annular area between the motor and production casing (**larger annular clearances**).

ESP- Motor Shrouded Configuration







7" (Twin-Seal) Hydrow II *ESP*- PKR

